Mobile and Autonomous Robotics

(UE23CS343BB7)

[Elective-4]

Specialization
Machine Intelligence & Data Science (MIDS)
Cyber Security & Connected Systems (CSCS)

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

Mobile and Autonomous Robots
(UE23CS343BB7)

Faculty Team

RR Campus

Dr. Ashok Kumar Patil	(ashokkumarp@pes.edu)
Dr. Adithya Balasubramanyam	(adithyab@pes.edu)
Dr. Roopa Ravish	(rooparavish@pes.edu)

EC Campus

Dr. Gokul Kannan	(gokul@pes.edu)
Prof. Naga Lakshmi	(nagalakshmi.s.r@pes.edu)
Dr. Mannar Mannan	(mannar.mj@pes.edu)
Prof. Ruby Dinakar	(rubydinakar@pes.edu)

PES UNIVERSITY

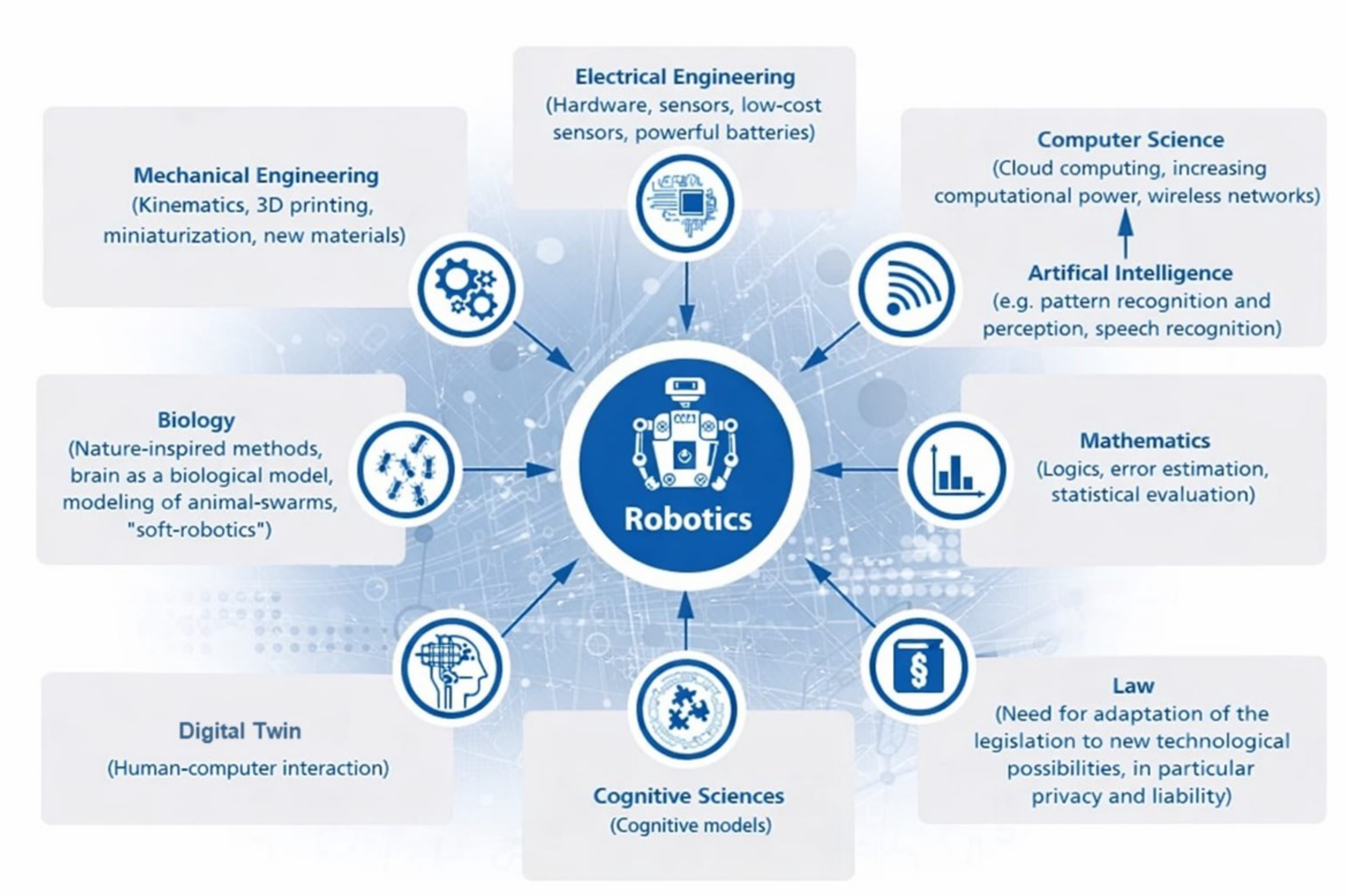
Autonomous Robots

- Key questions in Autonomous robotic
  1. What is around me?
  2. Where am I?
  3. Where am I going?
  4. How do I get there?

- Alternatively, these questions correspond to
  1. Sensor Interpretation: what objects are there in the vicinity?
  2. Localization: find your own position on a map (given or built autonomously)
  3. Map building: how to integrate sensor information and your own movement?
  4. Path planning: decide the actions to perform for reaching a target position

Mobile and Autonomous Robots

Robotics - A Multidisciplinary Approach



Mobile and Autonomous Robots

Applications for Robotic Technology:

- Industrial/assembly/inspection
- Search and rescue
- Hazardous operation
- Medical Applications
- Assistive/Rehabilitation
- Education
- Transportation
- Entertainment

Course Objectives:

- Understanding the past, present, and future of autonomous mobile robotics. Getting exposed to the current state-of-the-art of scientific literature.
- The course aims to teach the theoretical and practical fundamentals involved in designing and operating autonomous robots.
- The introductory discussions cover subtopics like robot perception, planning, and control.
- Other major topics include designing robot parts, integrating sensors, analyzing motion kinematics, simulation testing using ROS2, handling unmodeled environmental and social factors.

Course Outcomes:

- Students will gain a comprehensive understanding of autonomous robots and their applications.
- Understand the principles of robot manipulators, end-effectors, and identify the sensors used in robot control.
- Students will understand the ROS architecture and communication protocols and use ROS packages for robot software control.
- They will be able to use path planning and obstacle avoidance algorithms for robot navigation.
- Ultimately, students will be able to apply this knowledge to design and develop their own autonomous robot systems.

Course Content:

Unit 1: Introduction to Robotics and Locomotion: Robotics overview: Past, Present and Future; Robot Hardware and Software. AI in Robotics: Machine Learning Basics for Robotics. Locomotion: The basics of SOTA locomotion systems, Legged Robots, Wheeled Robots, Aerial Robots. Robot Kinematics, ROS overview: ROS/ROS2 for robotics, ROS architecture and communication protocols, ROS packages for robot hardware control.

Unit 2: Perception : Introduction to Perception, Sensors for Robots, Visual and inertial measurements: Gyroscope, accelerometers, IMU, GPS, Range sensors, camera vision and LiDAR. Fundamentals of Computer Vision and Image Processing and filtering, Feature Extraction: Object detection and Place Recognition, Stereo vision and 3D perception, Feature Extraction Based on Range Data (Laser, Ultrasonic).

Course Content:

Unit 3: Localization: Introduction and Challenges of Localization, To Localize or Not to Localize, Belief Representation, Map representation, Probabilistic Map-Based Localization: Markov localization, Kalman Filter localization, Autonomous Map Building: SLAM, EKF SLAM, Particle filter SLAM, GraphSLAM, Open challenges in SLAM.

Unit 4: Planning and Navigation : Introduction Path Planning and Navigation, Path planning: Graph search and Potential field path planning. Obstacle avoidance: Bug algorithm, Vector field histogram, bubble band technique, Curvature velocity techniques, Dynamic window approaches. Navigation Architectures. Revisiting Navigation using Reinforcement Learning and Imitation learning: A Robotics Perspective, Applications and Social Implications.

Recommended Materials

- Robot Operating System (ROS): http://wiki.ros.org/ROS/Tutorials
- ROS2 tutorials: https://docs.ros.org/en/foxy/Tutorials.html

Evaluation Pattern (Tentative)

	Marks to be conducted for	Marks to be scaled to
ISA 1 (Units 1 and 2) MCQ	40	10
ISA 2 (Units 3 and 4) MCQ	40	10
Lab&Assignments/Mini-Project	100+50	40
ESA (Units 1–4)	100	40
Total		**100**

* Subject to approval

Banana problem (Lab): 10 marks (conducted for 20 marks)
Orange problem (Assignment + Worksheet): 5 + 10 marks (conducted for 80 marks)
Jackfruit problem (Mini-Project): 15 marks (conducted for 50 marks)

Total: 40 marks

Mobile and Autonomous Robots

Rubrics for Mini-Project

Criteria \ Marks	5-Marks	4-Marks	3-Marks	2-Marks	1/0-Mark
Simulation phase- 1	An error free simulation as mentioned in the abstract. Answering 5 of the questions.	An error free simulation as mentioned in the abstract. Answering 4 or 3 of the questions.	An error free simulation as mentioned in the abstract. Answering 3 or 2 of the questions.	Errors or buggy simulation. Answering 4 or 3 of the questions.	Errors or buggy simulation and Answering 1 or 2 of the questions (1mark). If not answering almost any question (0 marks).
Simulation phase- 2/ Demonstration	An error free simulation and demonstration as mentioned in the abstract with a few last-minute modifications as instructed to check flexibility.	An error free simulation and demonstration as mentioned in the abstract.	An error free simulation and demonstration as mentioned in the abstract with a few errors.	An error free simulation and no demonstration.	No simulation and no demonstration.
Novelty and Viva phase 2	Novelty and 5 questions answered	Novelty and Answered 4 questions.	Novelty and Answered 3 questions.	No novelty and Answered 2 questions	No novelty and Answered 1 question.
Error Handling	Error handling in simulation and demonstration such that functioning is not disrupted majorly with an emergency kill/stop switch.	Error handling in simulation and demonstration with small errors still not handled and with an emergency kill/stop switch.	Error handling in simulation and emergency kill/stop switch but no error handling in demonstration	No error handling but kill/stop switch exists.	No error handling and no kill/stop switch exists.
Robot Interface/ Interaction	Ease of customization. Ease of use. Ability to control sensors, movements and actions. Reliability.	Ease of use. Ability to control sensors, movements and actions. Reliability.	Ease of use. Ability to control movements and actions. Reliability.	Ease of use. Reliability.	No UI.

Mobile and Autonomous Robotics

Books:

Prescribed Textbook:

1. Introduction to Autonomous Mobile Robots (Intelligent Robotics and Autonomous Agents series) second edition by Roland Siegwart, Illah Reza Nourbakhsh, Davide Scaramuzza

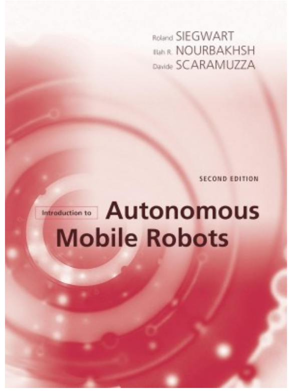
2. Probabilistic Robotics By Sebastian Thrun, Wolfram Burgard and Dieter Fox. (Intelligent Robotics and Autonomous Agents series); 1st Edition

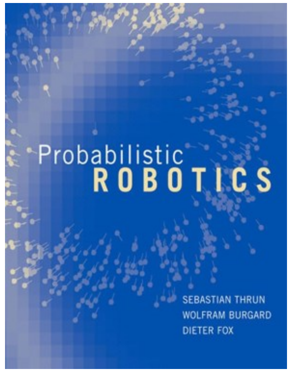
References:

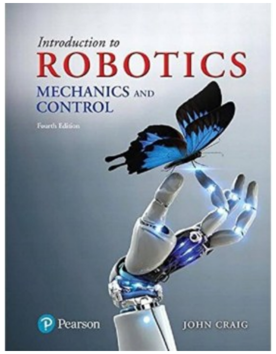
1. Introduction to Autonomous Robots: Nikolaus Correll, Magellan Scientific, 2016.

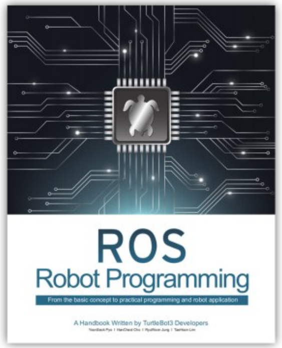
2. ROS Robot Programming, ROBOTIS Co., Ltd. From the basic concept to practical programming and robot application. YoonSeok Pyo, HanCheol Cho, RyuWoon Jung, and TaeHoon Lim.

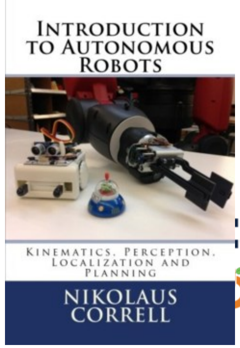
3. Introduction to Robotics: Mechanics and Control 4th Edition, John Craig, ISBN-13: 978-0133489798, Pearson; 4th edition











Mobile and Autonomous Robotics

MARS Lab

Robotic Arm

THANK YOU

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

https://www.iot.pes.edu/mars

Mobile and Autonomous Robotics

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Introduction to Autonomous Robots

Autonomous Robots, challenges and applications

Session – 1.1

What is a Robot…?

What is Robotics…?

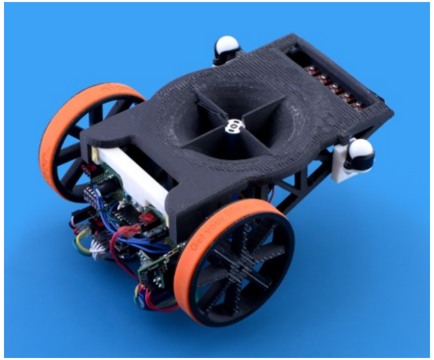
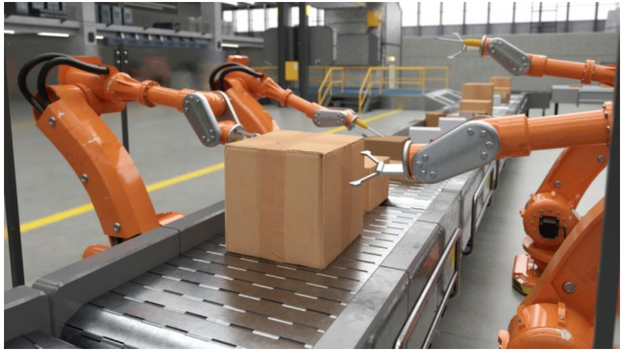
Why is Robotics needed…?



What is a Robot…?

What is Robotics…?

Why is Robotics needed…?

Introduction to Autonomous Robotics

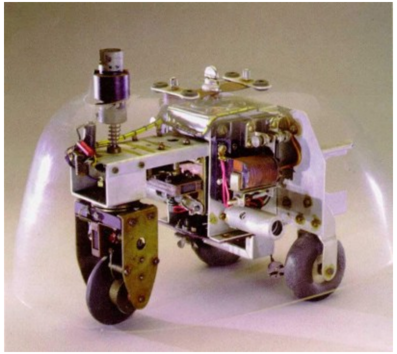
Definition of Autonomous Robots

- Autonomous robots are robotic systems that possess the capability to perform tasks and make decisions without continuous human intervention.
- These robots rely on a combination of sensors, actuators, and onboard computational systems, often integrating advanced algorithms and artificial intelligence (AI) techniques.
- Autonomous robots combine hardware and software elements to perceive, interpret, and act in their environment without direct human control.

- The technical sophistication lies in the integration of sensors, processing units, control algorithms, actuators, and communication systems to achieve a high degree of autonomy and adaptability in diverse applications.

Introduction to Autonomous Robots

- The **first autonomous robots** environment were known as **Elmer and Elsie**, which were constructed in the late 1940s by W. Grey Walter.
- The first robots in history that were programmed to "think" the way biological brains do and meant to have free will.
- **Historic** examples include **space probes**.
- **Modern** examples include **self-driving vacuums** and **cars**.
- Industrial robot arms that work on assembly lines inside factories **may** also be considered autonomous robots, though their **autonomy is restricted** due to a **highly structured environment** and their **inability to locomote.**



Characteristics of Autonomous Robots

- **Perception:** They can sense their environment using a variety of sensors, such as cameras, Lidar, and radar. This allows them to gather information about their surroundings, such as the location of obstacles, the presence of people, and the condition of the terrain.

- **Decision-making:** They can process the information they gather from their sensors and use it to make decisions about their actions. This may involve following a pre-programmed plan, adapting to changing conditions, or even learning from experience.

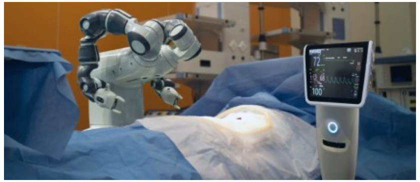
- **Actuation:** They can physically interact with their environment using motors, actuators, and other mechanisms. This allows them to move around, manipulate objects, and perform tasks.

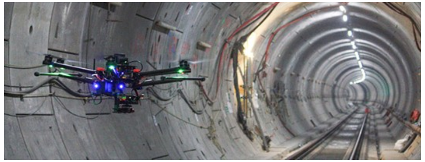
Mobile and Autonomous Robotics

Introduction to Autonomous Robots

Applications of Autonomous Robots

- **Manufacturing**: They can be used to perform tasks such as assembly, welding, and painting.
- **Logistics**: They can be used to move materials around warehouses and other facilities.
- **Healthcare**: They can be used to deliver medical supplies, perform surgery, and provide companionship to patients.
- **Agriculture**: They can be used to plant and harvest crops, weed fields, sprying pestisides and monitor livestock.
- **Exploration**: They can be used to explore dangerous or remote environments, such as underwater or in space.





Robots on the Rise

Autonomous robots are rapidly evolving and hold immense potential to revolutionize various aspects of our lives.

1. **Enhancing Efficiency and Productivity:**
    - **Logistics and Warehousing**: Imagine fleets of autonomous robots seamlessly navigating warehouses, optimizing inventory management, and efficiently transporting goods, significantly boosting productivity and reducing labor costs.
    - **Manufacturing**: Autonomous robots can tirelessly handle repetitive tasks like assembly, welding, and painting, improving accuracy and consistency while freeing up human workers for more strategic roles.
    - **Agriculture**: From autonomous tractors precisely planting and harvesting crops to drones monitoring field health and optimizing irrigation, robots can revolutionize agriculture, leading to increased yields and resource efficiency.

Robots on the Rise

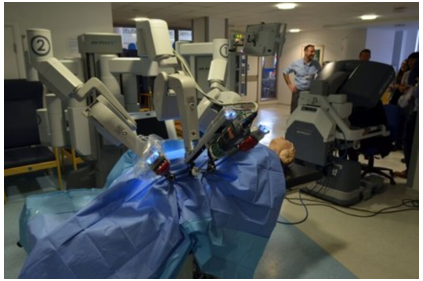
2. **Improving Safety and Security:**
   - **Hazardous Environments**: Robots can perform dangerous tasks in environments like disaster zones, nuclear reactors, or deep-sea exploration, minimizing risks for human personnel.
   - **Security and Surveillance**: Autonomous robots equipped with advanced sensors can patrol sensitive areas, deter crime, and provide real-time security insights, enhancing public safety.
   - **Search and Rescue**: Robots can navigate tight spaces and hazardous conditions during search and rescue operations, locating victims and providing critical assistance.

Robots on the Rise

3. **Expanding Human Capabilities**:

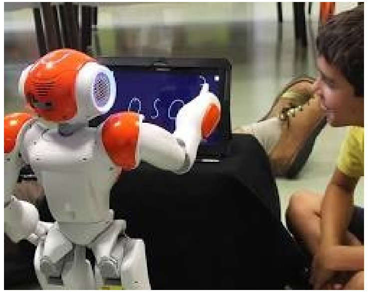
   - **Healthcare:** Surgical robots can assist doctors with minimally invasive procedures, offering greater precision and reducing risks for patients. Assistive robots can aid people with disabilities, enhancing their independence and quality of life.

   - **Space Exploration:** Rovers and probes equipped with advanced AI can explore distant planets and moons, gathering valuable data and paving the way for future human missions.



Robots on the Rise

4. **Transforming Daily Life:**
   - **Domestic Assistance:** Robots can handle household chores like cleaning, cooking, and laundry, freeing up time for leisure and family activities.
   - **Elderly Care:** Assistive robots can provide companionship, monitor health, and offer reminders for medication, enabling elderly individuals to live independently for longer.
   - **Personalization and Entertainment:** Educational robots can personalize learning experiences for children, while robots designed for entertainment can provide companionship and interactive experiences.



Challenges

Autonomous projects, despite their exciting potential, face a multitude of challenges that need to be addressed before they can be widely adopted. These challenges can be broadly categorized into **technical, ethical,** and **social spheres**.

- **Technical Challenges**:
  - **Perception and Decision-Making**: Accurately perceiving and understanding complex environments in real-time remains a hurdle. Factors like varying lighting, unpredictable weather, and dynamic obstacles can confuse sensors and algorithms, leading to errors in judgement.
  - **Robustness and Reliability**: Autonomous systems need to be robust enough to handle unexpected situations and operate reliably even in the face of hardware or software failures. Ensuring fail-safe mechanisms and redundancy is crucial for safety and trust.
  - **Data Security and Privacy**: The vast amount of data collected by autonomous systems raises concerns about privacy and security. Robust cybersecurity measures are essential to prevent hacking and misuse of sensitive data.

Challenges

- ### Ethical Challenges:
  - **Moral Dilemmas:** Autonomous systems may face situations where they need to make moral decisions, such as choosing who to harm in an unavoidable accident. Defining ethical frameworks and guiding principles for such situations is critical.
  - **Job Displacement:** Automation through robots raises concerns about job displacement, particularly in repetitive and manual labor sectors. Strategies for reskilling and creating new job opportunities need to be considered.
  - **Bias and Discrimination:** Algorithms used in autonomous systems can perpetuate existing biases and discrimination based on factors like race, gender, or socioeconomic status. Mitigating bias and ensuring fairness in algorithms is crucial.

Challenges

- ### Social Challenges:

    - **Public Acceptance and Trust:** Gaining public trust and acceptance for autonomous systems, especially in safety-critical applications like self-driving cars, requires addressing concerns about safety, transparency, and accountability.
    - **Regulatory Frameworks:** Developing clear and comprehensive regulations for the development, testing, and deployment of autonomous systems is essential to ensure responsible and ethical use.
    - **Infrastructure and Accessibility:** Adapting existing infrastructure and ensuring equitable access to the benefits of autonomous technology for all communities are crucial considerations.

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Introduction to Autonomous Robotics

Historical overview and future of ARs

A Whirlwind Tour of Autonomous Robots: A Historical Overview

- **1912:** Leonardo Torres Quevedo builds El Ajedrecista, the first truly autonomous machine capable of playing chess! This electromechanical marvel could checkmate a human opponent in an endgame.
- **1948-1949:** Enter Elsie and Elmer, the brainchild of William Grey Walter. These "tortoise" robots explored their environment using light sensors and simple behaviors, marking the dawn of truly autonomous robots.
- **1950s:** George Devol invents Unimate, the first industrial robot. This programmable arm ushered in a new era of automation in factories.
- **1960s:** SHAKEY, a robot developed at Stanford Research Institute, takes things up a notch with cameras, tactile sensors, and rudimentary planning capabilities. It navigates obstacle courses and even cleans up spills!
- **1970s:** Rise of the "pick-and-place" robots! These industrial workhorses become commonplace, grabbing and positioning objects with increasing precision.
- **1980s-1990s:** AI enters the scene! Robots get smarter, with learning algorithms and improved sensing allowing them to adapt to changing environments.

Mobile and Autonomous Robotics

Historical Overview and Future of ARs

- **2000s:** Sociable robots like Aibo and Paro emerge, designed to interact and respond to human emotions. The Roomba conquers homes, keeping floors clean autonomously.

- **2010s-Present:** The boom of self-driving cars, drones, and delivery robots! Advances in sensors, computing power, and algorithms push the boundaries of autonomy.

- **The Future:** Where will robots take us? From exploring Mars to assisting surgeons, the possibilities are endless. Ethical considerations and responsible development remain crucial as we navigate this exciting frontier.

Autonomous Robots: Future

- **Integration with AI and Machine Learning:** Continued integration of artificial intelligence (AI) and machine learning (ML) technologies will enhance the decision-making capabilities of autonomous robots. This will enable robots to learn and adapt to new environments and tasks.
- **Human-Robot Collaboration:** Increased focus on developing robots that can collaborate seamlessly with humans in shared workspaces. This includes robots working alongside human workers in manufacturing, healthcare, and other industries, augmenting human capabilities.
- **Robotic Swarms:** Advances in swarm robotics will lead to the development of coordinated groups of autonomous robots working together to accomplish complex tasks. Swarm robotics can be applied in areas such as search and rescue missions, environmental monitoring, and agriculture.

Autonomous Robots: Future

- **Autonomous Vehicles and Transportation:** The automotive industry is expected to witness widespread adoption of autonomous vehicles, including self-driving cars, drones, and delivery robots. This has the potential to revolutionize transportation, making it safer and more efficient.
- **AI-Powered Robotics in Healthcare:** Increased utilization of autonomous robots in healthcare settings, performing tasks such as surgery, patient care, and medication delivery. AI algorithms will contribute to diagnostics and treatment planning.
- **Advanced Sensing Technologies:** Continued development of advanced sensors, such as improved LiDAR and computer vision systems, will enhance the perception capabilities of autonomous robots. This is crucial for safe and efficient navigation in various environments.

Autonomous Robots: Future

- **Research in Soft Robotics:** Advances in soft robotics, enabling the development of robots with flexible and adaptable structures. This is particularly relevant for applications requiring interactions with delicate objects or environments.
- **Customization and Adaptability:** Development of robots with increased customization and adaptability, allowing them to be easily programmed and configured for a wide range of tasks.
- **Energy Efficiency and Sustainability:** Emphasis on designing energy-efficient autonomous robots and incorporating sustainable technologies. This includes optimizing power consumption and exploring renewable energy sources.
- **Ethical Considerations and Regulation:** Increased attention to ethical considerations and regulatory frameworks surrounding the deployment of autonomous robots. Ensuring responsible and ethical use, addressing concerns about job displacement, privacy, and safety.

Mobile and Autonomous Robotics (UE23CS343BB7)

Introduction to Autonomous Robotics

Basic Concepts and Terminology

Basic Concepts and Terminologies

- **Sensors:** Devices that gather information from the robot's environment. Examples include cameras, LiDAR, radar, sonar, and touch sensors.
- **Actuators:** Mechanisms that enable the robot to interact with its environment. Examples include motors, servos, and grippers.
- **Control Algorithms:** Sets of instructions that govern the robot's behavior, including navigation, decision-making, and task execution.
- **Navigation:** The process by which a robot moves from one location to another, often involving path planning and obstacle avoidance.
- **Localization:** Determining the robot's position within its environment.
- **Mapping**: Creating a representation of the environment, often used in conjunction with localization for navigation purposes.
- **Path Planning**: Calculating the optimal path for a robot to reach its destination while avoiding obstacles.
- **Collision Avoidance:** Techniques and algorithms used to prevent the robot from colliding with obstacles during navigation.

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Basic Concepts and Terminologies

- **Human-Robot Interaction (HRI):** The study of interactions between humans and robots, including design principles for effective communication and collaboration.
- **Swarm Robotics:** A field of robotics that involves the coordination and collaboration of multiple robots to achieve a common goal.
- **Simultaneous Localization and Mapping (SLAM):** A technique used by robots to create a map of an unknown environment while simultaneously determining their own location within that environment.
- **Teleoperation:** Controlling a robot remotely from a distance, often used for tasks in hazardous or inaccessible environments.
- **Feedback Control System:** A system that uses feedback from sensors to adjust and maintain the desired state or behavior of the robot.
- **End-Effector:** The tool or device at the extremity of a robot's arm, used for performing specific tasks.
- **Task Planning:** The process of determining a sequence of actions to achieve a specific goal or task.

Perception:

- The process of gathering and interpreting information about the environment around the robot using sensors like cameras, LiDAR, and sonar.

- This allows the robot to understand its surroundings and make informed decisions.

Example: A robot arm using a camera to recognize an object and adjust its grasp accordingly.

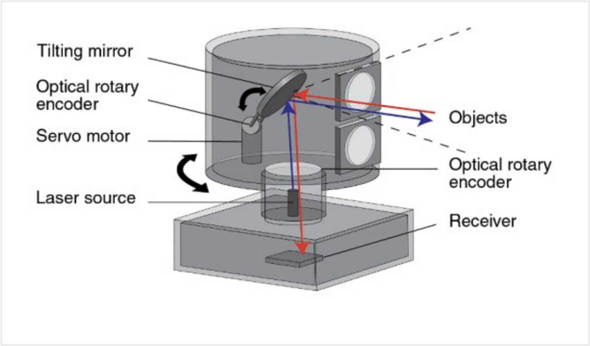
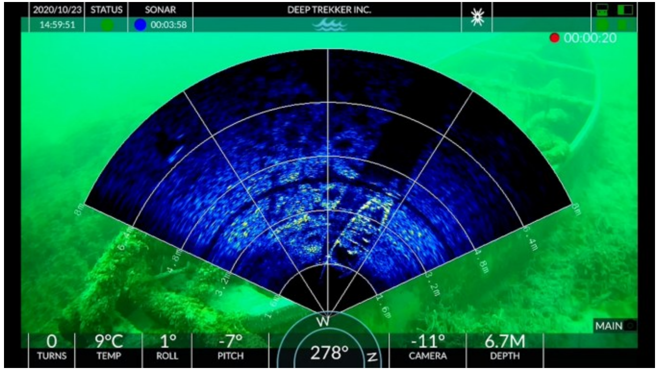


Image of a rotating LiDAR unit. Image courtesy of Renishaw.



Localization:

- The process of determining the robot's current position and orientation within its environment.
- This requires integrating data from various sensors like GPS, odometry (wheel rotations), and laser scanning.

  Example: A robot vacuum cleaner using its sensors to locate itself in a room and navigate around furniture.

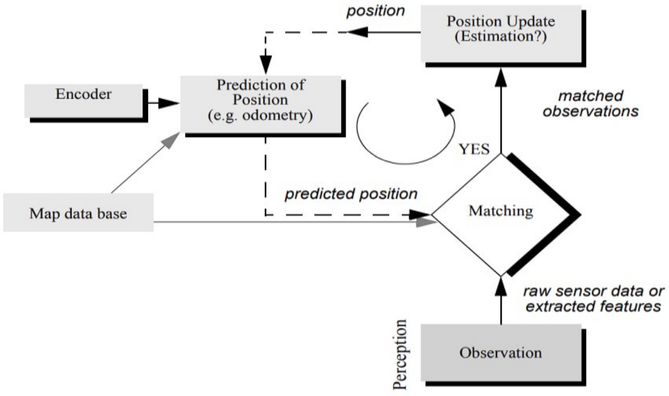


Figure 5.2
General schematic for mobile robot localization.

Path Planning:

- The process of determining a safe and efficient route for a robot to move from a starting point to a goal point, taking into account obstacles, terrain, and other constraints.
- Path planning is a strategic problem-solving competence, since the robot must decide what to do over the long term to achieve its goals.

Example: A delivery robot planning the best route through a city to complete its deliveries.

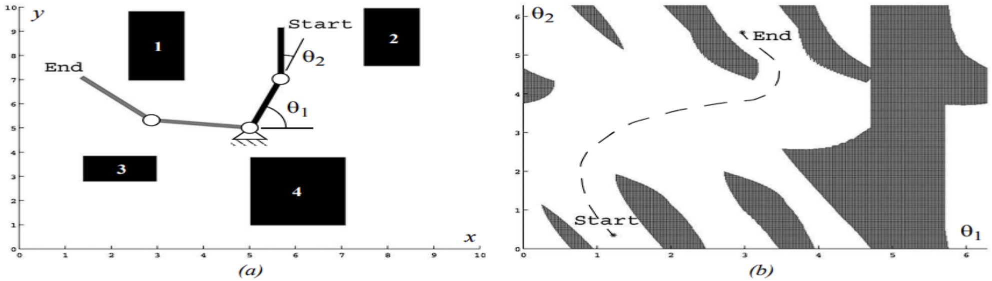


Figure 6.1
Physical space (a) and configuration space (b): (a) A two-link planar robot arm has to move from the configuration *start* to *end*. The motion is thereby constraint by the obstacles 1 to 4. (b) The corresponding configuration space shows the free space in joint coordinates (angle θ_1 and θ_2) and a path that achieves the goal.

Navigation:

- The act of guiding a robot through its environment along the chosen path determined by path planning.
- This involves real-time adjustments for unexpected obstacles or changes in the environment.

Example: A self-driving car navigating traffic and adhering to road rules while following its planned route.

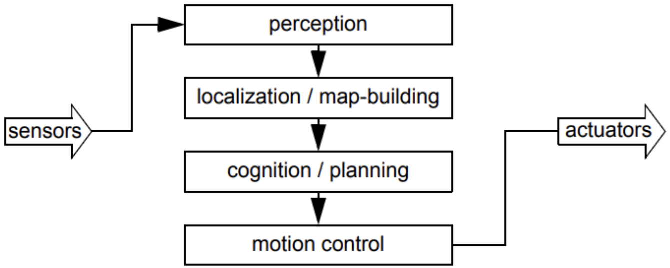


Figure 5.8
An architecture for map-based (or model-based) navigation.

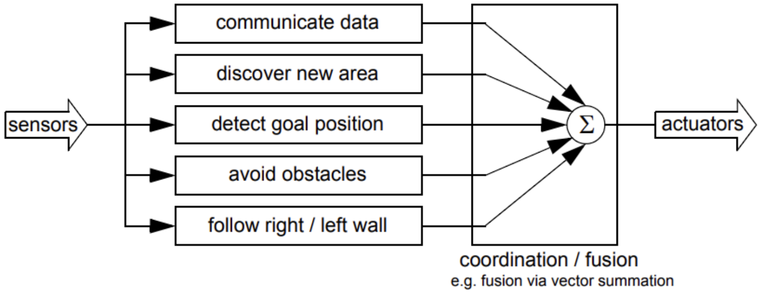


Figure 5.7
An architecture for behavior-based navigation.

THANK YOU

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

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UE23CS343BB7

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

Mobile and Autonomous Robotics

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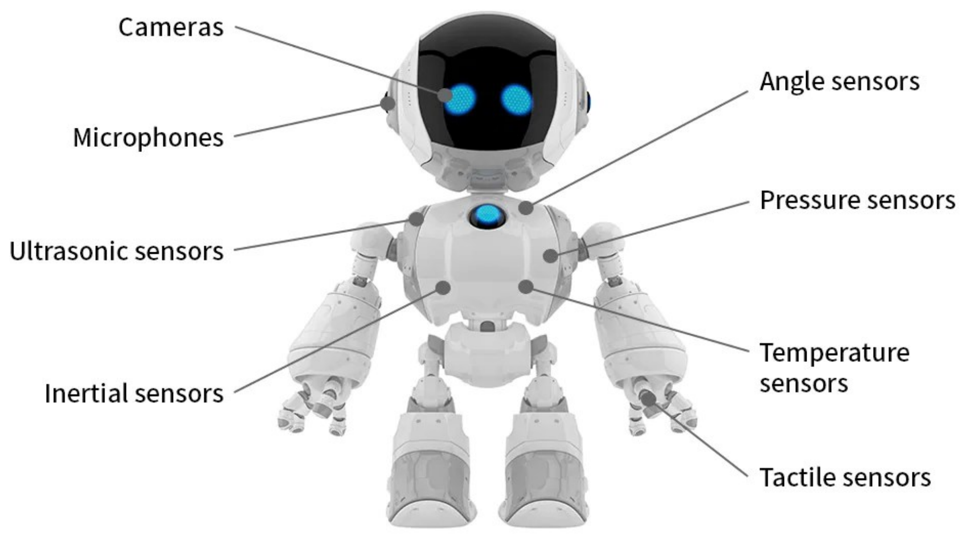
Introduction to Autonomous Robotics

Sensors and Actuators, Types and Applications

Session – 2.1

Sensors

- These are the robot's **eyes and ears** gathering information about the surrounding environment.
- Some sensors are used to measure simple values such as the internal temperature of a robot's electronics or the rotational speed of the motors.
- Other more sophisticated sensors can be used to acquire information about the robot's environment or even to measure directly a robot's global position.



Sensors

- Sensors are devices or instruments that are designed to detect and measure physical properties or changes in the environment and convert this information into signals or data that can be interpreted, displayed, or used for control purposes. Sensors play a crucial role in various fields, including robotics, industrial automation, healthcare, environmental monitoring, and more.

- The key characteristics of sensors include their ability to sense specific physical phenomena, such as light, temperature, pressure, sound, motion, or chemical composition. They act as the interface between the physical world and electronic systems, providing valuable input for monitoring and controlling processes.

- The information gathered by sensors can be analog or digital and is typically used to make informed decisions, trigger actions, or provide feedback in automated systems. Sensors are integral components in the development of smart devices, IoT (Internet of Things) applications, and autonomous systems where real-time data is essential for efficient and responsive operation.

Sensor Classification

Sensors clasified based on two important functional axes: **proprioceptive/exteroceptive** and **passive/active.**

- **Proprioceptive** sensors measure values internal to the system (robot), for example, motor speed, wheel load, robot arm joint angles, and battery voltage.

- **Exteroceptive** sensors acquire information from the robot's environment, for example, distance measurements, light intensity, and sound amplitude. Hence exteroceptive sensor measurements are interpreted by the robot in order to extract meaningful environmental features.

- **Passive** sensors measure ambient environmental energy entering the sensor. Examples of passive sensors include temperature probes, microphones, and CCD or CMOS cameras.

- **Active** sensors emit energy into the environment, then measure the environmental reaction. Because active sensors can manage more controlled interactions with the environment, they often achieve superior performance. However, active sensing introduces several risks: the outbound energy may affect the very characteristics that the sensor is attempting to measure.

Introduction to Autonomous Robotics

The sensor classes in table are arranged in ascending order of complexity and descending order of technological maturity.

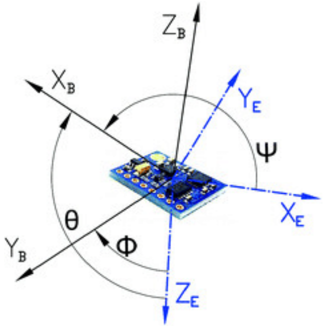
- **Tactile sensors** and **proprioceptive sensors** are critical to virtually all mobile robots and are well understood and easily implemented.
- **Commercial quadrature encoders**, for example, may be purchased as part of a gear-motor assembly used in a mobile robot.
- At the other extreme, visual interpretation by means of one or more **CCD/CMOS cameras** provides a broad array of potential functionalities, from obstacle avoidance and localization to human face recognition.

Table 4.1
Classification of sensors used in mobile robotics applications

General classification (typical use)	Sensor Sensor System	PC or EC	A or P
Tactile sensors (detection of physical contact or closeness; security switches)	Contact switches, bumpers Optical barriers Noncontact proximity sensors	EC EC EC	P A A
Wheel/motor sensors (wheel/motor speed and position)	Brush encoders Potentiometers Synchros, resolvers Optical encoders Magnetic encoders Inductive encoders Capacitive encoders	PC PC PC PC PC PC PC	P P A A A A A
Heading sensors (orientation of the robot in relation to a fixed reference frame)	Compass Gyroscopes Inclinometers	EC PC EC	P P A/P
Acceleration sensor	Accelerometer	PC	P
Ground beacons (localization in a fixed reference frame)	GPS Active optical or RF beacons Active ultrasonic beacons Reflective beacons	EC EC EC EC	A A A A
Active ranging (reflectivity, time-of-flight, and geometric triangulation)	Reflectivity sensors Ultrasonic sensor Laser rangefinder Optical triangulation (1D) Structured light (2D)	EC EC EC EC EC	A A A A A
Motion/speed sensors (speed relative to fixed or moving objects)	Doppler radar Doppler sound	EC EC	A A
Vision sensors (visual ranging, whole-image analysis, segmentation, object recognition)	CCD/CMOS camera(s) Visual ranging packages Object tracking packages	EC	P

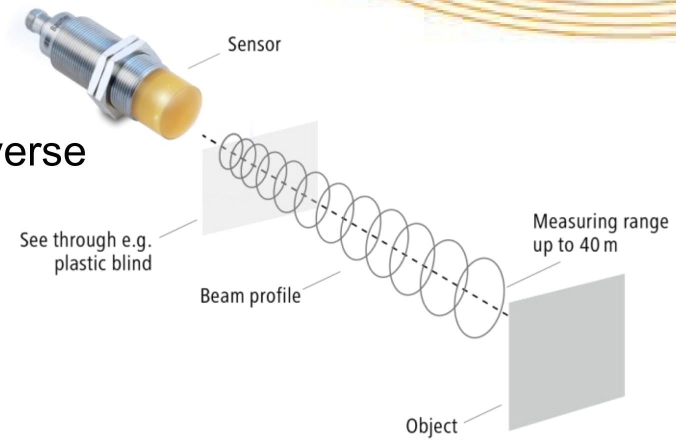
A, active; P, passive; P/A, passive/active; PC, proprioceptive; EC, exteroceptive.

Sensors: Types and Applications



- **Inertial Measurement Unit (IMU):**
  - Type: Gyroscopes and accelerometers.
  - Applications: Measurement of acceleration, orientation, and angular velocity. Used for navigation, balancing, and motion control in robots.

- **Lidar (Light Detection and Ranging):**
  - Type: Laser-based ranging sensor.
  - Applications: 3D mapping of the environment, obstacle detection and avoidance, localization, and navigation.



- **Radar (Radio Detection and Ranging):**
  - Type: Radio frequency-based ranging sensor.
  - Applications: Detection of objects at longer ranges, especially in adverse weather conditions. Used for obstacle detection and navigation.

Sensors: Types and Applications

- **Ultrasonic Sensors:**
  - Type: Sound wave-based ranging sensor.
  - Applications: Proximity sensing, obstacle detection, and navigation in close quarters. Commonly used in robotics for short-range applications.

- **Camera Systems:**
  - Type: Optical sensors capturing visual information.
  - Applications: Object recognition, scene interpretation, navigation, and mapping. Cameras are essential for computer vision applications in robotics.

- **GPS (Global Positioning System):**
  - Type: Satellite-based positioning system.
  - Applications: Outdoor localization, waypoint navigation, and mapping. GPS is commonly used for robots operating in large outdoor environments.

- **Wheel Encoders:**
  - Type: Rotary encoders on robot wheels.
  - Applications: Measurement of wheel rotations, speed control, and odometry for estimating robot position and movement.

Sensors: Types and Applications

- **Force/Torque Sensors:**
  - Type: Sensors measuring force and torque.
  - Applications: Interaction with the environment, object manipulation, and force feedback. Used in robotic arms and grippers for delicate and precise tasks.

- **Touch Sensors:**
  - Type: Tactile sensors.
  - Applications: Detection of contact or pressure. Used for object manipulation, grasping, and sensing physical interactions.

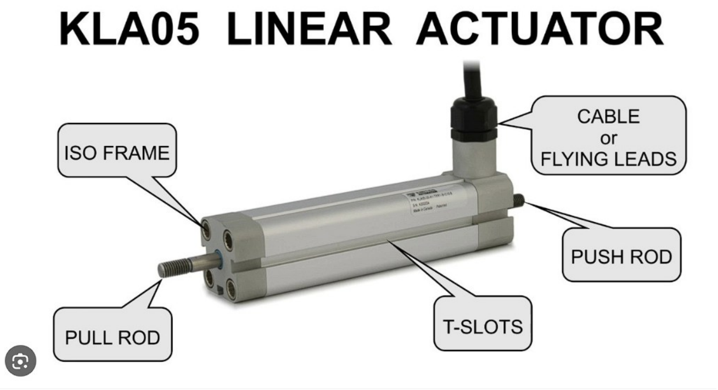
- **Gas and Chemical Sensors:**
  - Type: Sensors detecting specific gases or chemicals.
  - Applications: Environmental monitoring, detecting gas leaks, and ensuring safety in industrial or hazardous environments.

Actuators

- Actuators are devices or components in a system that are responsible for converting energy into mechanical movement or action.

- In the context of robotics, automation, and mechatronics, actuators play a crucial role in controlling and manipulating physical objects or systems.

- They are responsible for carrying out specific tasks based on the signals received from sensors or a control system.



Demonstration of the KLA05 - A Shape Memory Alloy Actuator
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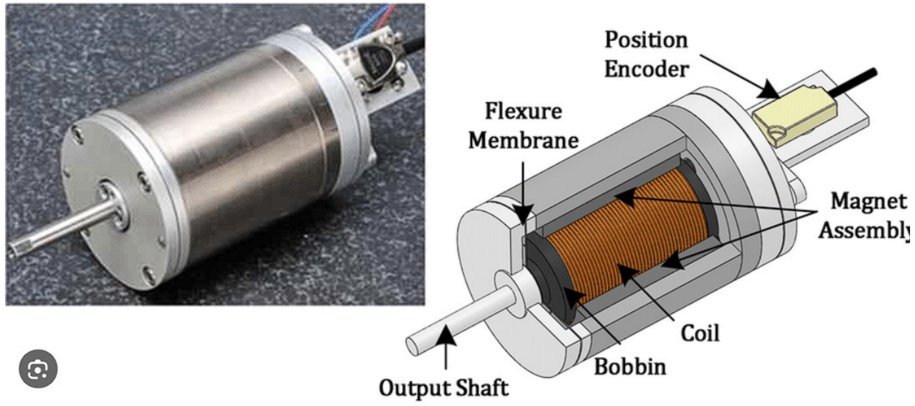

Flexure based electromagnetic linear actuator | Download Scientific Diagram
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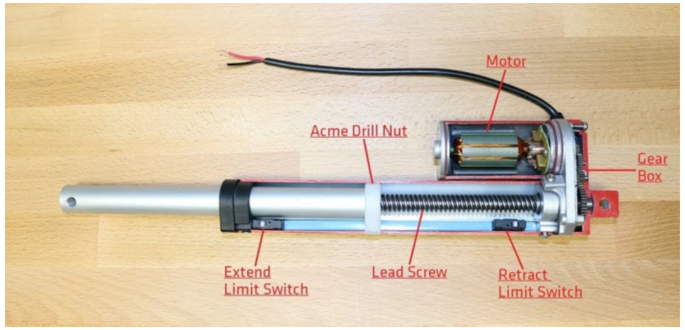
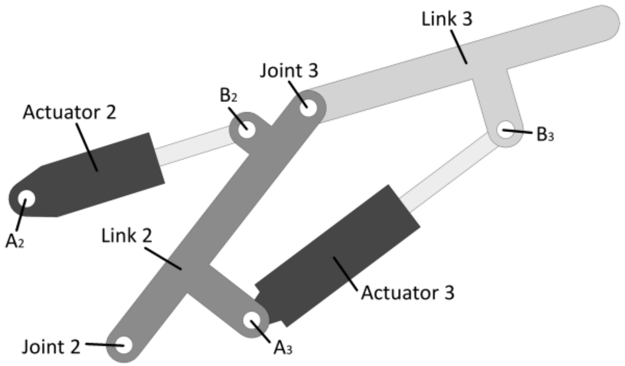
Actuators

- Actuators are devices or components within a system that are responsible for converting control signals or input into physical action, motion, or work. In other words, they are mechanisms that produce a mechanical response in response to an external command or stimulus. Actuators play a crucial role in various engineering and control applications, enabling the control and manipulation of physical processes.

- The primary purpose of an actuator is to effect a change in the system by generating motion, force, or some other form of mechanical output. Actuators are essential in systems where controlled movement or operation is required, such as in robotics, industrial automation, automotive systems, aerospace, and many other fields.

- Actuators come in various types, each suited for specific applications and operating principles. Common types of actuators include electric motors, pneumatic cylinders, hydraulic cylinders, piezoelectric devices, and more. The choice of actuator depends on factors such as the required force, speed, precision, environmental conditions, and the nature of the task the actuator needs to perform.

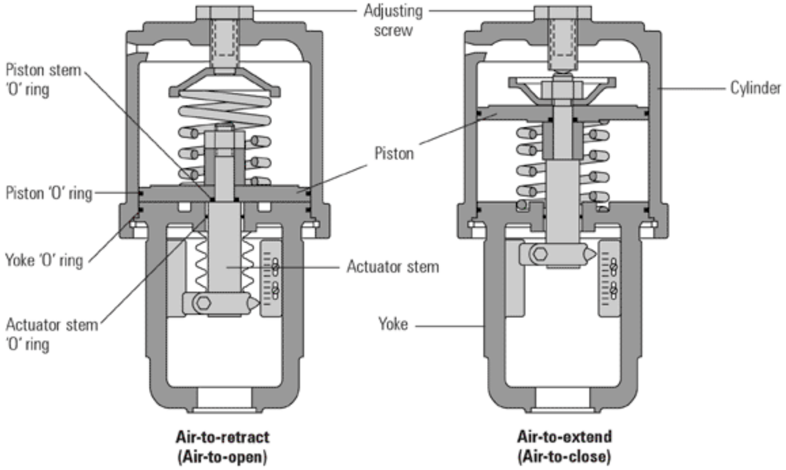




Actuators: Types and Applications

- **Electric Motors:**
  - Type: DC motors, AC motors, stepper motors, etc.
  - Applications: Robotics, industrial automation, consumer electronics, HVAC systems, and many more. Electric motors are widely used for their versatility and ability to provide controlled rotational motion.

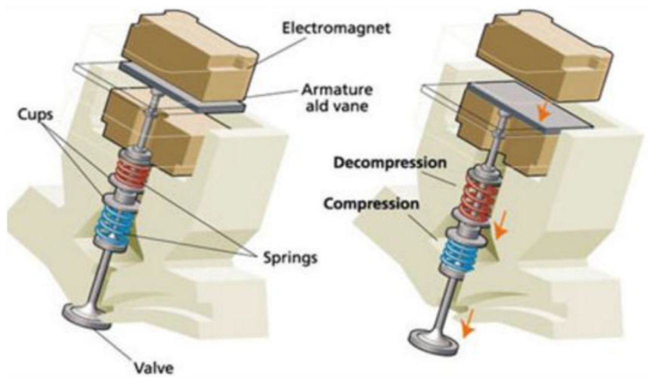
- **Pneumatic Actuators:**
  - Type: Pneumatic cylinders, pneumatic motors.
  - Applications: Automation systems, robotics, manufacturing equipment. Pneumatic actuators use compressed air to generate linear or rotary motion.



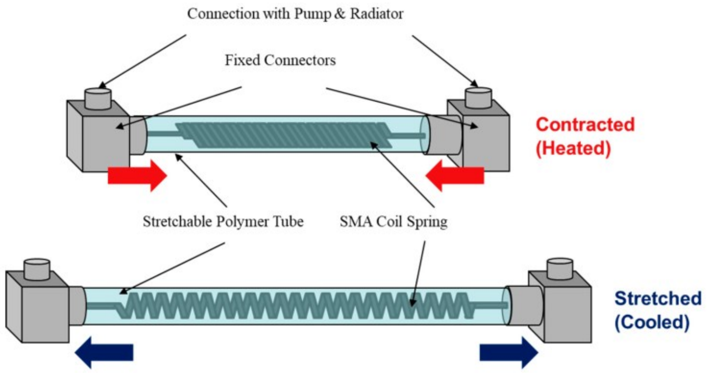
- **Hydraulic Actuators:**
  - Type: Hydraulic cylinders, hydraulic motors.
  - Applications: Heavy machinery, construction equipment, aerospace systems. Hydraulic actuators use pressurized fluid (usually oil) to generate motion.

Actuators: Types and Applications

- **Electromagnetic Actuators:**
  - Type: Solenoids, voice coils.
  - Applications: Automotive systems, valves, relays, robotics. Electromagnetic actuators use the magnetic field generated by electric current to produce motion.

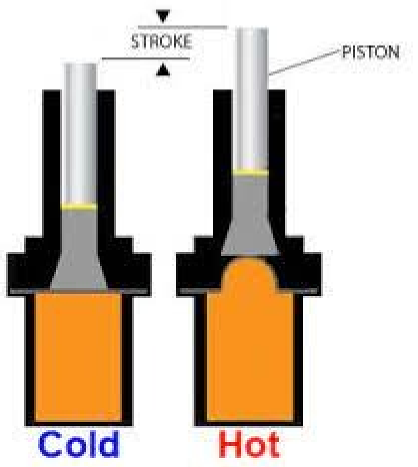


- **Shape Memory Alloy (SMA) Actuators:**
  - Type: Alloys that exhibit shape memory effect.
  - Applications: Biomedical devices, robotics, aerospace. SMAs can change shape in response to temperature changes or stress.

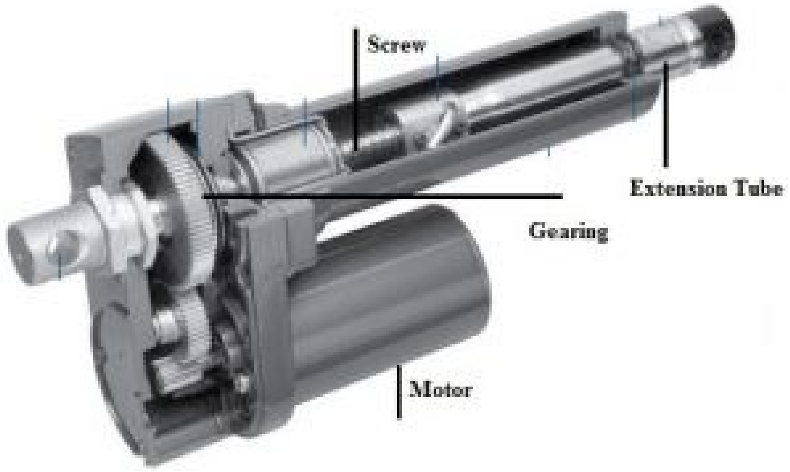


Actuators: Types and Applications

- **Thermal Actuators:**
  - Type: Devices that use heat for actuation.
  - Applications: Microelectromechanical systems (MEMS), microvalves, microactuators. Thermal actuators often use thermal expansion for generating motion.



- **Mechanical Actuators:**
  - Type: Screws, gears, levers, and other mechanical devices.
  - Applications: Various mechanical systems and machinery. Mechanical actuators directly convert rotational or linear motion into mechanical work.



Motors and Controllers

- A motor is a device that converts electrical energy into mechanical energy, producing motion or movement. Motors are essential components in various applications, powering machines and systems across different industries. The basic principle of a motor involves the interaction of magnetic fields and electric currents to generate mechanical force.

- In autonomous robotics, motors and controllers play crucial roles in enabling a robot to move, navigate, and interact with its environment. These components work together to execute control algorithms and achieve desired actions.

Motors

- DC Motors:
  - Type: Direct Current Motors (brushed or brushless).
  - Applications: DC motors are commonly used in robotic systems for their simplicity and controllability. They provide rotational motion and are used in various applications, including wheel propulsion and joint movement.
- Servo Motors:
  - Type: Electric motors with built-in feedback mechanisms.
  - Applications: Servo motors are used for precise control of angular position. They are often employed in robotic arms, grippers, and other applications requiring accurate positioning.
- Stepper Motors:
  - Type: Motors that move in discrete steps.
  - Applications: Stepper motors are used when precise positioning is essential. They find applications in robotics for tasks such as controlling joint movements and providing accurate navigation.
- Actuators:
  - Type: Various actuators, including pneumatic, hydraulic, and piezoelectric.
  - Applications: Actuators are used for specific tasks that require controlled linear or rotational motion. Pneumatic and hydraulic actuators may be used for tasks like lifting or manipulating objects.

Controllers

- Controllers are devices or software systems that manage and regulate the behavior of a robot.
- They are responsible for processing sensor data, making decisions, and generating commands to control the robot's actuators (such as motors) to achieve desired tasks or movements.
- Controllers play a crucial role in enabling robots to operate autonomously or in response to external stimuli.
- Examples : Siemens SIMATIC S7 series, Allen-Bradley PLCs, Arduino boards (e.g., Arduino Uno) and Raspberry Pi (when used as a microcontroller), ABB IRC5, Fanuc R-30iB, etc.

Controllers

- **Microcontrollers/Microprocessors:**
  - Function: Embedded processors that execute control algorithms.
  - Applications: Microcontrollers are the brains of the robot, responsible for processing sensor data, making decisions, and sending control signals to motors and actuators.
- **Motor Controllers:**
  - Function: Electronic devices that regulate the speed and direction of motors.
  - Applications: Motor controllers interpret signals from the microcontroller and adjust the voltage or current supplied to the motors accordingly. This allows precise control of motor speed and direction.
- **Feedback Systems:**
  - Function: Sensors (encoders, gyroscopes) providing feedback on motor or system status.
  - Applications: Feedback systems are crucial for closed-loop control. Encoders, for example, provide information about the motor's position, allowing the controller to make real-time adjustments.
- **Motion Planning and Control Algorithms:**
  - Function: Algorithms that determine the robot's motion and behavior.
  - Applications: These algorithms, implemented in software, enable the robot to navigate autonomously, avoid obstacles, and perform tasks. They work in conjunction with sensor data and the control system.

Power Sources

- Autonomous robots require a reliable power source to operate independently.
- The choice of power source depends on the specific requirements of the robot, its intended application, and the environmental conditions it will encounter.

Here are some common power sources for autonomous robots:

- **Battery Power:** Lithium-ion (Li-ion) Batteries ,Lithium Polymer (LiPo) Batteries, Nickel Metal Hydride (NiMH) Batteries, Lead-Acid Batteries,etc.
- **Fuel Cells**: Hydrogen Fuel Cells
- **Solar Power :** Photovoltaic Cells (Solar Panels)
- **Internal Combustion Engines :** Gasoline Engines, Diesel engines, gas turbines, etc.

Power Sources and Management

- Power sources and management are critical aspects of any autonomous system, including autonomous robots. Efficient and reliable power sources, along with effective power management, are essential for ensuring the sustained operation and autonomy of the robot.

- A power source refers to any device, system, or means that provides electrical or mechanical energy to operate devices, machines, or systems. Power sources are essential components in various applications, from small electronic devices to large industrial machinery. They supply the energy needed to drive motors, electronics, sensors, and other components within a system.

- Effective power management in autonomous robotics involves a combination of hardware and software strategies to optimize energy consumption, extend operational durations, and ensure reliable performance. It also includes monitoring and maintaining the health of power sources, especially rechargeable batteries, to maximize their lifespan.

Mobile and Autonomous Robotics

Power Sources

- **Batteries**:
  - Rechargeable batteries, such as lithium-ion (Li-ion) or lithium-polymer (LiPo) batteries, are commonly used in robotics. They provide a portable and high-energy-density power source.
  - Applications: Autonomous robots often rely on batteries for mobility and portability.
- **Fuel Cells**:
  - Hydrogen fuel cells and other types of fuel cells convert chemical energy directly into electrical energy. They can provide longer operational durations compared to batteries.
  - Applications: Used in some robotic applications where extended endurance is required, such as unmanned aerial vehicles (UAVs) and underwater robots.

- **Solar Power**:
  - Photovoltaic cells convert sunlight into electrical energy. Solar panels can be integrated into the robot's structure to harness energy from the environment.
  - Applications: Common in outdoor and environmental monitoring robots, as well as some UAVs.

- **Tethered Power**:
  - Robots are connected to an external power source through a tether (cable). This allows for continuous operation without concerns about battery life.
  - Applications: Often used in applications where constant power supply is critical, such as industrial inspection robots.

Power Management

- **Voltage Regulation**:
    - Function: Maintaining a stable voltage level to ensure proper operation of electronic components.
    - Components: Voltage regulators or power management ICs.
- **Current Limiting**:
    - Function: Controlling the amount of current flowing through the system to prevent damage to components.
    - Components: Current limiters, fuses.
- **Battery Management Systems (BMS):**
    - Function: Monitoring and managing the charging and discharging of batteries to optimize performance, prevent overcharging, and enhance battery life.
    - Components: Battery management ICs, temperature sensors.
- **Energy Harvesting:**
    - Function: Capturing and utilizing ambient energy sources, such as vibrations, heat, or light, to supplement or recharge the main power source.
    - Components: Energy harvesting modules, converters.

Power Management

- **Power Distribution:**
  - Function: Distributing power to different components within the robot efficiently.
  - Components: Power distribution boards, circuit breakers.
- **Sleep Modes and Low-Power States:**
  - Function: Putting non-essential components or the entire system into low-power states during periods of inactivity to conserve energy.
  - Components: Software algorithms, hardware mechanisms for power gating.
- **Emergency Shutdown Systems:**
  - Function: Activating a shutdown mechanism in critical situations to prevent damage to the system or ensure safety.
  - Components: Emergency shutdown switches, software-controlled shutdown procedures.

THANK YOU

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

Mobile and Autonomous Robots (UE23CS343BB7)

Introduction to Autonomous Robots

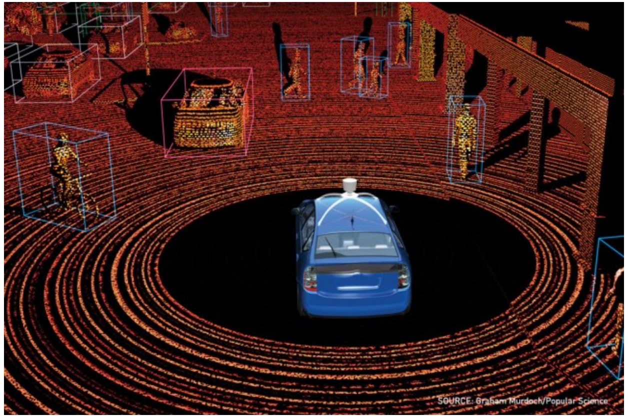
Range Finders, Encoders, Vision Sensors

Introduction to Autonomous Robots

Range Finders

Range Finders

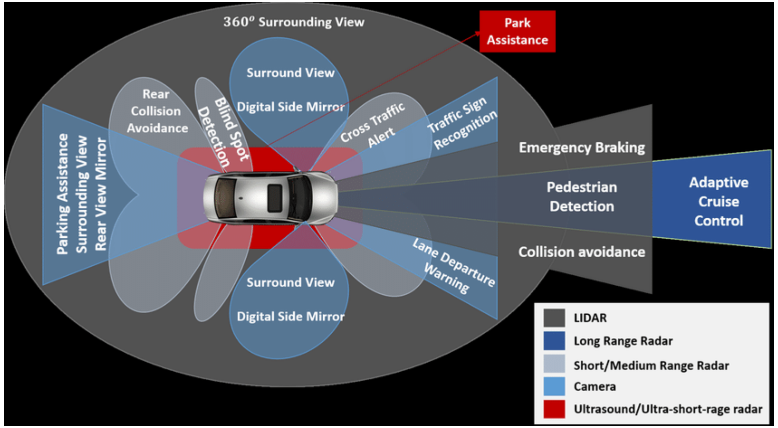
- Autonomous robots thrive in a captivating realm where they navigate the world independently.
- They play the role of unsung heroes, providing crucial information about the environment.
- Act like whispering guides, constantly pinging the environment to unveil secrets of distance.
- These devices empower autonomous machines, granting them the confidence and precision needed for navigation.
- These are the robot's sixth sense, enhancing their ability to perceive the surroundings effectively.
- Range finders continuously interact with the environment, returning vital information about the location of objects and their distances.



Range Finders

Types of Rangefinders for Autonomous Robots:

- **Lidar:** Powerful and long-range, ideal for outdoor navigation and obstacle detection.
- **Ultrasonic**: Accurate for short-range tasks like object manipulation and proximity sensing.
- **Radar:** Penetrates harsh environments and excels in low-visibility conditions.
- **Vision-based:** Cameras can also estimate distance through triangulation or depth sensors.



Range Finders

Applications:

- *Obstacle Avoidance:* Imagine a self-driving car zipping down a highway. Lidar rangefinders, like spinning laser eyes, scan the road ahead, detecting other vehicles, pedestrians, and even potholes. This vital information is fed to the car's brain, allowing it to adjust its course and avoid collisions.

- *Precision Manipulation*: A robotic arm in a factory needs to pick up a delicate component without crushing it. Ultrasonic rangefinders, emitting silent sound waves, accurately measure the distance to the object, ensuring the robot arm grasps it gently yet firmly.

- Mapping and Exploration: A Mars rover ventures into the unknown. Laser rangefinders map the surrounding terrain, creating a 3D image of the Martian landscape, guiding the rover's exploration and ensuring safe navigation.

- Search and Rescue: In a disaster zone, drones equipped with radar rangefinders pierce through dust and debris, locating survivors trapped under rubble or in remote areas. This precious information can guide rescue teams to those in need.

Introduction to Autonomous Robots

Encoders

Encoders

In the realm of robotics and automation, encoders serve as essential translators, transforming the language of motion into precise digital signals that machines can comprehend and act upon. They are the tireless observers, meticulously tracking movement and providing crucial feedback for control and precision.

What Encoders Do:

- **Measure Position and Motion:** They gauge the position, speed, direction, and rotation of various moving parts within a system, such as wheels, motors, or robot joints.
- **Provide Feedback:** This information is relayed to controllers and systems that regulate and adjust movements, ensuring accuracy, synchronization, and efficiency.
- **Enable Closed-Loop Control**: By incorporating real-time feedback, encoders allow for adaptive adjustments to maintain desired performance, even in the face of disturbances or changing conditions.

Encoders

Types of Encoders:

- **Rotary Encoders:**
  - **Incremental Encoders**: Generate pulses as they rotate, indicating relative position changes.
    - **Optical**: Use a light source and photodetector to sense rotation.
    - **Magnetic**: Employ magnetic fields and sensors to track movement.
  - **Absolute Encoders:** Provide a unique digital code for each distinct position, enabling absolute position tracking without reference to a starting point.
- **Linear Encoders:**
  - **Optical:** Utilize a linear scale with graduations read by a light sensor.
  - **Magnetic:** Operate similarly to magnetic rotary encoders, but along a linear path.

Encoders

Applications of Encoders:

- **Robotics:**
  - Measure joint angles and velocities for precise control of robot movements.
  - Track wheel rotations for accurate navigation and obstacle avoidance.
- **Industrial Automation**:
  - Monitor motor speed and position in assembly lines, conveyor systems, and CNC machines.
  - Ensure precise positioning of tools and machinery.
- **Medical Devices:**
  - Control the movement of medical imaging equipment, surgical robots, and prosthetics.
- **Automotive Systems**:
  - Track wheel speed for anti-lock braking systems (ABS) and traction control.
  - Monitor steering wheel position for power steering and lane departure warning systems.
- **Printing and Packaging:**
  - Control the movement of print heads and rollers for precise alignment and speed.
- **Consumer Electronics:**
  - Track the rotation of knobs and dials in audio equipment, game controllers, and other devices.

Choosing the Right Encoder:

- **Resolution**: The level of detail needed in position measurement.
- **Accuracy:** The degree of precision required.
- **Operating Speed**: The maximum speed at which the encoder can track movement.
- **Environmental Conditions**: Resistance to dust, moisture, vibration, or other factors.
- **Cost:** The overall budget for the application.

Introduction to Autonomous Robots

Vision Sensors

Vision sensors

Understanding the world through sight is crucial for humans, and it's no surprise that we've strived to equip machines with similar capabilities. This journey toward robotic vision involves two key steps:

- **Capturing the world:** Building sensors that mimic our eyes, gathering light data and transforming it into digital images. This forms the raw material for the robot's "vision."
- **Making sense of the scene:** Processing these digital images is where the magic happens. Extracting key information like depth, motion, color, features, and even recognizing entire scenes allows the robot to interpret the world in a meaningful way.

Vision sensors

Types of vision sensors:

- **Cameras:**
  - Standard Cameras: Most common, providing color or grayscale images at varying resolutions and frame rates. Used for object recognition, navigation, and tracking.
  - Depth Cameras: Capture depth information along with the image, enabling 3D perception. Crucial for tasks like obstacle avoidance and robot-human interaction.
  - Thermal Cameras: Detect heat radiation, making them useful in low-light conditions or for identifying objects based on their temperature.
- **Time-of-Flight (ToF) Sensors:**
  - Similar to LiDAR, but utilizes modulated light instead of pulsed lasers. Offers high-resolution depth data at shorter ranges, making them suitable for indoor applications and object manipulation tasks
- **Event Cameras:**
  - Capture changes in light intensity rather than complete images. Offer high temporal resolution, making them ideal for fast-moving objects and dynamic environments.

Applications of Vision Sensors:

- **Navigation and Obstacle Avoidance:** Robots navigating complex environments rely on cameras, LiDAR, and ToF sensors to map their surroundings, detect obstacles, and plan safe paths.
- **Object Recognition and Manipulation:** Vision sensors help robots identify objects, their orientation, and even grasp them accurately.
- **Human-Robot Interaction:** Cameras and depth sensors enable robots to understand human gestures and expressions, facilitating natural interaction.
- **Inspection and Quality Control:** Vision systems are used in automated inspection tasks, detecting defects and ensuring product quality.
- **Autonomous Vehicles:** Self-driving cars rely heavily on cameras, LiDAR, and radar for obstacle detection, lane tracking, and traffic analysis.

Choosing the Right Vision Sensor:

- **Application:** Consider the specific task and environment the robot will operate in.
- **Resolution and Accuracy:** Determine the level of detail and precision needed for the task.
- **Field of View:** Choose a sensor with a suitable field of view to capture the necessary information.
- **Cost and Power Consumption:** Balance sensor capability with budget and power constraints.

The future of vision sensors:

- Miniaturization and improved processing power are leading to smaller, more efficient sensors.
- Advancements in AI and machine learning are enabling better object recognition and scene understanding.
- Fusion of data from multiple sensors is creating a richer and more robust perception of the environment.

THANK YOU

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

Mobile and Autonomous Robots (UE21CS343BB7)

ROS2 Overview and Architecture

Introduction

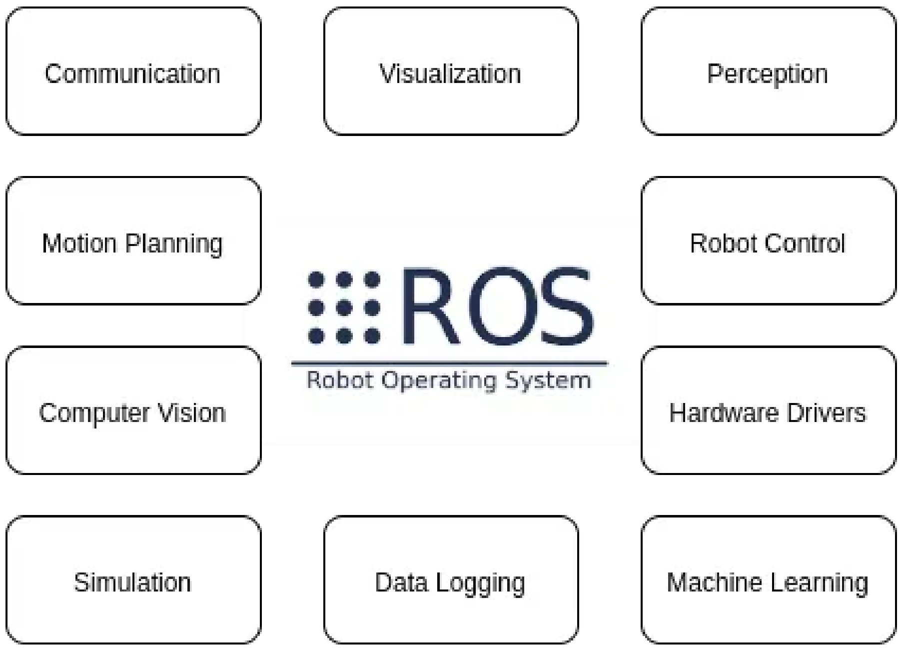
Robot Operating System (ROS) is a set of open source algorithms, hardware driver software and tools developed to develop robot control software. Even though it has operating system in its name it is not an operating system. It is

- Communication System (Publish Subscribe and Remote Method Invocation),
- Framework & Tools (Build system & dependency management, Visualization, Record and Replay)
- Ecosystem (Language bindings, Drivers, libraries and simulation (Gazebo)).

Introduction

- The first version of ROS was mostly used in academic projects. ROS 2 was developed so that it can be used in commercial projects. The biggest change that came with ROS2 was the selection of the DDS middleware for the communication layer.

- ROS includes mature open source libraries to be used for navigation, control, motion planning, vision and simulation purposes.
- The 3D visualization tool called RVIZ is an important tool used with ROS. Similarly, the simulation tool called Gazebo is seen as a useful tool for robot developers.
- Apart from this, Open CV is a library used for detection purposes in ROS 2. In addition, we see that the QT graphic library is also used for the user interface in ROS 2 projects and is an add-on available for this purpose.

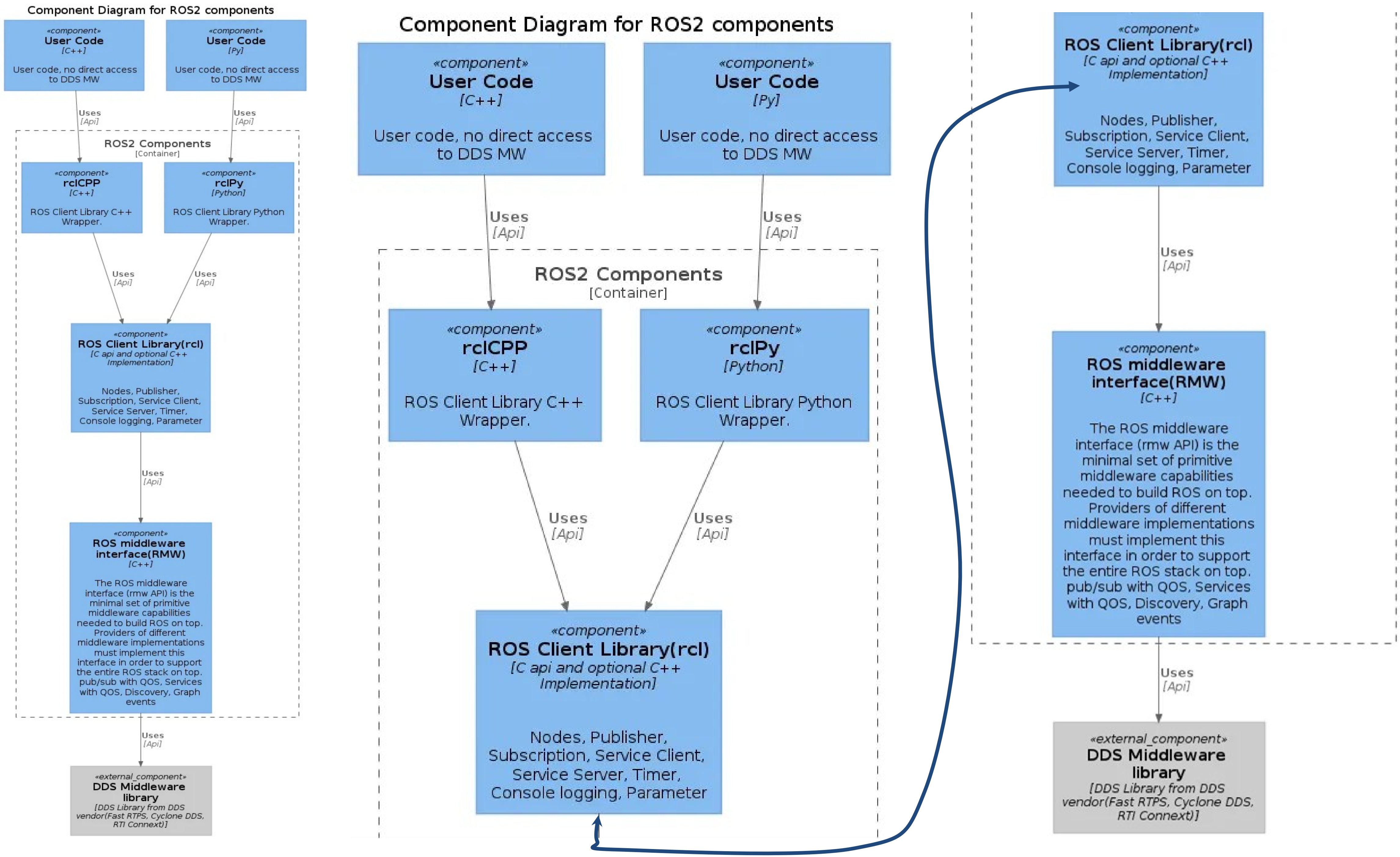


ROS 2 Architecture

The ROS 2 has distributed real-time system architecture. Sensors in robots, motion controllers, detection algorithms, artificial intelligence algorithms, navigation algorithms, etc are all components (called as node) of this distributed architecture. The DDS middleware selected with ROS 2 for data exchange enables these components to communicate with each other in a distributed environment.

ROS 2 Client Library

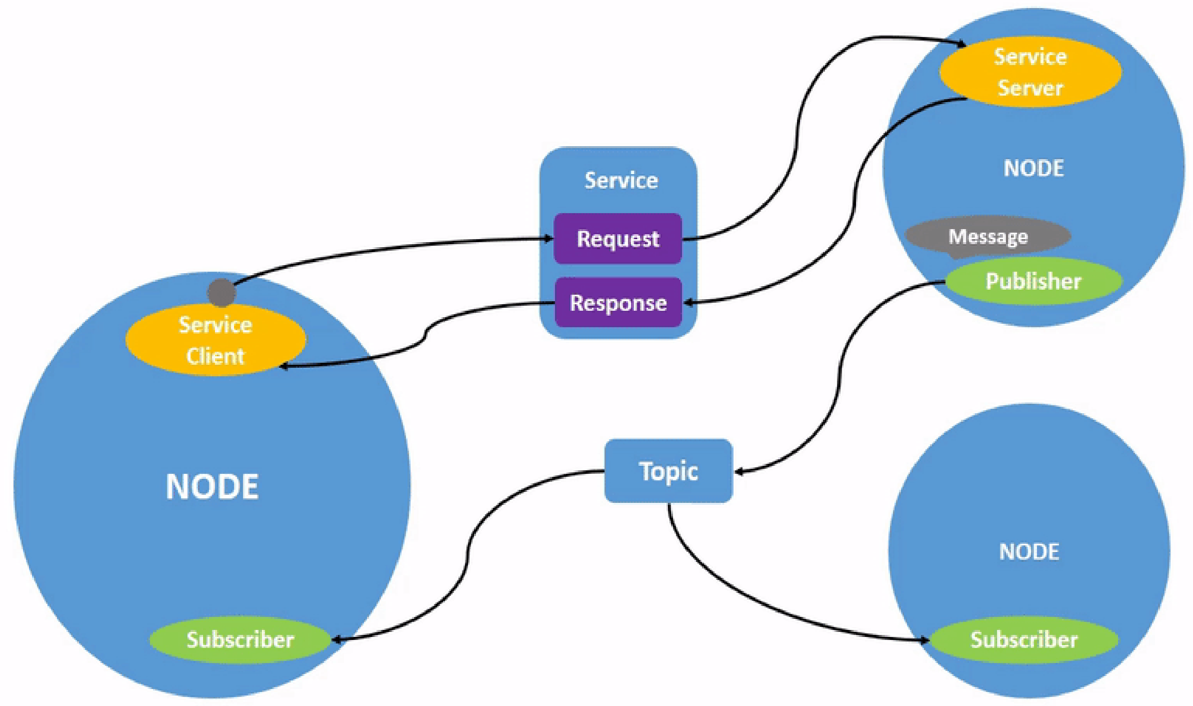
ROS2 applications access ROS 2 features through the ROS Client Library (RCL) client library. The Rcl library is written in C language and on it there are Rclcpp client libraries for C ++ language and Rclpy for Python language. There are independently written ROS 2 client libraries in other languages such as Java, Go. The client library is primarily provided with the standard interface required to exchange data with Topic and Service approaches over ROS 2. In addition, the ROS2 library has capabilities to provide operating system abstraction and ready-made micro-architectural structures.


Component Diagram for ROS2 components
«component»
User Code
[C++]
User code, no direct access to DDS MW
«component»
User Code
[Py]
User code, no direct access to DDS MW
Uses
[Api]
ROS2 Components
[Container]
«component»
rclCPP
[C++]
ROS Client Library C++ Wrapper.
«component»
rclPy
[Python]
ROS Client Library Python Wrapper.
«component»
ROS Client Library(rcl)
[C api and optional C++ Implementation]
Nodes, Publisher, Subscription, Service Client, Service Server, Timer, Console logging, Parameter
«component»
ROS middleware interface(RMW)
[C++]
The ROS middleware interface (rmw API) is the minimal set of primitive middleware capabilities needed to build ROS on top. Providers of different middleware implementations must implement this interface in order to support the entire ROS stack on top. pub/sub with QOS, Services with QOS, Discovery, Graph events
«external_component»
DDS Middleware library
[DDS Library from DDS vendor(Fast RTPS, Cyclone DDS, RTI Connext)]

Mobile and Autonomous Robots

ROS 2 Graph Structure

- The node, topic, message structure, and discovery form the basic distributed architecture of ROS 2. This structure is called a graph in ROS 2 terminology. Let's get into the details of these architectural basic structures.



Node

- Distributed applications are designed as units known as Nodes.
- In a robot system, sensors (Lidar, camera) motion controllers (motors providing motion), algorithm components (route planner) can be nodes each.
- ROS 2 separates the node concept from the operating system level process structure. If desired, multiple nodes can be created within a process and these nodes can communicate independently with other nodes.
- All nodes in the system can be run on a single computer or they can be distributed and run across multiple computers.

Node

Nodes communicate with each other via Topic, Service Invocation and Actions. The topic based communication has publish subscribe architecture where the data produced and published by a node is subscribed by more than one node and similarly, one topic data can be produced to more than one node. Topics defined in ROS 2 exactly match DDS topics. Service calls use a remote method invocation approach from an application developer perspective. Actions are an approach used for long service calls. The client requests an action with goal, the action server starts the long-running process, publishes intermediate results, and reports the final result when the goal is achieved. These service and actions are also implemented using topics over the DDS middleware.

Decentralized architecture of Topic structure creates advantage in terms of both performance and fault tolerance. In addition, rich quality of service features provided by the DDS middleware have also been a factor that raised the ROS 2 architecture to higher levels.

Messages

The data types used in topic communication are called message (Message). These Message data structures basically match the DDS data types (Type). A sample camera message is given below:

```
sensor_msgs/CameraInfo
Header header
 uint32 seq
 time stamp
 string frame_id
uint32 height
uint32 width
RegionOfInterest roi
 uint32 x_offset
 uint32 y_offset
 uint32 height
 uint32 width
float64[5] D
float64[9] K
float64[9] R
float64[12] P
```

The messages defined in ROS 2 are actually of great importance for interoperability.

Communication Patterns (Topic, Service, Action)

With the help of these standard messages, cameras developed by different companies can be quickly integrated into the system and can communicate with standard messages regardless of the model camera. In this case, the Node component developed for the relevant camera will convert the camera's special communication protocol messages into standard DDS data. Preliminary studies can be performed using the camera simulator in the Gebazo simulation.

ROS 2 also effectively uses the quality of services (QOS) defined in the DDS middleware. For example, while continuously updated sensor data is sent in best effort, a command sent to the robot to do a job can be sent reliably. Matching the rich quality of service features in DDS with the topics is an important design decision. Ready-made service quality profiles that can be used according to the type of data are defined and the service quality related to newly defined topics is determined by selecting one of these profiles.

The QOS profiles defined in ROS 2 are known by service, sensor data, parameters and default. For example, the parameters use the reliable service quality value while the sensor data uses the best effort QOS value.

Discovery of applications in a distributed environment

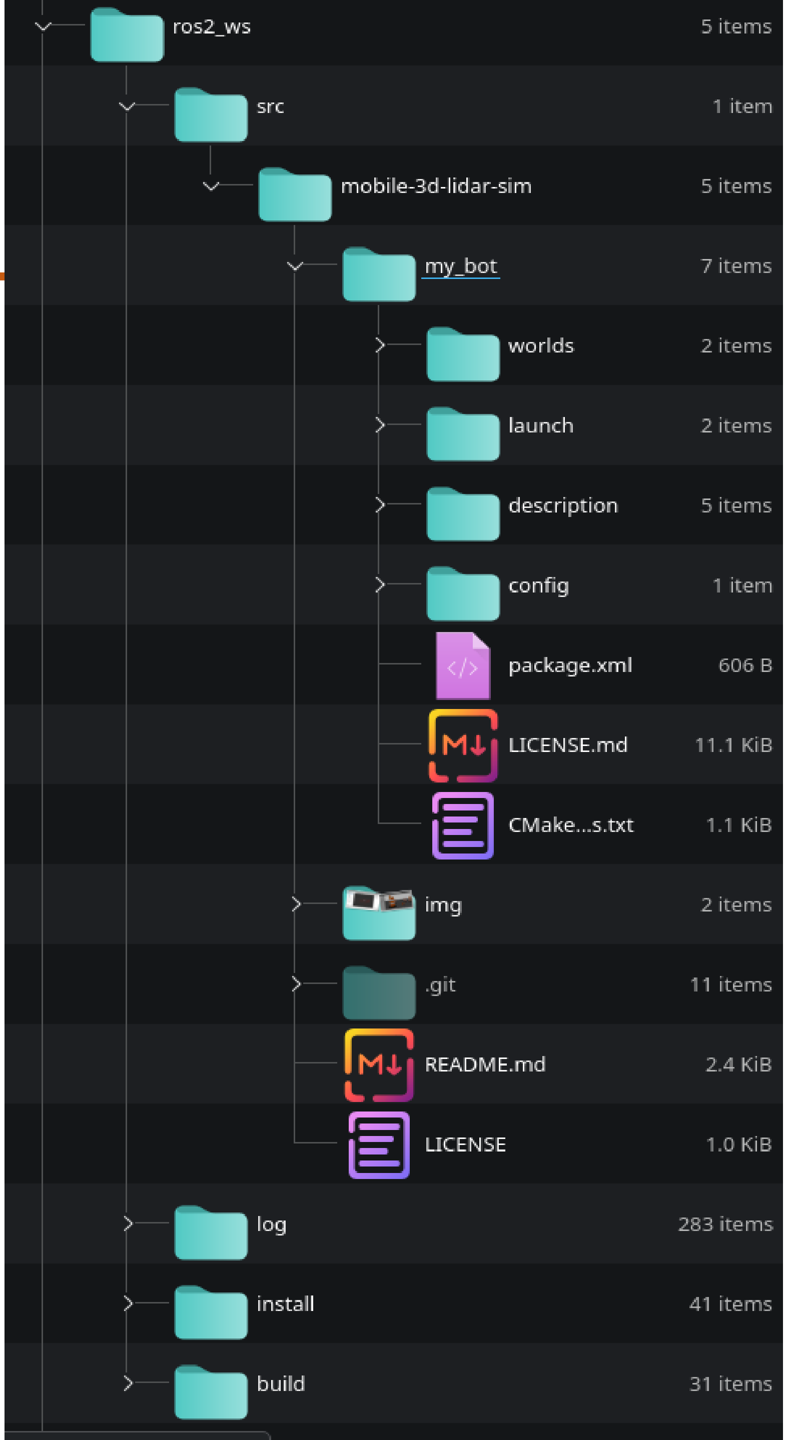
In ROS 1, the discovery mechanism and message transmission were communicating through a service called Roscore. DDS middleware does not need a broker to distribute data. In addition, applications in the DDS middleware dynamically discover each other through special DDS topics. As a result, DDS has switched ROS 2 to a decentralized architecture by meeting the discovery, distribution and data encoding needs of ROS 2. So, in the new architecture, the Roscore service has been retired.

Mobile and Autonomous Robots

ROS Filesystem

The ROS 2 File System consists of ROS 2 packages – the smallest build part in ROS 2.

Usually, a ROS File System consists of hundreds of different packages. To navigate efficiently through your ROS system, ROS provides different management tools. The most common tools are described in the example of the ROS package turtlesim.



ROS Filesystem

1. Workspace

Colcon is a build tool for ROS 2. A workspace is a **folder**, where you can modify, build, and install packages. It is the place to create **your packages and nodes or modify existing ones** to fit your application. In the further tutorials, you will work in your workspace.

2. ROS packages

Two types of ROS 2 packages exist: binary packages and build-from-source packages.

ROS binary packages are in Ubuntu provided as debian packages, which can be managed via **apt commands**.

Build-from-source packages can be divided into packages provided by ROS 2 and your developments. Both have to be placed into the **src folder** of a ROS 2 workspace.

THANK YOU

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering, RR Campus
ashokkumarp@pes.edu

Mobile and Autonomous Robotics

UE22CS343BB7

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

Mobile and Autonomous Robotics

UE23CS343BB7

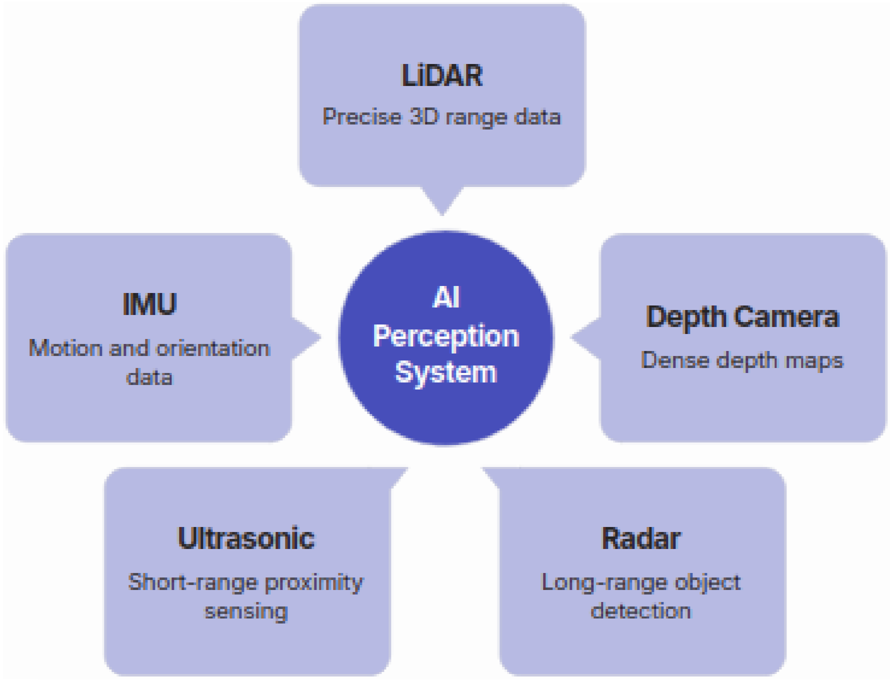
Introduction to Autonomous Robotics

AI-Powered Robotics

AI Perception Systems

AI perception systems are crucial for autonomous robots, allowing them to interpret and understand their surroundings in real-time. Through advanced algorithms and sensor integration, robots can process complex sensory data to make informed decisions and navigate dynamic environments.

- **Computer Vision:** AI-powered visual recognition enables robots to identify objects, people, and scenes, including facial recognition, gesture interpretation, and understanding the context of visual data.
- **Object Detection and Tracking:** Robots can precisely locate and follow specific objects within their environment, differentiating between static and moving elements crucial for interaction and navigation.
- **Sensor Fusion:** AI combines data from various sensors (e.g., cameras, lidar, radar, SONARS) to create a comprehensive and robust understanding of the environment, overcoming limitations of individual sensors.
- **Environmental Mapping and Navigation:** Through techniques like Simultaneous Localization and Mapping (SLAM) and deep learning, robots can build detailed maps of their surroundings, localize themselves within these maps, and plan efficient paths.



Mobile and Autonomous Robotics

AI-Enhanced Sensing

AI-enhanced sensing revolutionizes how autonomous systems perceive their environment by integrating advanced artificial intelligence capabilities directly into sensor data processing. This enables robots and other smart devices to not only detect physical properties but also to interpret, analyze, and act upon this information with unprecedented intelligence and accuracy.

Neural Networks for Interpretation: Artificial neural networks are employed to process raw sensor data, enabling systems to recognize complex patterns, classify objects, and understand contextual information that traditional sensors alone cannot. This allows for more sophisticated interpretations of the environment.

Real-time Data Processing: AI algorithms facilitate rapid analysis of vast streams of sensor data, making real-time decision-making possible. This is crucial for dynamic environments where quick responses are necessary for navigation, interaction, and safety.

Intelligent Filtering of Inputs: AI systems can intelligently filter out noise, irrelevant data, or redundant information from multiple sensors, ensuring that only the most pertinent and accurate data is used for further processing. This improves efficiency and reliability, reducing computational load and enhancing the clarity of environmental perception.

Sensor Technologies in Robotics

Autonomous systems rely on a diverse array of sensors to perceive their surroundings. Each sensor type provides unique data crucial for AI-driven interpretation, enabling robots to navigate, interact, and make intelligent decisions.

Li-DAR

Li-DAR uses pulsed lasers to measure distances, generating highly accurate 3D point clouds. AI processes this data for semantic segmentation, precise localization, and obstacle detection, essential for autonomous navigation and mapping.

Depth Cameras

These cameras provide per-pixel distance information, creating detailed depth maps. AI leverages depth data for 3D object recognition, gesture tracking, and accurate grasping, crucial for human-robot interaction and object manipulation.

RGB Cameras

Standard color cameras capture visual data rich in texture and color. AI models, particularly CNNs, utilize this for broad object detection, scene understanding, facial recognition, and visual odometry, providing critical contextual information.

Machine Learning Models for Robotics

Machine Learning approaches enable robots to learn from data, adapt to new situations, and make intelligent decisions in complex environments.

Supervised Learning for Task Execution: In supervised learning, robots are trained using labeled datasets where inputs are mapped to desired outputs. This approach is used for tasks such as object recognition, path planning, and precise manipulation, allowing robots to perform specific actions based on learned patterns and examples.

Unsupervised Learning for Pattern Discovery: Unsupervised learning algorithms allow robots to discover hidden patterns and structures in unlabeled data. This is particularly useful for tasks like anomaly detection, clustering of sensor data, and creating internal representations of the environment without explicit human guidance, fostering greater autonomy.

Reinforcement Learning for Autonomous Decision-Making: Reinforcement learning (RL) enables robots to learn optimal behaviors through trial and error, interacting with their environment to maximize a reward signal. RL is crucial for complex tasks requiring sequential decision-making, such as navigation in dynamic environments, human-robot interaction, and adaptive control, allowing robots to autonomously learn and refine their strategies.

Deep Learning in Robotics

Deep neural networks enable robots to learn complex behaviors and perform advanced tasks, pushing the boundaries of autonomous systems.

Convolutional Neural Networks (CNN) for Vision: CNNs are vital for robots to process and understand visual information from cameras. They enable tasks such as object recognition, scene understanding, and real-time obstacle detection, allowing robots to navigate and interact with their environment effectively.

Recurrent Neural Networks (RNNs) for Sequential Decision-Making: RNNs are well-suited for processing sequential data, making them crucial for tasks requiring memory and context over time. In robotics, RNNs are used for predicting future states, controlling robot movements based on past interactions, and enabling complex behaviors like human-robot collaboration and adaptive manipulation.

Transformer Models for Advanced Reasoning: Emerging transformer models bring sophisticated reasoning capabilities to robotics, allowing robots to understand complex instructions, engage in natural language interaction, and perform high-level planning. These models facilitate more human-like cognitive abilities, leading to more intelligent and versatile robotic applications.

Natural Language Processing in Robotics

Natural Language Processing (NLP) is a crucial field enabling robots to understand, interpret, and generate human language, facilitating more intuitive and effective interaction with their human counterparts.

- **Human-Robot Interaction**: NLP allows robots to engage in natural conversations, understand user queries, and respond appropriately, making interaction more intuitive and user-friendly.

- **Voice Commands and Control:** Robots can process spoken instructions, enabling hands-free operation and complex task execution through voice interfaces, enhancing accessibility and efficiency.

- **Instruction Understanding**: NLP helps robots interpret diverse and often ambiguous human instructions, allowing them to translate high-level commands into actionable steps for task completion.

- **Collaborative Communication:** Facilitates seamless teamwork between humans and robots, enabling shared understanding of goals, progress, and environmental cues through verbal communication.

AI-Powered Decision Making

AI empowers robots to make autonomous, intelligent decisions in complex and dynamic environments. This capability is fundamental for advanced robotic systems, enabling them to adapt, learn, and collaborate effectively.

Decision Trees

Hierarchical models that guide robots through a series of choices based on conditions, leading to specific actions or classifications. Ideal for transparent and explainable decision paths in varied scenarios.

Bayesian Networks

Probabilistic graphical models representing causal relationships between variables. They enable robots to make informed decisions under uncertainty, updating beliefs as new sensor data arrives.

Multi-Agent Systems

Frameworks where multiple autonomous AI entities interact to achieve common goals. Essential for collaborative robotics, allowing robots to coordinate actions, share information, and solve complex tasks collectively.

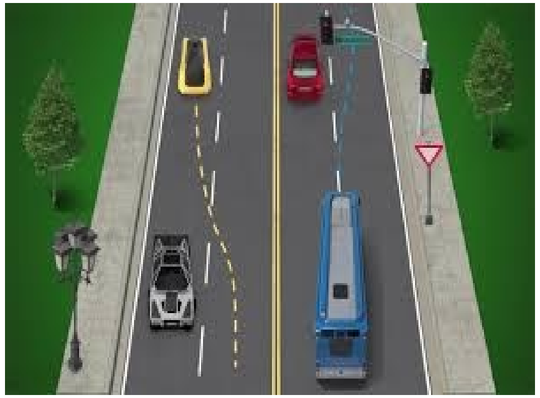
Edge AI and On-Robot Processing

Edge AI deploys AI models directly on robots, enabling local data processing and decision-making without reliance on cloud infrastructure This approach is especially valuable for autonomous systems operating in dynamic or constrained environments.

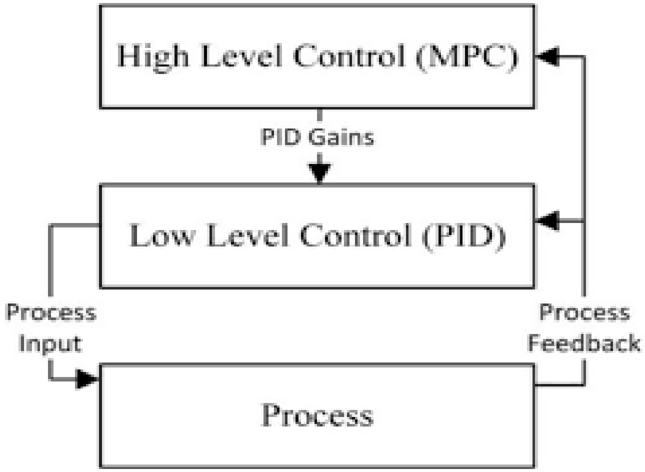
- **Real-time Inference and Reduced Latency**: Local on-robot inference minimizes latency, enabling rapid responses to environmental changes. This is essential for time-critical tasks such as navigation, obstacle avoidance, and manipulation.
- **Enhanced Privacy and Security**: Processing data locally avoids transmitting sensitive information to external servers, reducing privacy risks and improving security—crucial for robots in confidential or safety-critical settings.
- **Improved Autonomy and Cloud Independence**: On-board AI allows robots to function reliably in low-connectivity or offline environments, increasing robustness and ensuring uninterrupted operation in remote or hazardous areas.
- **Optimized Resource Utilization**: Although it requires capable on-board hardware, Edge AI reduces bandwidth usage and cloud dependency, lowering operational costs when handling high-volume sensor data.

AI-Optimized Control Systems

Integrating artificial intelligence with classical and modern control algorithms revolutionizes robotic motor control by improving efficiency, precision, and adaptability. By combining proven control strategies—such as PID, LQR, and MPC—with AI, robots move beyond fixed, pre-programmed behavior toward intelligent, self-optimizing control. This hybrid approach is essential for advanced autonomous systems.

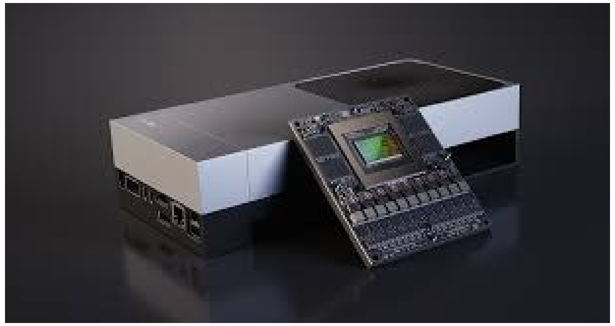


Traditional controllers like PID ensure stability and fast response, LQR provides optimal control with state-space formulations, and MPC enables constraint-aware, predictive decision-making. AI enhances these controllers by automatically tuning gains, adapting cost functions, and handling nonlinearities and uncertainties.
Adaptive Algorithms: AI enables real-time adaptation of control parameters for PID, LQR, or MPC, allowing robots to sustain optimal performance under changing loads, environmental disturbances, or system degradation.
Predictive and Intelligent Control: When combined with Model Predictive Control (MPC), AI improves system modeling and disturbance estimation, enabling proactive motor adjustments that optimize trajectories, reduce overshoot, and enhance responsiveness.
Learning-based Optimization: Using reinforcement learning and data-driven methods, AI continuously refines control strategies layered over PID, LQR, or MPC, resulting in improved energy efficiency, reduced wear, lower maintenance requirements, and higher task execution accuracy.
This fusion of classical control theory and AI creates robust, adaptive, and high-performance robotic control systems suitable for complex real-world environments



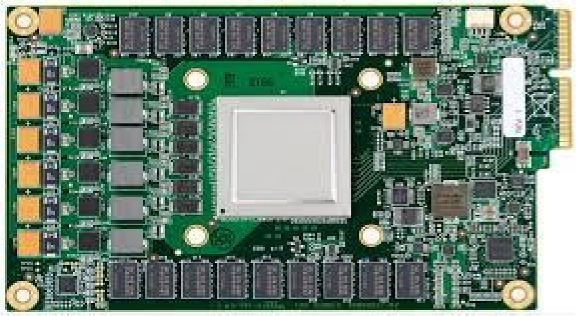
AI Computing Platforms

Advancements in AI computing platforms are crucial for modern robotics, enabling powerful on-board machine learning capabilities, real-time inference, and autonomous decision-making.



GPU (Graphics Processing Unit)

- Highly parallel architecture ideal for deep learning training & inference
- Excels in computer vision, SLAM, object detection
- Used in robots requiring high compute throughput
- Examples: NVIDIA Jetson Nano / Xavier / Orin/ Thor, desktop RTX GPUs



TPU (Tensor Processing Unit)

- Specialized hardware for tensor operations and neural networks in robotic workloads
- Delivers high performance-per-watt, ideal for on-board ML inference on robots
- Best suited for fixed, optimized AI models deployed in autonomous and mobile robots
- Examples: Google Coral Edge TPU (on-robot inference), Cloud TPU (training & large-scale deployment)

NPU (Neural Processing Unit)

- Designed specifically for low-power AI inference
- Enables real-time perception on mobile and embedded robots
- Efficient for vision, speech, and sensor fusion
- Examples: Qualcomm Snapdragon RB5, Huawei Ascend NPU, ARM Ethos-U

AI Applications in Robotics

AI plays a crucial role in enhancing the capabilities of robots across various industries and applications, enabling them to perform complex tasks with greater autonomy and intelligence.

Autonomous Driving & Mobile Robots

Collaborative Robots (Cobots)

Medical & Healthcare Robotics

Agricultural Robotics (AgriBots)

Ecological Robotics (EcoBots)

Defence

AI Robots in Health Care

AI robots combine artificial intelligence and robotics to assist clinicians, improve efficiency, support patients, and perform precise or repetitive tasks autonomously or semi-autonomously. They enhance quality, safety and accessibility of care, especially in high-demand medical environments.

1) Surgical Assistance Robots– da Vinci Surgical System: AI robots like da Vinci are used in minimally invasive surgeries. They translate the surgeon's hand movements into micro-precise instrument actions. The system provides 3D visualization, reduces blood loss, shortens recovery time, and improves precision in complex procedures such as prostatectomies and cardiac repairs.

2) Hospital Support Robots – Moxi (Diligent Robotics): Moxi is designed to support clinical staff by performing logistics tasks such as fetching supplies, delivering samples, or transporting materials. By taking over routine work, Moxi frees nurses and technicians to focus more on direct patient care.

3) DNA Origami Nanobots with Decision Rules – They are tiny\y programmable DNA structures that can recognize specific disease markers and then change shape to release drugs. While not AI in the conventional software sense, many research efforts combine machine learning models to design and optimize their behavior based on biological inputs, effectively making DNA nanobots "smart."

AI Robots in Collaborative Environments (Cobots)

Collaborative robots (Cobots) are AI-enabled robots designed to work safely alongside humans in shared workspaces. Unlike traditional industrial robots, cobots use AI, sensors, computer vision, and force feedback to understand human actions, adapt to changes, and assist rather than replace workers.

1) Universal Robots- Machines which use AI-based vision and force control for assembly, pick-and-place, machine tending, and inspection. They can be taught by demonstration and automatically slow or stop when humans approach, making them ideal for SMEs.

2) ABB YuMi – Human-Robot Teamwork: YuMi is built specifically for close human collaboration. AI-driven motion planning and vision allow YuMi to assist in precision electronics and small-parts assembly while maintaining high safety through real-time collision detection.

3) FANUC CR Series – These cobots are widely used in packaging, inspection, and material handling. Force sensing enables immediate response to contact, ensuring safe operation in shared environments.

4) Foxconn Lights-Out Manufacturing Units - It has deployed robot-only production lines in several Chinese plants. These facilities replaced large numbers of manual assembly workers with robotic arms and automated inspection systems, especially for repetitive, high-volume tasks

Agricultural Robotics (AgriBots)

Agricultural robots (AgriBots) are robotic systems designed to assist or automate farming operations such as planting, weeding, spraying, monitoring, and harvesting. They use sensing, perception, and adaptive control to operate in unstructured outdoor environments, improving productivity and sustainability in agriculture.

1) John Deere Autonomous Tractors - John Deere deploys self-driving tractors capable of plowing, seeding, and spraying with minimal human intervention. These machines follow predefined paths and adapt to field conditions, enabling precise large-scale farming operations.

2) Blue River Technology – Precision Weeding RobotsTheir robotic systems selectively remove weeds while preserving crops. Instead of spraying chemicals across the entire field, the robot targets only unwanted plants, reducing herbicide usage and environmental impact.

3)Agrobot– Harvesting RobotsAgrobot develops robotic systems for delicate harvesting tasks such as strawberries. These robots identify ripe produce and pick it carefully without damaging the fruit.

4) Drones & Mobile Robots for Crop Monitoring - Mobile ground robots and aerial drones are used to inspect crop health, monitor soil conditions, and detect stress or disease early—helping farmers take timely action and improve yield.

AI Robots in Defence (Military Robotics)

Defence robots are unmanned systems used to reduce risk to soldiers by performing tasks like surveillance, reconnaissance, logistics, and explosive disposal. They rely on perception and autonomy to operate in complex environments while keeping humans in control for critical decisions.

1) Unmanned Aerial Systems for ISR MQ-9 Reaper:
Used for long-endurance intelligence gathering, wide-area surveillance, and real-time tracking. It provides persistent aerial monitoring and sends live feeds to command units for situational awareness. It uses AI to analyze live video to automatically detect and track targets, and combines sensor data (thermal + vision + GPS/radar) to generate faster alerts and clearer battlefield awareness.

2) EOD Robots (Bomb Disposal / Hazard Response) TALON QinetiQ
EOD robots safely inspect suspicious objects, identify potential threats using camera/vision sensing, and assist operators with object recognition and navigation in cluttered environments. This reduces the need for humans to approach dangerous areas while enabling safer, faster bomb assessment and handling.

3) Humanoid Robots in Warfare - SLAUGHTERBOTS
Humanoid robots are being explored for high-risk missions like urban reconnaissance and operating in hazardous zones where human-like mobility is useful. They use perception and autonomy for balance, terrain navigation, object recognition, and manipulation—helping operators perform tasks while keeping soldiers out of direct danger.

4) AI-powered underwater drones - ZROVs in Warfare – REMUS (HII) / Saab Seaeye Sabertooth
UUVs / ROVs are used for underwater reconnaissance, mine detection, and covert inspection in high-risk maritime zones. They operate in low-visibility environments using sonar-based perception (side-scan sonar, multibeam sonar), sensor fusion, and autonomy for stable navigation, obstacle avoidance, seabed mapping, and target identification. These systems help navies monitor ports, track underwater threats, and neutralize hazards without putting divers or crewed submarines at risk.

THANK YOU

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

Mobile and Autonomous Robotics (UE23CS343BB7)

Understanding Robot Motion

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

Understanding Locomotion in a Robot

- A mobile robot needs locomotion mechanisms that enable it to move unbounded throughout its environment.
- But there are a large variety of possible ways to move, and so the selection of a robot's approach to locomotion is an important aspect of mobile robot design.
- There are robots that can walk, jump, run, slide, skate, swim, fly, roll.
- Mobile robots generally locomote either using **wheeled locomotion**, a well-known human technology for vehicles, or using a small number of **articulated legs**, the simplest of the biological approaches to locomotion.

Robotic Motion

Most of these locomotion mechanisms have been inspired by their biological counterparts.

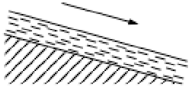
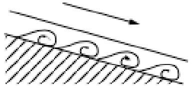
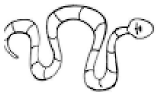
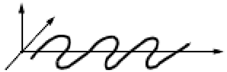
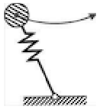
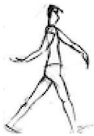
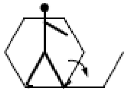
Type of motion	Resistance to motion	Basic kinematics of motion
Flow in a Channel 	Hydrodynamic forces	Eddies 
Crawl 	Friction forces	 Longitudinal vibration
Sliding 	Friction forces	 Transverse vibration
Running 	Loss of kinetic energy	Periodic bouncing on a spring 
Walking 	Loss of kinetic energy	Rolling of a polygon (see figure 2.2) 

Figure 2.1
Locomotion mechanisms used in biological systems.

Understanding Locomotion in a Robot

- On **flat surfaces wheeled locomotion** is one to two orders of magnitude more efficient than legged locomotion.
- But as the surface becomes soft, wheeled locomotion accumulates inefficiencies due to rolling friction, whereas legged locomotion suffers much less because it consists only of point contacts with the ground.
- In effect, the efficiency of wheeled locomotion depends greatly on **environmental qualities**, particularly the flatness and hardness of the ground, while the efficiency of legged locomotion depends on the **leg mass and body mass**, both of which the robot must support at various points in a legged gait.

Robotic Motion : Key issues

Key issues for locomotion: Locomotion and manipulation share the same core issues of stability, contact characteristics, and environmental type:

- **Stability**
  - number and geometry of contact points
  - center of gravity
  - static/dynamic stability
  - inclination of terrain

- **Characteristics of contact**
  - contact point/path size and shape
  - angle of contact
  - Friction

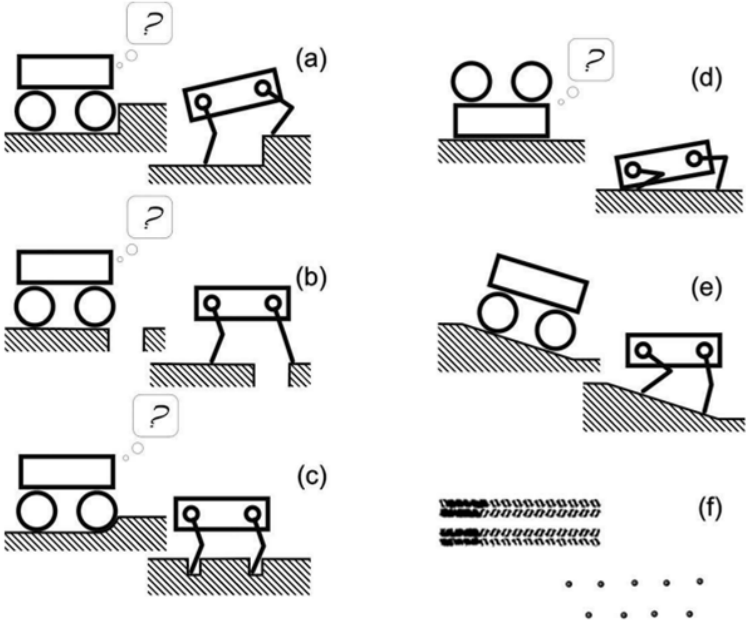
- **Type of environment**
  - structure
  - medium (e.g., water, air, soft or hard ground)

Legged Mobile Robots

- Legged locomotion is characterized by a series of point contacts between the robot and the ground.
- The key advantages include adaptability and maneuverability in rough terrain,
- Only a set of point contacts is required, the quality of the ground between those points does not matter as long as the robot can maintain adequate ground clearance.
- Another advantage of legged locomotion is the potential to manipulate objects in the environment with great skill.

Legged Mobile Robots

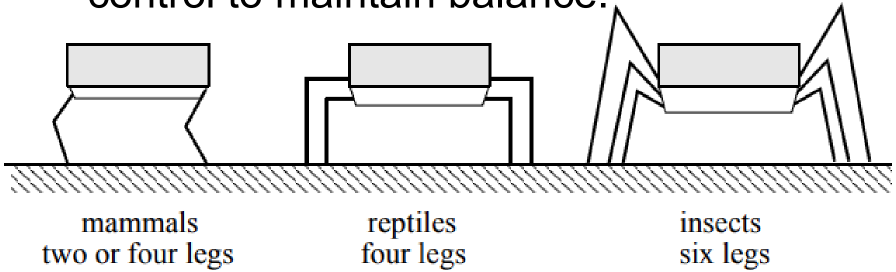
- The main disadvantages of legged locomotion include power and mechanical complexity.
- The leg, which may include several degrees of freedom, must be capable of sustaining part of the robot's total weight, and in many robots it must be capable of lifting and lowering the robot.
- High maneuverability will only be achieved if the legs have a sufficient number of degrees of freedom to impart forces in a number of different directions.



Legged Mobile Robots

Leg configurations and stability: A number of different leg configurations have been successful in a variety of organisms.

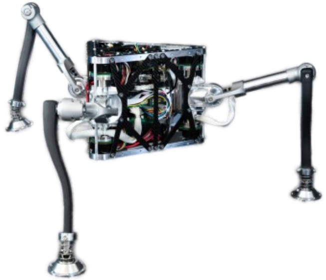
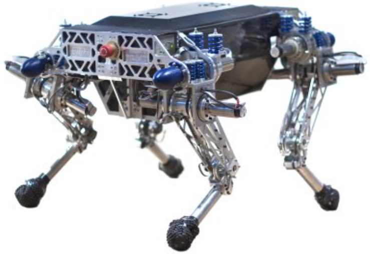
- Large animals, such as mammals and reptiles, have four legs, whereas insects have six or more legs.
- In some mammals, the ability to walk on only two legs has been perfected. Especially in the case of humans, balance has progressed to the point that we can even jump with one leg.
- This exceptional maneuverability comes at a price: much more complex active control to maintain balance.



Arrangement of the legs of various animals.

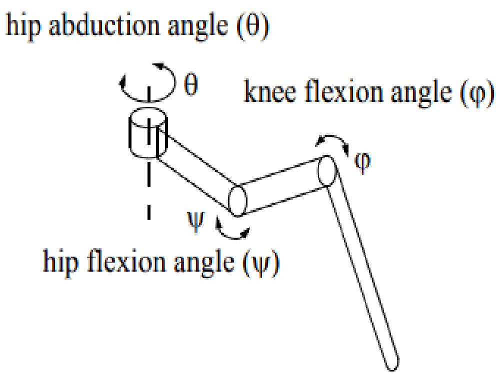
Legged Mobile Robots

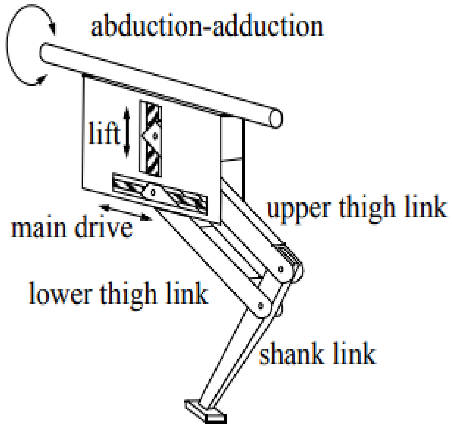
- In contrast, a creature with three legs can exhibit a static, stable pose provided that it can ensure that its center of gravity is within the tripod of ground contact.
- Static stability, demonstrated by a three-legged stool, means that balance is maintained with no need for motion.
- In order to achieve static walking, a robot must have at least four legs, moving one of them at a time.
- For six legs, it is possible to design a gait in which a statically stable tripod of legs is in contact with the ground at all times .


Legged Mobile Robots

- In the case of legged mobile robots, a minimum of two degrees of freedom is generally required to move a leg forward by lifting the leg and swinging it forward. More common is the addition of a third degree of freedom for more complex maneuvers, resulting in legs.
- the human leg has more than seven major degrees of freedom(8 complex joint)
- Adding degrees of freedom to a robot leg increases the maneuverability of the robot, both augmenting the range of terrains on which it can travel and the ability of the robot to travel with a variety of gaits.





Wheeled Mobile Robots

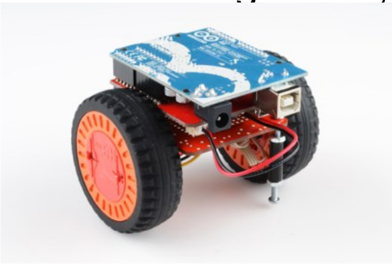
- The wheel has been by far the most popular locomotion mechanism in mobile robotics and in man-made vehicles in general.
- It can achieve very good efficiencies and it does so with a relatively simple mechanical implementation.
- Balance is not usually a research problem in wheeled robot designs, because wheeled robots are almost always designed so that all wheels are in ground contact at all times.
- Thus, three wheels are sufficient to guarantee stable balance, although two-wheeled robots can also be stable.
- When more than three wheels are used, a suspension system is required to allow all wheels to maintain ground contact when the robot encounters uneven terrain.

Wheeled Mobile Robots

- Wheeled robot research tends to focus on the problems of traction and stability, maneuverability, and control.
- The standard wheel and the castor wheel have a primary axis of rotation and are thus highly directional. To move in a different direction, the wheel must be steered first along a vertical axis.
- The Swedish wheel and the spherical wheel are both designs that are less constrained by directionality than the conventional standard wheel.

Wheeled Mobile Robots

- The minimum number of wheels required for static stability is two.
- Under ordinary circumstances such a solution requires wheel diameters that are impractically large.
- Dynamics can also cause a two-wheeled robot to strike the floor with a third point of contact, for instance, with sufficiently high motor torques from standstill.
- Conventionally, static stability requires a minimum of three wheels, with the additional caveat that the center of gravity must be contained within the triangle formed by the ground contact points of the wheels. The choice of wheel types for a mobile robot is strongly linked to the choice of wheel arrangement, or wheel ge y

Mobile and Autonomous Robotics

Wheeled Mobile Robots

Table 2.1
Wheel configurations for rolling vehicles

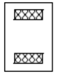
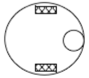
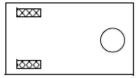
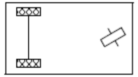
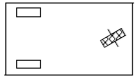
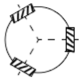
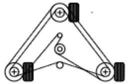
# of wheels	Arrangement	Description	Typical examples
2		One steering wheel in the front, one traction wheel in the rear	Bicycle, motorcycle
		Two-wheel differential drive with the center of mass (COM) below the axle	Cye personal robot
3		Two-wheel centered differential drive with a third point of contact	Nomad Scout, smartRob EPFL
		Two independently driven wheels in the rear/front, one unpowered omnidirectional wheel in the front/rear	Many indoor robots, including the EPFL robots Pygmalion and Alice
		Two connected traction wheels (differential) in rear, one steered free wheel in front	Piaggio minitrucks
		Two free wheels in rear, one steered traction wheel in front	Neptune (Carnegie Mellon University), Hero-1
		Three motorized Swedish or spherical wheels arranged in a triangle; omnidirectional movement is possible	Stanford wheel Tribolo EPFL, Palm Pilot Robot Kit (CMU)
		Three synchronously motorized and steered wheels; the orientation is not controllable	"Synchro drive" Denning MRV-2, Georgia Institute of Technology, I-Robot B24, Nomad 200

Table 2.1
Wheel configurations for rolling vehicles

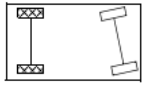
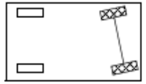
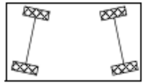
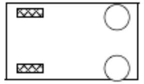
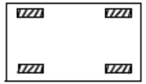
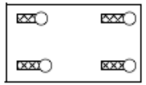
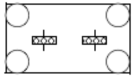
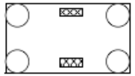
# of wheels	Arrangement	Description	Typical examples
4		Two motorized wheels in the rear, two steered wheels in the front; steering has to be different for the two wheels to avoid slipping/skidding.	Car with rear-wheel drive
		Two motorized and steered wheels in the front, two free wheels in the rear; steering has to be different for the two wheels to avoid slipping/skidding.	Car with front-wheel drive
		Four steered and motorized wheels	Four-wheel drive, four-wheel steering Hyperion (CMU)
		Two traction wheels (differential) in rear/front, two omnidirectional wheels in the front/rear	Charlie (DMT-EPFL)
		Four omnidirectional wheels	Carnegie Mellon Uranus
		Two-wheel differential drive with two additional points of contact	EPFL Khepera, Hyperbot Chip
		Four motorized and steered castor wheels	Nomad XR4000

Table 2.1
Wheel configurations for rolling vehicles

# of wheels	Arrangement	Description	Typical examples
6		Two motorized and steered wheels aligned in center, one omnidirectional wheel at each corner	First
		Two traction wheels (differential) in center, one omnidirectional wheel at each corner	Terregator (Carnegie Mellon University)

Icons for the each wheel type are as follows:	
	unpowered omnidirectional wheel (spherical, castor, Swedish)
	motorized Swedish wheel (Stanford wheel)
	unpowered standard wheel
	motorized standard wheel
	motorized and steered castor wheel
	steered standard wheel
	connected wheels

Robot dynamics

Robot dynamics

Robot dynamics is concerned with the relationship between the forces acting on a robot mechanism and the accelerations they produce. Typically, the robot mechanism is modelled as a rigid-body system, in which case robot dynamics is the application of rigid-body dynamics to robots.

The two main problems in robot dynamics are:

- **Forward dynamics:** given the forces, work out the accelerations.
  - Forward dynamics is also known as "direct dynamics," or sometimes simply as "dynamics." It is mainly used for simulation.
- **Inverse dynamics:** given the accelerations, work out the forces.
  - Inverse dynamics has various uses, including: on-line control of robot motions and forces, trajectory design and optimization, design of robot mechanisms, and as a component in some forward-dynamics algorithms.

Dynamics Algorithms

There are three main algorithms used in robot dynamics:

- The **recursive Newton-Euler algorithm (RNEA)**. This algorithm solves the inverse dynamics problem, and has a computational complexity of O(n).
- The **articulated-body algorithm (ABA)**. This algorithm solves the forward dynamics problem, and has a computational complexity of O(n).
- The **composite-rigid-body algorithm (CRBA)** is a computational algorithm used in the context of robot dynamics and control. It is employed for the efficient computation of the mass matrix in the equations of motion for a robotic system.

Forward and Inverse Kinematics

Kinematics is the most basic study of how mechanical systems behave.

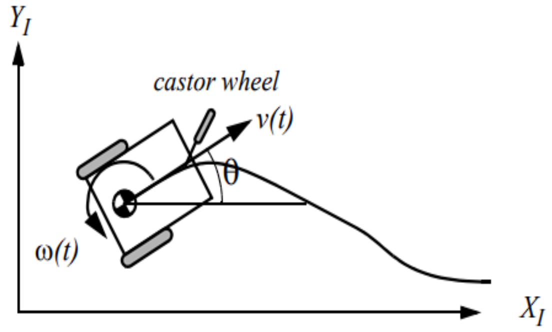
Forward kinematics

- In the fantastical world of robotics, forward kinematics plays a crucial role, acting as the translator between a robot's joint configurations and its final position and orientation.
- Think of it as the decoder ring, unraveling the mystery of how a robot's internal movements translate into its external actions.
- It's the foundation for many robotic tasks, from picking up objects and navigating obstacles to writing words or painting masterpieces.

How it works?

- Each joint in a robot is like a variable in a complex equation. By inputting the values of these joint angles, we can solve for the position and orientation of the end effector using mathematical formulas specific to the robot's structure.

- These formulas often involve concepts like transformation matrices and vector rotations, which can seem daunting at first. But think of it as mapping coordinates on a map – you take the robot's starting point (joint angles) and use the map (kinematic equations) to find its final destination (end effector pose).



The future of forward kinematics:

- As robots become more complex and interact with the world in increasingly sophisticated ways, forward kinematics will continue to evolve.
- Advancements in machine learning and artificial intelligence could lead to more efficient and adaptive kinematic algorithms.

Inverse Kinematics

- While forward kinematics takes the reins, guiding us from joint angles to robot poses, inverse kinematics flips the script. It tackles the puzzle of finding the specific joint angles needed to place the robot's end effector (think of it as its "hand" or working tool) in a desired position and orientation.

- Inverse kinematics is the mathematical process of calculating the joint angles of a robot needed to achieve a specific desired position and orientation of its end effector.

Why it's important:

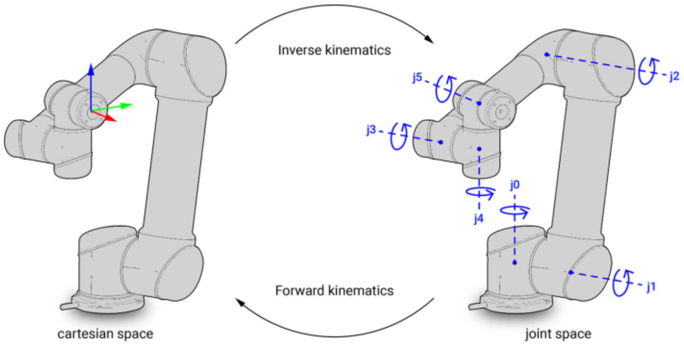
- It's crucial for planning robot movements and executing precise tasks. From robotic arms picking up objects to self-driving cars navigating roads, inverse kinematics makes it all possible.
- Without it, robots would be like puppets without strings, unable to perform controlled and purposeful actions.

Inverse Kinematics

How it works:

- Unlike the straightforward calculations of forward kinematics, inverse kinematics often involves complex optimisation algorithms. These algorithms search for the "best" combination of joint angles that satisfies the desired pose of the end effector while considering joint limitations and other constraints.

- Solving these equations can be computationally expensive, but the payoff is significant – precise and controlled robot movements.



THANK YOU

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

Mobile and Autonomous Robots (UE23CS343BB7)

LOCOMOTION MECHANISMS FOR ROBOTS

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

Locomotion Mechanisms for Robots

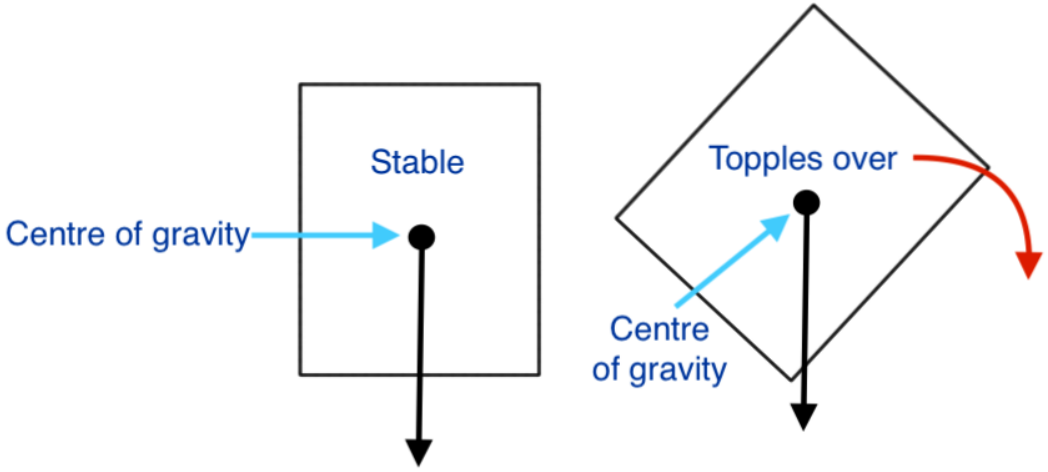
- A robot needs locomotion mechanisms that enable it to move unbounded through-out its environment. These movements include the ability to walk, run, jump, slide, skate, swim, fly and roll.
- Mobile robots generally locomote either using wheeled mechanisms or using articulated legs, the simplest of the biological approaches to locomotion.
- Legged locomotion requires higher degrees of freedom and therefore greater mechanical complexity than wheeled locomotion.
- On the other hand, wheels are more suited to flat ground due to minimum rolling friction. But as the surface becomes soft, wheeled locomotion accumulates inefficiencies due to rolling friction whereas legged locomotion suffers much less because it consists only of point contacts with the ground.

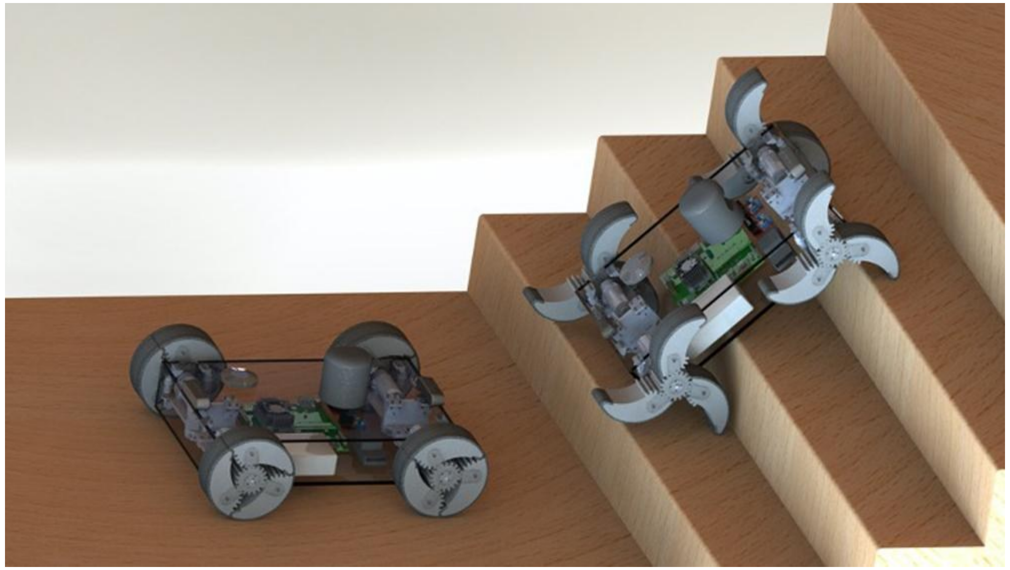
Key Issues for Locomotion

1. Stability:
- number and geometry of contact points
- centre of gravity
- static/dynamic stability
- inclination of terrain

2. Characteristics of Contact:
- contact point/path size and shape
- angle of contact
- friction

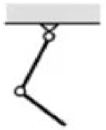
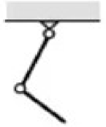
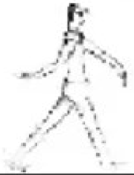
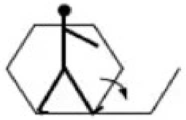
3. Type of Environment:
- structure
- medium (water, air, soft or hard ground)





Types of Motion

- Biological locomotion is evolved to deal with harsh environments and is desirable to imitate.
- However, mechanical complexity is achieved in biological systems through **structural replication and cell division**. Imitating self-replicating artefacts like biological cells can not be fabricated in laboratory easily.
- **Energy generation and storage** mechanisms in biological systems cannot be easily applied to man-made systems.

Running 	Loss of kinetic energy	Oscillatory movement of a multi-link pendulum 
Jumping 	Loss of kinetic energy	Oscillatory movement of a multi-link pendulum 
Walking 	Gravitational forces	Rolling of a polygon (see figure 2.2) 

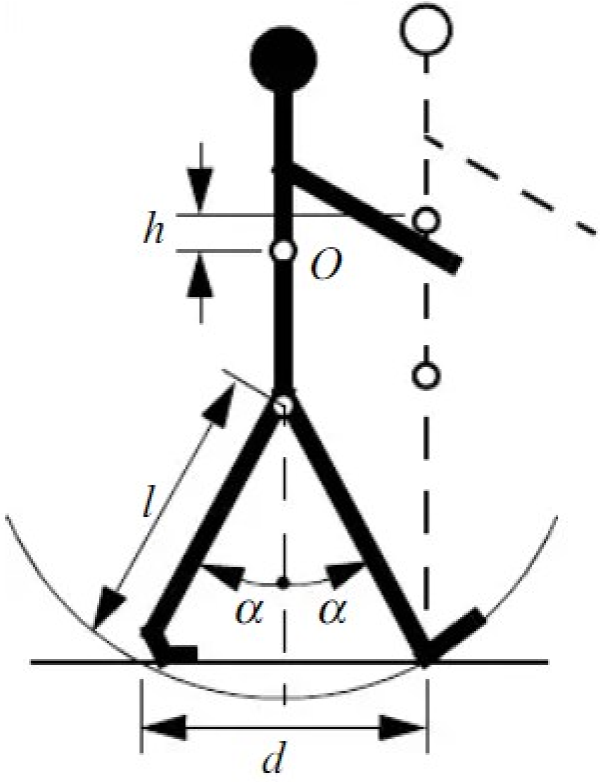


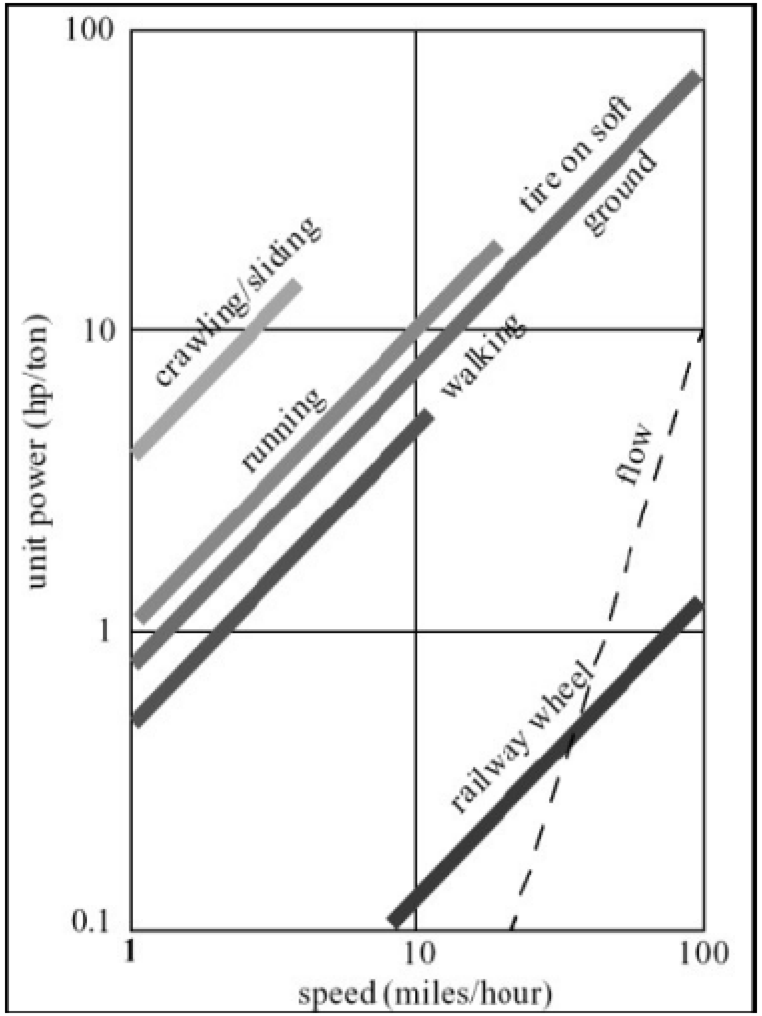
Figure 2.2

Biped walking system approximated by rolling polygon
With side = length d = span of step
As step size decreases, polygon approaches circle/wheel with radius *l*.

Specific Power vs. Attainable Speed

Locomotion Mechanisms:

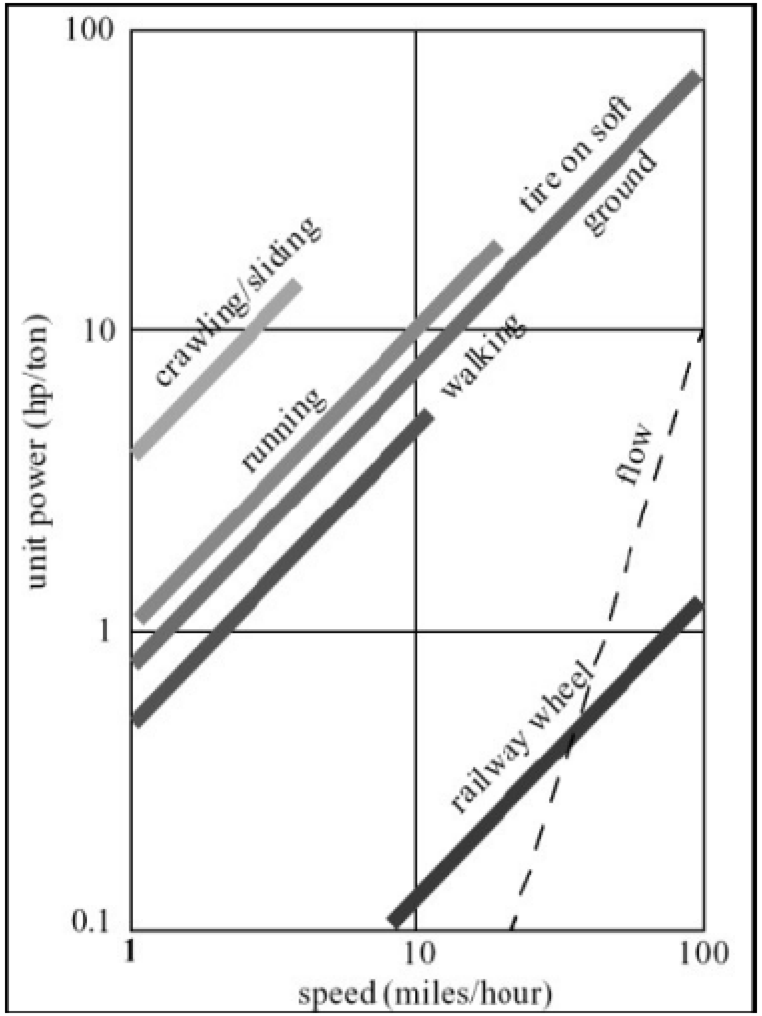
- **Crawling/Sliding:**
  - Very high specific power (inefficient) with low speed, suitable for extreme terrain or low mobility requirements.
- **Running and Walking:**
  - Moderate efficiency; walking is slightly more efficient than running at lower speeds.
  - Speeds are limited compared to wheeled locomotion.
- **Wheels on Soft Ground:**
  - Rolling friction increases dramatically on soft surfaces, reducing efficiency.
- **Railway Wheel (on hard steel surface):**
  - The most efficient locomotion mechanism due to minimal rolling friction, allowing high speed and low specific power.
- **Flow-based Locomotion (e.g., fluids, conveyor belts):**
  - Theoretical upper bound of efficiency, with some overlap with railway wheel at higher speeds.



Power consumption of different locomotion mechanisms

Specific Power vs. Attainable Speed

- **Flat Surfaces:**
  - Railway wheels dominate in efficiency due to reduced energy loss from friction.
- **Soft Surfaces:**
  - Tires on soft ground lose efficiency, while legged locomotion (walking/running) remains more stable.
- **Efficiency Drop for Wheels:**
  - As the surface becomes softer, wheels lose their energy advantage over legs.
- **Speed Range:**
  - Railway wheels achieve high speeds efficiently (10–100 mph).
  - Walking and running are limited to lower speeds (~1–10 mph).



Power consumption of different locomotion mechanisms

Types of Locomotion

1. **Terrestrial:**
Walking, galloping, running, hopping, crawling, sliding, rolling

2. **Flight:**
Flapping wing, gliding, perching

3. **Aquatic:**
Swimming (undulatory, flagellar, ciliar), jet propulsion

LOCOMOTION MECHANISMS FOR ROBOTS

Legged Mobile Robots

Legged Mobile Robots

- A legged robot is well suited for rough terrain and irregular grounds.
- It is characterized by a series of point contacts between the robot and the ground.
- Although legs allow for high adaptability and manoeuvrability, they consume more power.
- A walking robot is capable of crossing a hole or chasm so long as its reach exceeds the width of the hole.
- Quality of ground does not matter as long as robot maintains adequate ground clearance.
- Another potential is to manipulate objects in the environment with great skill.
- However, they are mechanically complex and require sufficient number of degrees of freedom to impart forces in a number of different directions to achieve maneuverability.

Legged Mobile Robots

https://youtu.be/beE5CpxiCDw?si=THp8yOPMSQWKAr3q

Leg Configurations

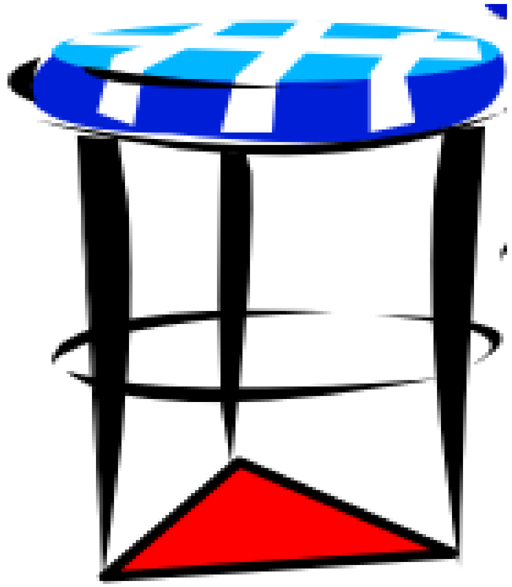
- Large animals, such as mammals and reptiles, have four legs, whereas insects have six or more legs.
- In case of humans, balance has progressed to the point that we can even jump with one leg. This exceptional maneuverability comes at a price: much more complex active control to maintain balance.
- In contrast, a creature with three legs can exhibit a static, stable pose provided that its centre of gravity is within the tripod of ground contact.
- A small deviation from stability (e.g., gently pushing the stool) is passively corrected toward the stable pose when the upsetting force stops.
- To achieve static walking, a robot must have at least six legs and be able to lift them. In such a configuration, it is possible to design a gait where a statically stable tripod of legs is in contact with the ground at all times.

Stability

- The primary disadvantages of additional joints and actuators are energy, control, and mass.
- Additional actuators require energy and control, and they also add to leg mass, further increasing power and load requirements on existing actuators.
- Thus, stability is a very important issue for a robot. It should not overturn.
- There are 2 types; **static and dynamic stability**.
- Static stability means that the robot is stable, with no need of motion at every moment of time. A statically stable mechanism will not fall even when all of its joints freeze.
- A dynamically stable robot instead requires constant motion to prevent it from falling.

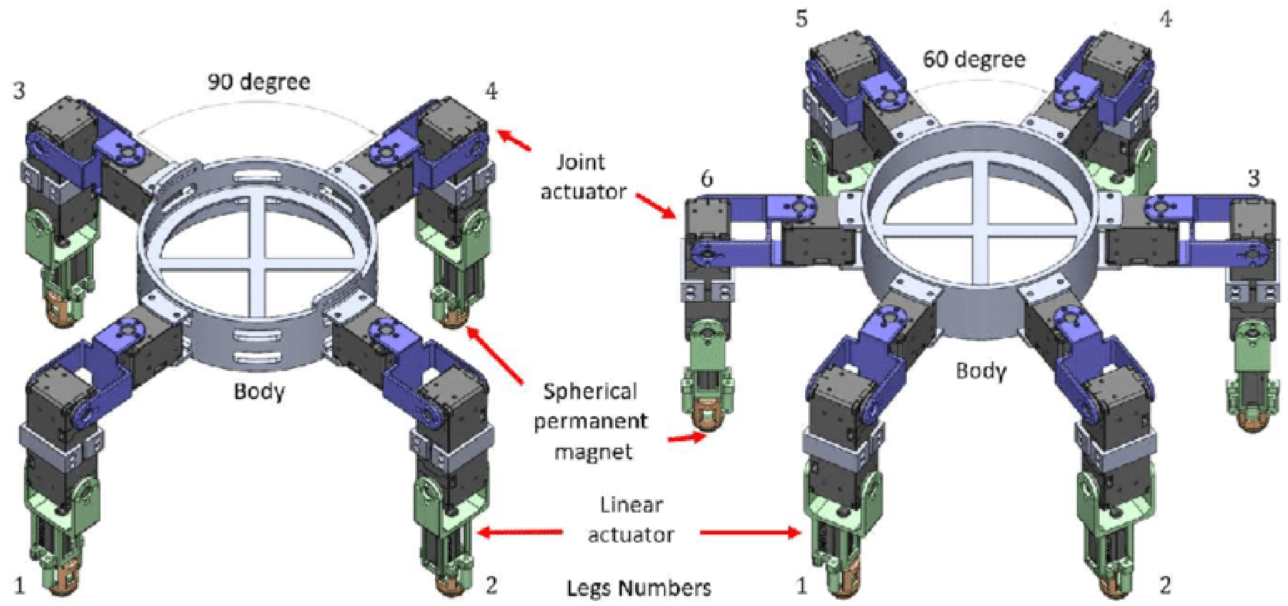
Static Stability

- Consider the three-legged stool.
- Balance is maintained as long as the **centre of mass** is completely within the red triangle, which is set by the stools' footprints. This triangle is called ***support polygon***.
- The support polygon is the convex hull which is set by the ground contact points. If there are more ground contact points, the polygon can be a quadrangle or a pentagon or a different geometrical figure.
- Static stability is given, when the centre of mass is completely within the support polygon and the polygon's area is greater than zero, therefore static stability requires at least three points of ground contact.



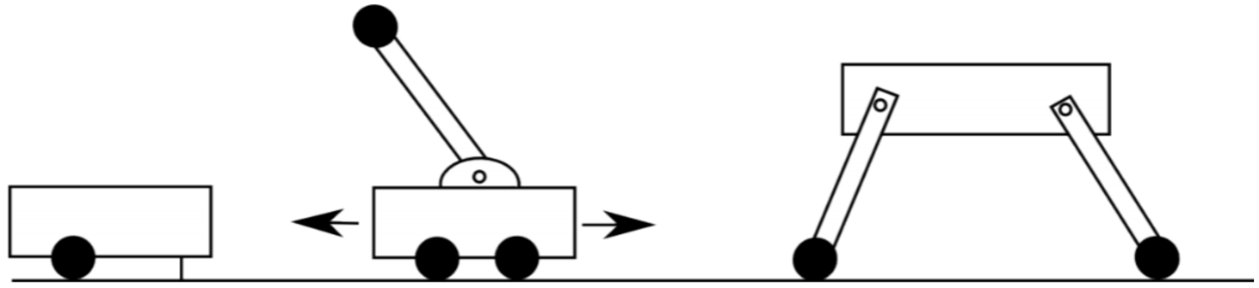
Static Stability

- To achieve statically stable, walking a robot must have a minimum number of four legs, because during walking at least one leg is in the air.
- Most robots which are able to walk static stable have six legs, because walking static stable with four legs means that just one leg can be lifted at the same time (lifting more legs will reduce the support polygon to a line), so walking becomes slow.



Dynamic Stability

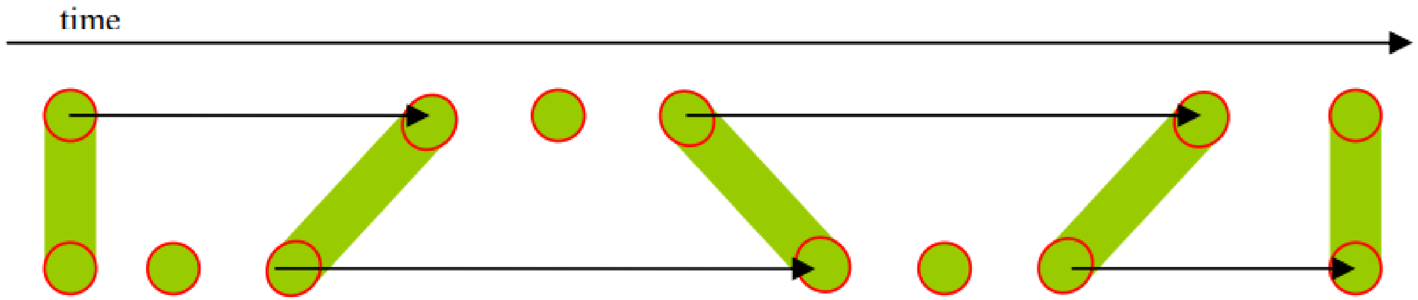
- A quadruped robot's feet span a rectangle. Once such a robot lifts one of its feet, this rectangle becomes a triangle. If the projection of the centre of mass of the robot along the direction of gravity is outside of this triangle, the robot will fall.
- A dynamically stable robot can overcome this problem by changing its configuration rapidly.
- An example of a purely dynamically stable robot is an inverted pendulum on a cart.
- Such a robot has no statically stable configurations and needs to keep moving all the time to keep the pendulum upright. While dynamic stability is desirable for high-speed, agile motions, robots should be designed so that they can easily switch into a statically stable configuration.



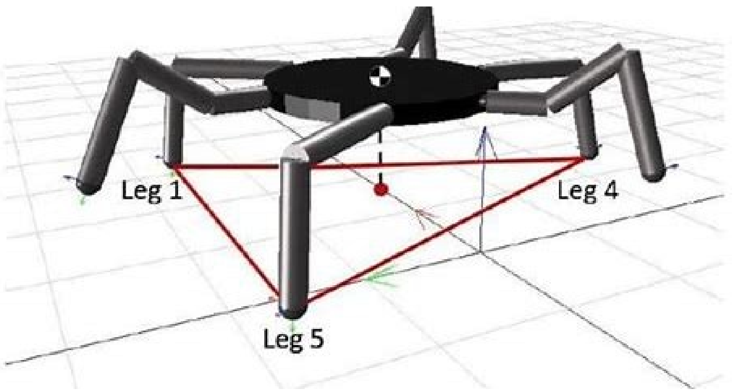
From left to right: statically stable robot. Dynamically stable inverted pendulum robot. Static and dynamically stable robot (depending on configuration).

Dynamic Stability

- Most two legged walking machines are dynamically stable for several reasons.
- Human like robots have relatively small footprints, because of this the support polygon is almost a line (in the double support phase, when both feet are connected with the ground) which is even reduced to a single point (in the single support phase, when just one foot has ground contact) during walking.
- Therefore the robot must actively balance itself to prevent overturning.
- In face of that the robots' centre of mass has to be shifted actively between the footprints



A bipedal robots' gait circle; the red circles indicate the footprints; the green area is the supporting area; the leg movement is expressed by the arrows



Degrees of freedom

- For legged mobile robots, a minimum of two degrees of freedom is required to move a leg forward by lifting the leg and swinging it forward.
- Addition of a third degree of freedom for more complex manoeuvres as shown below.
- Bipedal walking robots have added a fourth degree of freedom at the ankle joint.
- The ankle enables more consistent ground contact by actuating the pose of the sole of the foot.
- In general, adding degrees of freedom to a robot leg increases the manoeuvrability of the robot, both augmenting the range of terrains on which it can travel and the ability of the robot to travel with a variety of gaits.

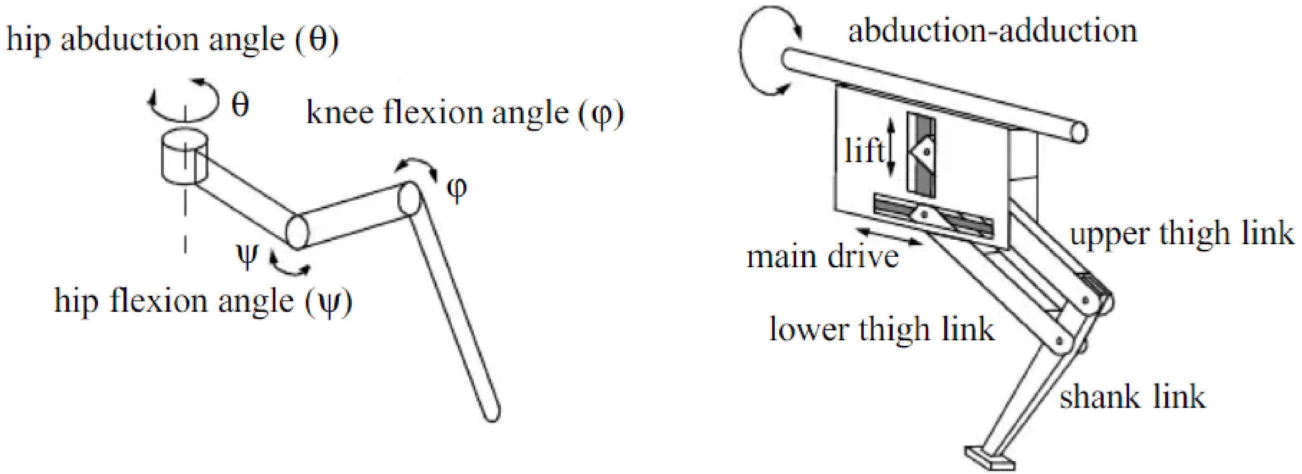
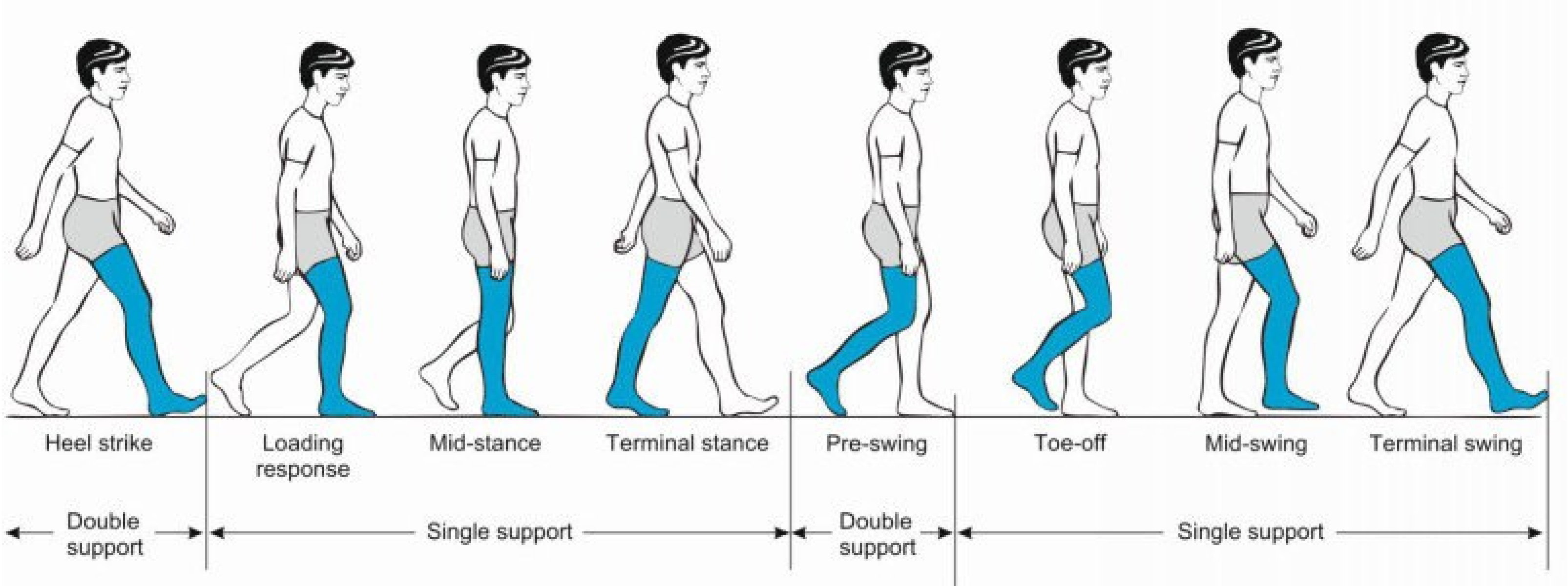


Figure 2.6
Two examples of legs with three degrees of freedom.

Leg Coordination

- The human leg has more than seven major degrees of freedom, combined with further actuation at the toes. More than fifteen muscle groups actuate eight complex joints.
- If the robot has more than one leg there is the issue of leg coordination for locomotion.
- The total number of possible gaits in which a robot can travel depends on the number on legs it has.
- The gait is a periodic sequence of lift and release events for each leg.



Leg Coordination

- If a robot has k legs, the number of possible events N is given by, $N=(2k-1)!$
- In case of a bipedal walking machine (k=2), the number of possible events is

$$N=(2k-1)! = (2*2-1)! = 3! = 6.$$

- The six distinct event sequences that can be combined for more complex sequences are:

1. both legs down – right down / left up – both legs down;
2. both legs down – right leg up / left leg down – both legs down;
3. both legs down – both legs up – both legs down;
4. right leg down / left leg up – right leg up / left leg down – right leg down / left leg up;
5. right leg down / left leg up – both legs up – right leg down / left leg up;
6. right leg up / left leg down – both legs up – right leg up / left leg down.

- Of course, this quickly grows quite large. For example, a robot with six legs has far more events: **N = 11! = 39916800**

Leg Coordination

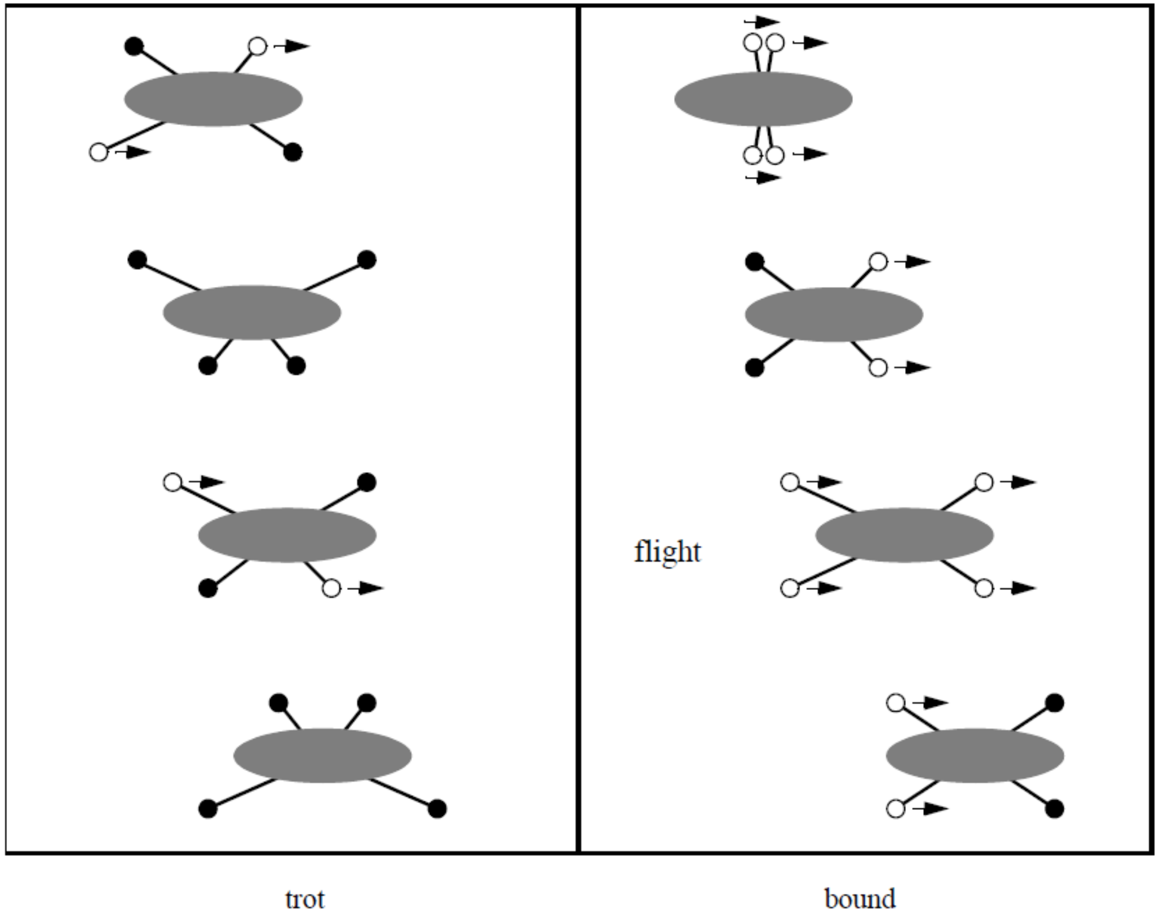


Figure 2.8
Two gaits with four legs.

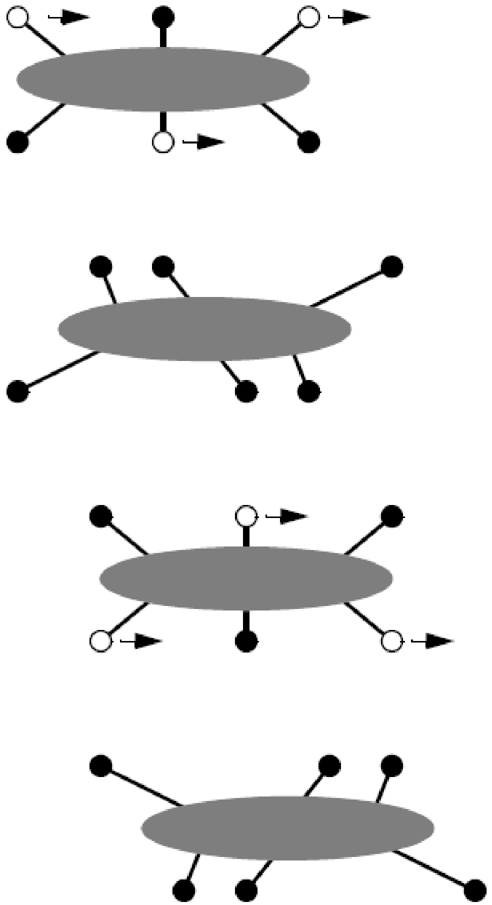


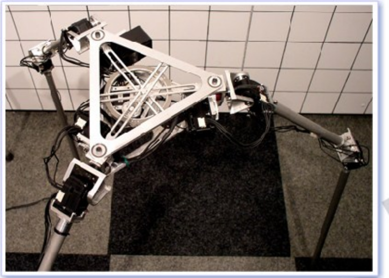
Figure 2.9
Static walking with six legs. A tripod formed by three legs always exists.

Examples of legged robot locomotion

One Leg

Bipedal

Tripedal



Quadrupedal

Hexapod

Many Legs

One Leg

- One leg is of course the minimum number of legs which a legged robot can have.
- A smaller number of legs **reduces body mass** of the robot and no leg coordination is needed.
- One-legged locomotion requires just a **single point of ground contact**; this makes the robot amenable to travel the roughest terrain.
- A hopping robot can dynamically cross a gap that is larger than its stride by taking a running start, whereas a multilegged walking robot that cannot run is limited to crossing gaps that are as large as its reach.
- The **major challenge** in creating a single-legged robot **is balance**.
- For a robot with one leg, static walking is not only impossible but static stability when stationary is also impossible. The robot must actively balance itself by either changing its centre of gravity or by imparting corrective forces. Thus, the successful single-legged robot must be dynamically stable.

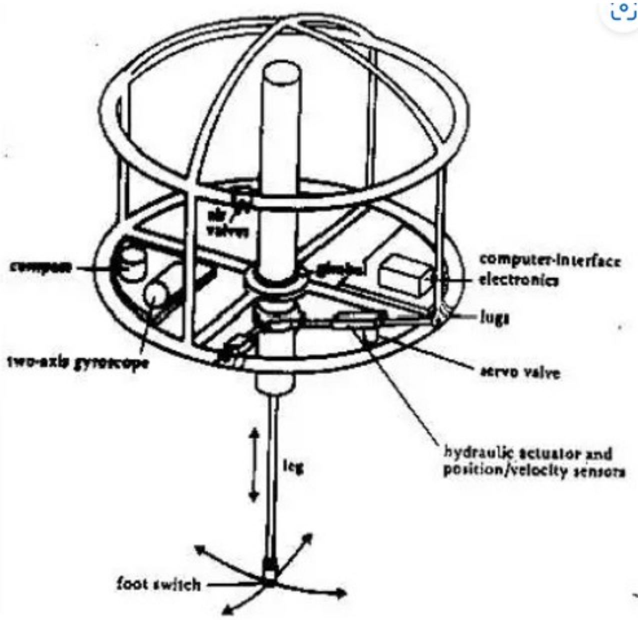
Raibert's Hopper – One Legged Robot Example

- Raibert's hopper is a one legged robot that needs to hop at all times and is unstable when stationary.
- Raiberts' hopper uses a simple controller, which divides the control problem into three independent parts. These parts are -

❑ **Hopping height**

❑ **Velocity**

❑ **Attitude**



Two Legs (Biped)

- **Movement Abilities of Two-Legged Robots**:

Two-legged robots can perform various actions like walking, running, jumping, dancing, and navigating stairs. However, maintaining stability remains a challenge due to the need for dynamic stability.

- **Zero Moment Point (ZMP) Approach:**

To ensure balance, the ZMP approach involves planning the position of the robot's footprints. It identifies the point on the ground where the robot must be based to maintain balance, with all momentums equal to zero.

Figure 7: QRIO (Sony)

Two Legs (Biped)

- **Weight Advantage of Two-Legged Locomotion:**

Two-legged robots benefit from reduced total weight compared to multi-legged robots. While this minimizes overall mass, it introduces the challenge of ensuring each leg can support the entire weight of the robot.

- **Anthropomorphic Shape in Bipedal Robots:**

Bipedal robots are designed with an anthropomorphic shape, resembling human dimensions. This characteristic makes them valuable for research in human-robot interaction, facilitating studies and developments in this field.

Figure 7: QRIO (Sony)

Four Legs (Quadruped)

- **Challenges in Four-Legged Robot Walking:**

While standing on four legs is inherently stable, walking presents difficulties because the robot's center of gravity needs active shifting throughout the gait to maintain stability.

- **Sony's Aibo - A Well-Known Four-Legged Robot:**

Sony's Aibo, a famous four-legged robot, incorporates features such as sound detection, a head sensor responding to taps, eye lights reflecting emotional states, and a color camera for object recognition based on color and movement.

Figure 11: Aibo (Sony) [10]

Four Legs (Quadruped)

- **Dynamic Stable Walking in Four-Legged Robots:**

Most four-legged robots utilize dynamic stable walking, as static stable walking demands a minimum of three ground contact points. Dynamic stability allows varying the number of ground contact points, ranging from zero during jumps to the total number of legs when stationary.

- **AIBO's Emulation of Learning and Maturation:**

AIBO's walking style and behavior simulate learning and maturation, providing a dynamic experience for the owner to observe and track over time.

Figure 11: Aibo (Sony) [10]

Four Legs (Quadruped) - Configuration

- The body of the robot is described by a polygon, the legs by straight lines and the footprints by empty or filled circles (empty if the leg is raised, filled if the leg has ground contact).
- The centre of the polygon is considered as the robots centre of mass, indicated the green point (this is of course a simplified assumption).
- The number one to five indicates a predefined set of leg position. That means that each leg can take one of these positions and it can be raised or set down.
- The total number of possible robot configurations is equal to $5^4 \times 2^4 = 10000$, but not all of these are stable.

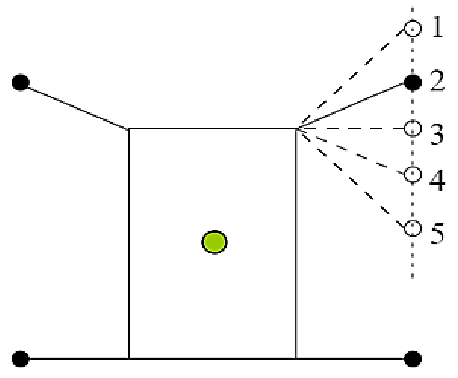


Figure 13 a): s2D+1 Model of a four Legged robot

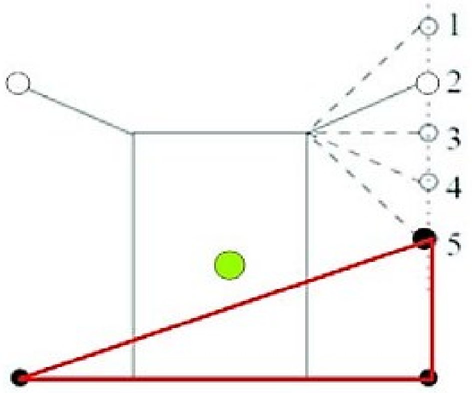


Figure 14 c) unstable configuration

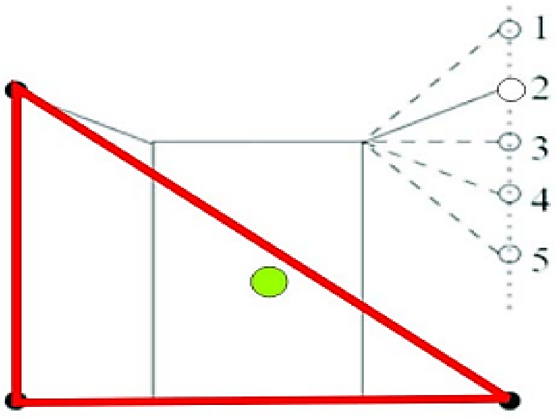
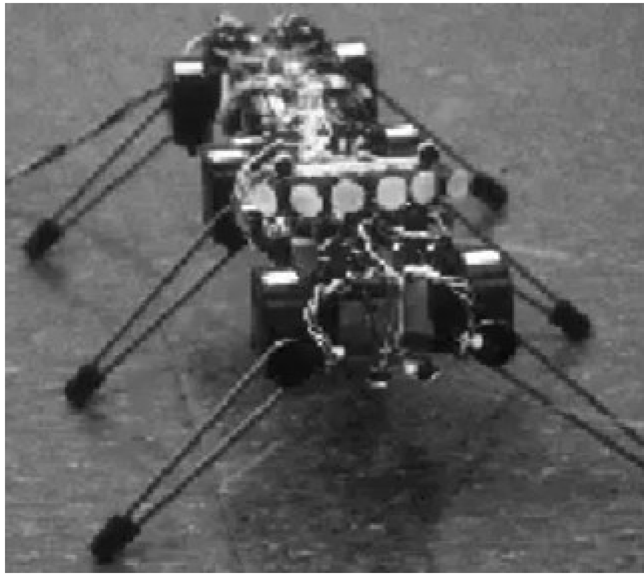


Figure 13 b): stable configuration

Six Legs (Hexapod)

- **Popularity of Six-Legged Locomotion:**

Six-legged locomotion is widely favored due to its concept of static stable walking, which simplifies control complexities associated with stability.



- **Tripod Gait in Six-Legged Locomotion:**

The tripod gait, a commonly used static stable walking pattern in six-legged locomotion, involves moving two exterior legs on one side and the inner leg on the other side simultaneously.

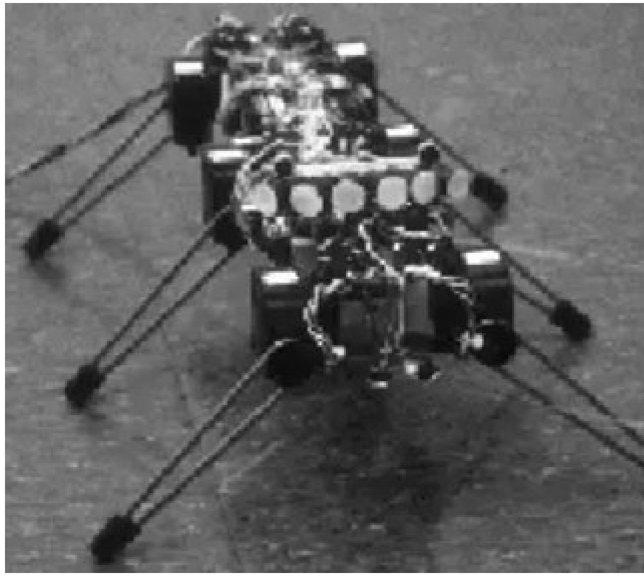
- **Control Complexity in Six-Legged Robots:**

Despite the advantage of static stable gaits reducing stability control concerns, the complexity arises in coordinating the three degrees of freedom in each of the six legs, increasing the challenge of leg coordination.

Six Legs (Hexapod)

- **Degrees of Freedom in Six-Legged Robots:**

Typically, each leg in six-legged robots has three degrees of freedom, including hip and knee flexion, as well as hip abduction, contributing to their range of motion.



- **Insects vs. Six-Legged Robots:**

Insects, highly successful terrestrial locomotors, navigate diverse terrains upside down using their six legs. Currently, artificial six-legged robots lag behind insects in locomotive abilities, despite having sufficient degrees of freedom.

Problems to think

Consider an eight-legged walking robot. Consider gaits in terms of lift/release events.

(a) How many possible events exist for this eight-legged machine?

(b) Specify two different statically stable walking gaits using the notation of figure 2.8.

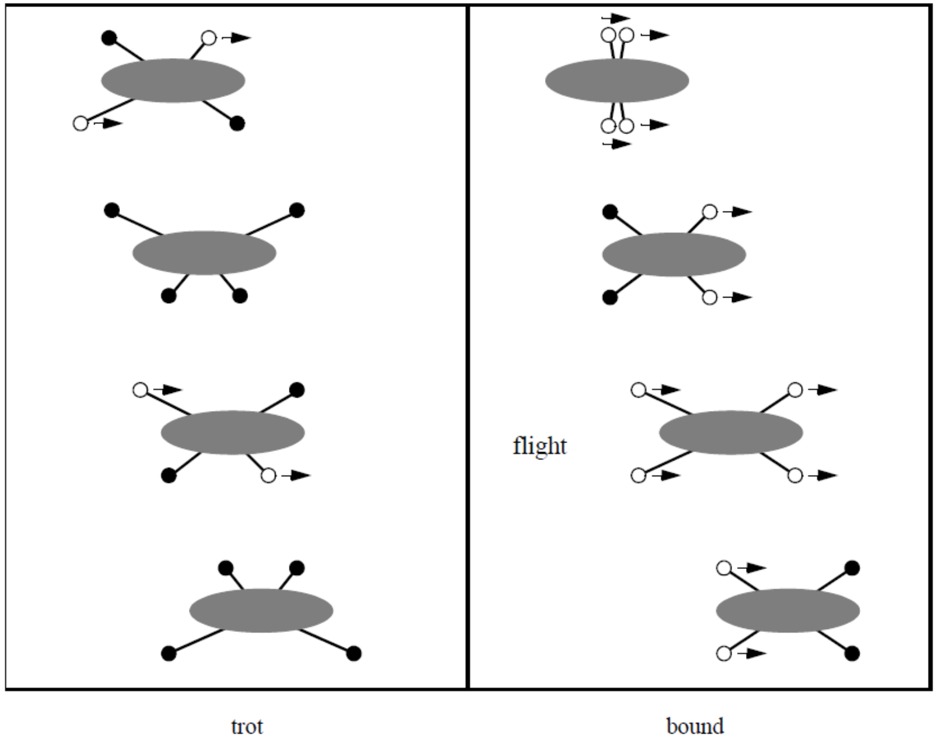


Figure 2.8
Two gaits with four legs.

THANK YOU

Mobile and Autonomous Robots (UE23CS343BB7)

LOCOMOTION MECHANISMS FOR ROBOTS

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

LOCOMOTION MECHANISMS FOR ROBOTS

Types of Locomotion:
Wheeled Robots

Wheeled Robots

- The most popular locomotion mechanism is wheeled locomotion.
- Reasons for this are the easy mechanical implementation of the wheel, there is no need of balance control if the vehicle has at least three or in some case two wheels and wheeled locomotion is relatively power efficient, even at high speeds.
- Wheeled robots are almost always designed so that all wheels are in ground contact at all times.
- Thus, three wheels are sufficient to guarantee stable balance, although, two-wheeled robots can also be stable. When more than three wheels are used, a suspension system is required to allow all wheels to maintain ground contact when the robot encounters uneven terrain
- Instead of balance, wheeled robot research tends to focus on the problems of traction and stability, manoeuvrability, and control.

Wheel Types

- In general there are four major classes of wheels.
- Figure a) shows the standard wheel with two degrees of freedom, these are rotation around the wheel axle and around the contact.
- Figure b) shows the castor wheel with two degrees of freedom, rotation around the wheel axle and the offset steering joint.
- Figure c) shows the Swedish 45° and Swedish 90° or omni wheel, which has three degrees of freedom: rotation around the contact point, around the wheel 13 axle and around the rollers.
- Figure d) shows the Ball or spherical wheel, this wheel is omnidirectional, but it is technical difficult to implement.

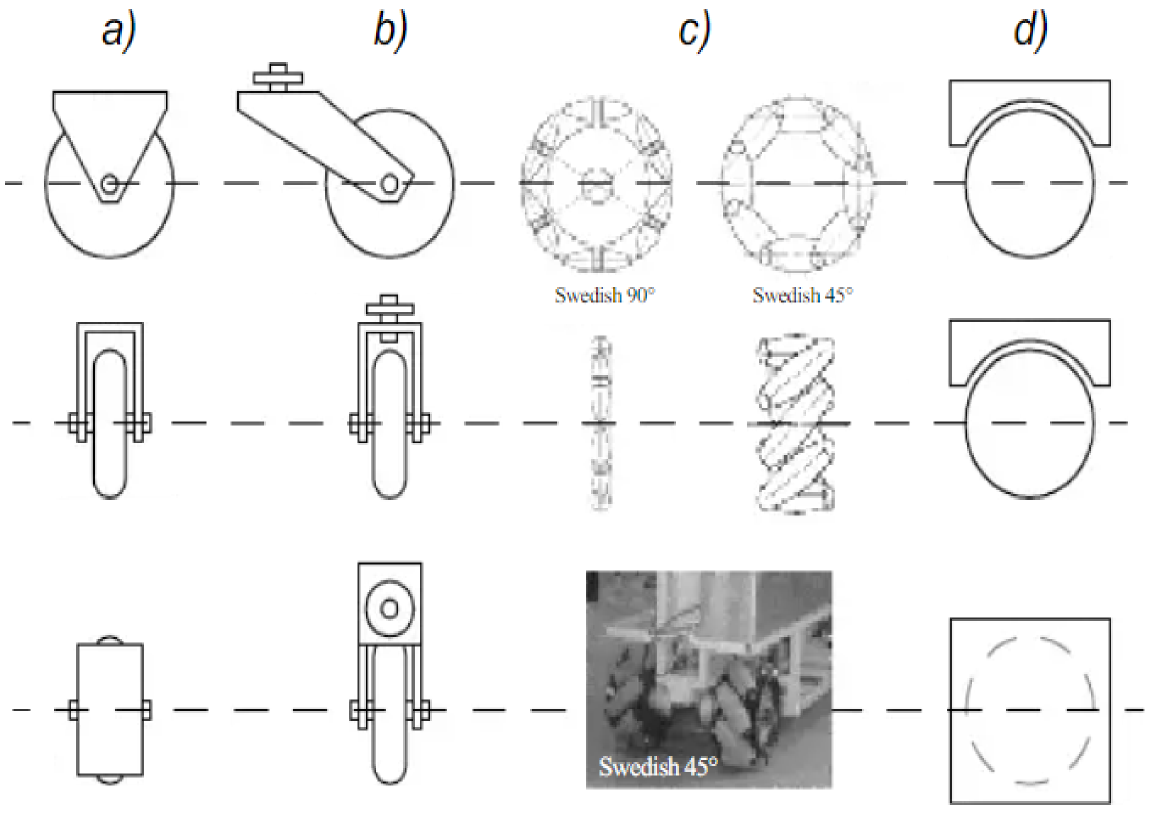


Figure : Four basic wheel types

Wheel Types

- **Advantages of Standard and Castor Wheels:**

Easy implementation, high load capacity, and tolerance to ground irregularities are key advantages of standard and castor wheels. However, these wheels are not inherently omnidirectional.

- **Steering Challenges in Non-Omnidirectional Wheels:**

To enable steering in vehicles using these wheels, the steerable wheel(s) must be first directed along a vertical axis and then moved around a horizontal axis.

- **Issues in Heavy Vehicles During Steering:**

For heavy vehicles at rest during steering, actively twisting the wheel around its vertical axis causes increased friction and scrubbing. This results in higher power consumption and reduced positioning accuracy for the vehicle.

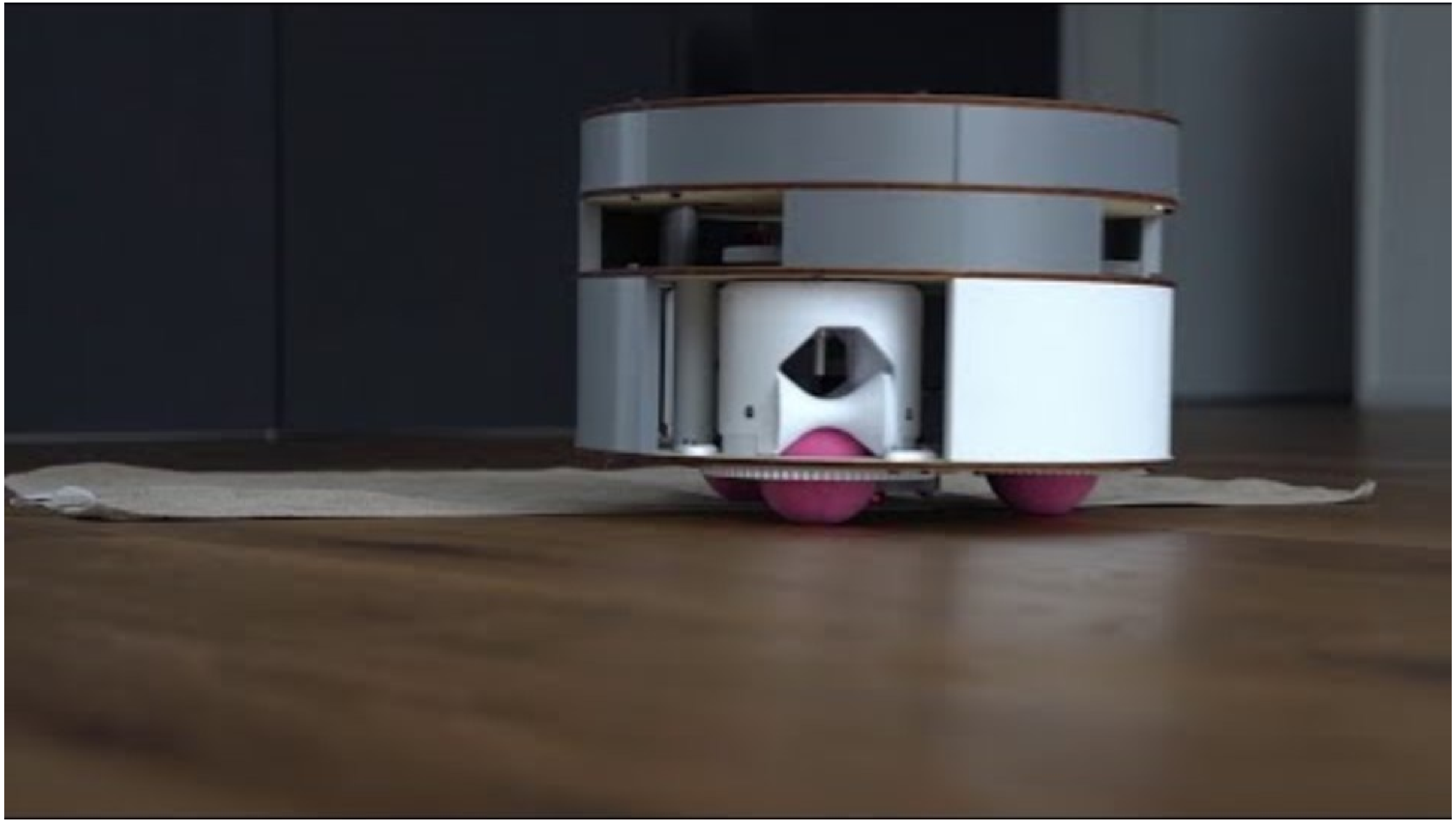
Wheel Design

- Standard wheels and castor wheels have a primary axis of rotation, making them highly directional.
- To change direction, standard wheels can be steered without side effects because the center of rotation passes through the contact patch with the ground. Castor wheels, however, rotate around an offset axis, which can cause forces on the robot chassis during steering.
- Swedish wheels function as normal wheels but offer low resistance in additional directions, such as perpendicular or at intermediate angles.
- Spherical wheels are truly omnidirectional and can be actively powered to spin in any direction. One mechanism for this design imitates a computer mouse, using actively powered rollers against the sphere's surface to impart rotational force.

Spherical Design

https://youtu.be/rHDCIng0Wak?si=8WSnkvs9z3NE0ZUG

Robotic Systems
Spherical Drive



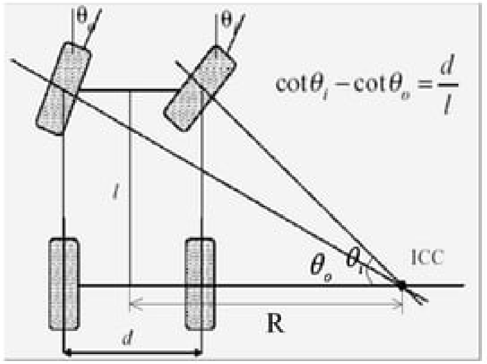
https://youtu.be/cvMBuUwgwwk?si=63gEkx6iw0IOQHex

Wheel Geometry

- Three fundamental characteristics of a robot :
  - Maneuverability,
  - Controllability,
  - Stability.

- One example of such a combination is the Ackermann wheel configuration of a car, with two steerable wheels in the front, two not steerable wheels in the rear; at least two wheels, connected by an axis, are motorized and maximizes controllability, stability and manoeuvrability in the same shared environment.

Car Drive (Ackerman Steering)



- Used in motor vehicles, the inside front wheel is rotated slightly sharper than the outside wheel (reduces tire slippage).
- Ackerman steering provides a fairly accurate dead-reckoning solution while supporting traction and ground clearance.
- Generally the method of choice for outdoor autonomous vehicles.

Wheel Geometry

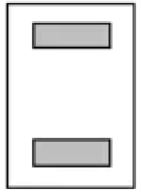
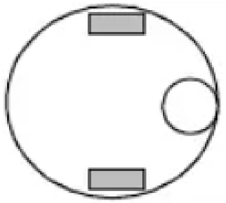
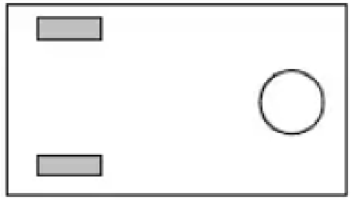
- In case of mobile robots there is not just one environment where all robots are designed for, different robots are designed for applications in a wide variety of situations. But there is no single wheel configuration that maximizes controllability, stability and manoeuvrability qualities for every environment.

- Table 2.1 gives an overview of wheel configurations ordered by the number of wheels.

- For instance, the two-wheeled bicycle arrangement has moderate maneuverability and poor controllability. Like a single-legged hopping machine, it can never stand still.

Wheel Geometry

Table 2.1

# of wheels	Arrangement	Description	Typical examples
2		One steering wheel in the front, one traction wheel in the rear	Bicycle, motorcycle
		Two-wheel differential drive with the center of mass (COM) below the axle	Cye personal robot
3		Two-wheel centered differential drive with a third point of contact	Nomad Scout, smartRob EPFL
		Two independently driven wheels in the rear/front, 1 unpowered omnidirectional wheel in the front/rear	Many indoor robots, including the EPFL robots Pygmalion and Alice

Maneuverability

- Maneuverability in robotics refers to the ability of a robotic system to move effectively and efficiently in its environment, adapting to obstacles, terrain, and other dynamic factors.
- It encompasses various aspects such as mobility, agility, and flexibility of movement.
- **Mobility:** Robots must move in various directions (forward, backward, sideways, and rotation) using mechanisms like wheels, tracks, legs, or flying capabilities.
- **Agility:** Robots need rapid direction and speed changes to avoid obstacles or respond to dynamic environments accurately.
- **Terrain Adaptability:** Some robots navigate challenging terrains such as rough surfaces or underwater environments using specialized locomotion mechanisms or adaptive features.
- **Control Algorithms:** Precise and coordinated robot motion relies on effective control algorithms that manage actuators and motors within the robot's body.

Maneuverability

- Some robots are omnidirectional, meaning that they can move at any time in any direction along the ground plane (x,y) regardless of the orientation of the robot around its vertical axis.

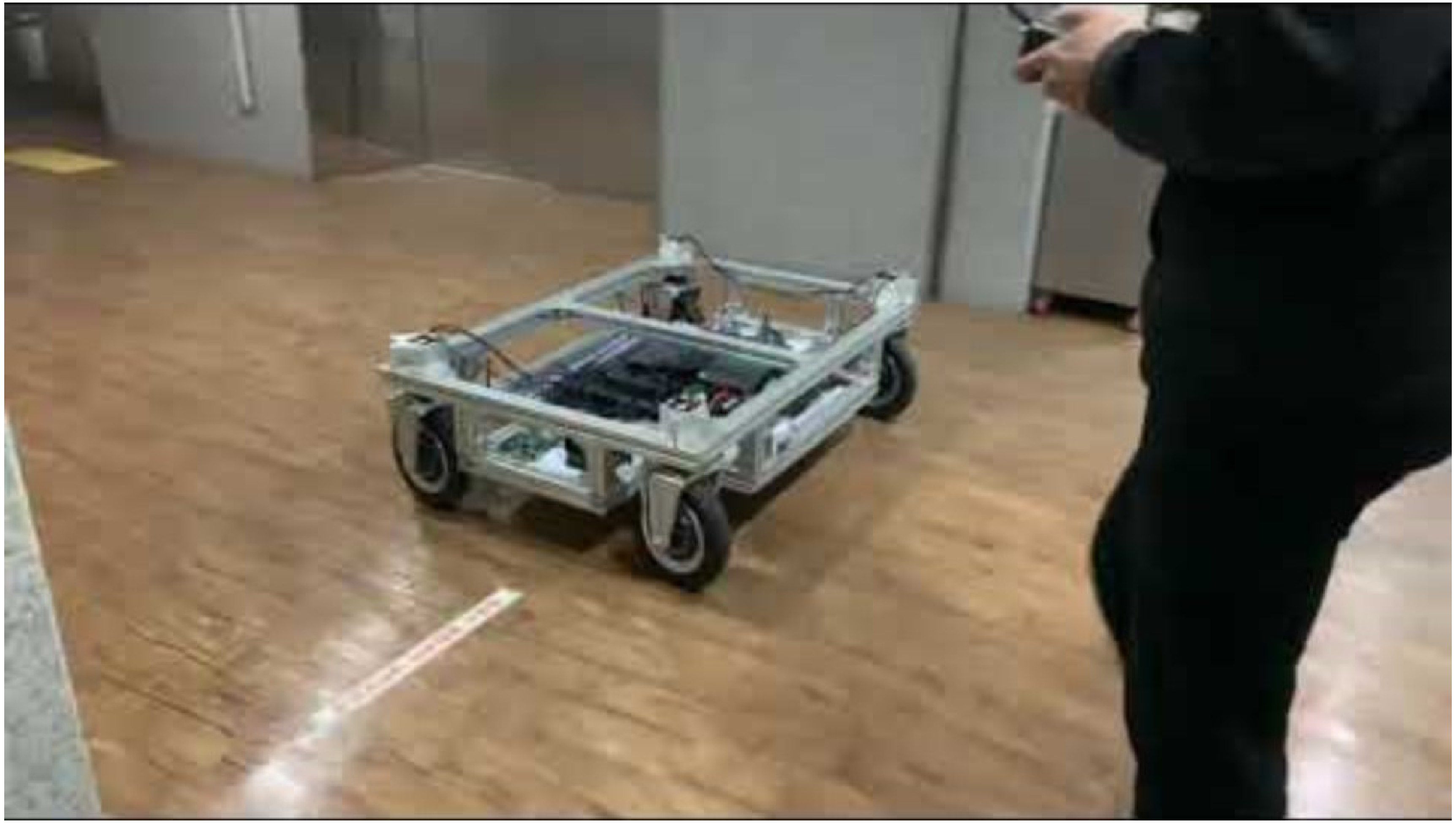
- But the ground clearance of these robots is limited due to omnidirectional wheels.

- This level of maneuverability requires wheels that can move in more than just one direction, and so omnidirectional robots usually employ Swedish or spherical wheels that are powered.

Maneuverability

- An interesting solution to the problem of omnidirectional navigation while solving this ground-clearance problem is the four-castor wheel configuration in which each castor wheel is actively steered and actively translated.
- The robot is truly omnidirectional because, even if the castor wheels are facing a direction perpendicular to the desired direction of travel, the robot can still move in the desired direction by steering these wheels. Because the vertical axis is offset from the ground-contact path, the result of this steering motion is robot motion.
- With a circular chassis and an axis of rotation at the center of the robot, such a robot can spin without changing its ground footprint.
- The most popular such robot is the two-wheel differential-drive robot where the two wheels rotate around the center point of the robot.

Maneuverability



https://youtu.be/HZudyDwEIJA?si=QRkNB7oNxe_g5xNy

Controllability

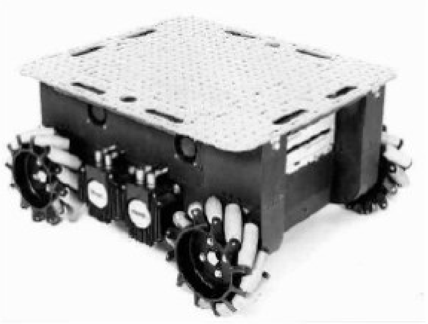
- Controllability is the degree to which a robot's behavior and movements can be precisely governed and directed by external inputs or control signals.
- It encompasses several key aspects:
- **Actuator Responsiveness:** The ability of the robot's actuators (motors, servos, etc.) to promptly and accurately execute commands from the control system.
- **Sensitivity to Control Inputs:** How effectively the robot responds to different levels of control signals or commands, allowing for fine adjustments in movement or behavior.
- **Feedback Control Systems:** Utilization of feedback loops to continuously monitor the robot's state (position, velocity, etc.) and adjust control signals accordingly, improving precision and robustness.
- **Predictability:** How reliably the robot's behavior can be anticipated and predicted based on control inputs, aiding in the development of precise and consistent control strategies.

Controllability

- Generally, there is an inverse correlation between controllability and maneuverability.
- If the vehicle is easy to control then it is less manoeuvrable; if it is high manoeuvrable, controlling is more difficult.
- The advantage of omnidirectional designs is the high manoeuvrability of the robot, but this advantage makes it more difficult to control the robot.

Controllability

- For example driving a robot which uses four powered Swedish wheels, like the Carnige Mellon Uranus robot, straight forward, all wheels must be driven with exactly the same speed, to move in a perfectly straight line.
- Even little errors in the speed of the wheels will cause mistakes in the desired travel path of the robot.
- At this point the benefit of Ackermann steering appears, because controlling such vehicles is much easier.
- Driving straight forward means just locking the steerable wheels and driving the motorized wheels. These are connected by an axis, so the speed of the drive wheels is always the same by actuating just one motor.



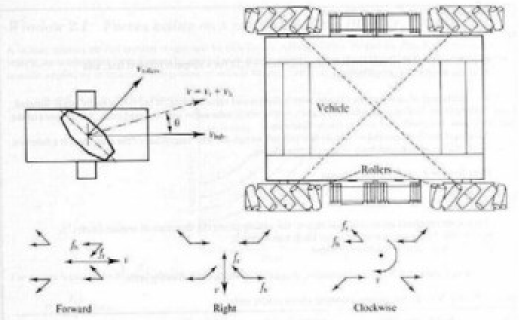


Figure 23: Carnegie Mellon Uranus robot [1]

Maneuverability

https://youtu.be/noqBUEgyQ8A?si=0-6YVbBQZzp4sllV

Robotic Systems
Synchro Drive

- Synchro Drive is a type of robotic drive system that utilizes synchronized movement of multiple wheels or tracks to achieve enhanced maneuverability and control.

- **Synchronized Movement:** Synchro Drive coordinates the movement of wheels or tracks on both sides of the robot, enabling smoother turns and more precise control than traditional systems.
- **Differential Steering:** Unlike traditional methods, Synchro Drive achieves turning by adjusting the relative speeds of wheels or tracks on each side while maintaining synchronized movement.
- **Enhanced Maneuverability:** Synchro Drive allows robots to navigate tight spaces and make sharp turns with agility and efficiency due to synchronized wheel or track movement.
- **Control Algorithms:** Synchro Drive systems utilize advanced algorithms to precisely control the speeds of individual wheels or tracks, ensuring smooth and coordinated motion.

Robotic Systems
Synchro Drive

- The synchro drive configuration (figure) is a popular arrangement of wheels in indoor mobile robot applications because, although there are three driven and steered wheels, only two motors are used in total.
- The one translation motor sets the speed of all three wheels together, and the one steering motor spins all the wheels together about each of their individual vertical steering axes.
- But the wheels are being steered with respect to the robot chassis, and therefore there is no direct way of reorienting the robot chassis. In fact, the chassis orientation does drift over time due to uneven tire slippage, causing rotational dead-reckoning error.

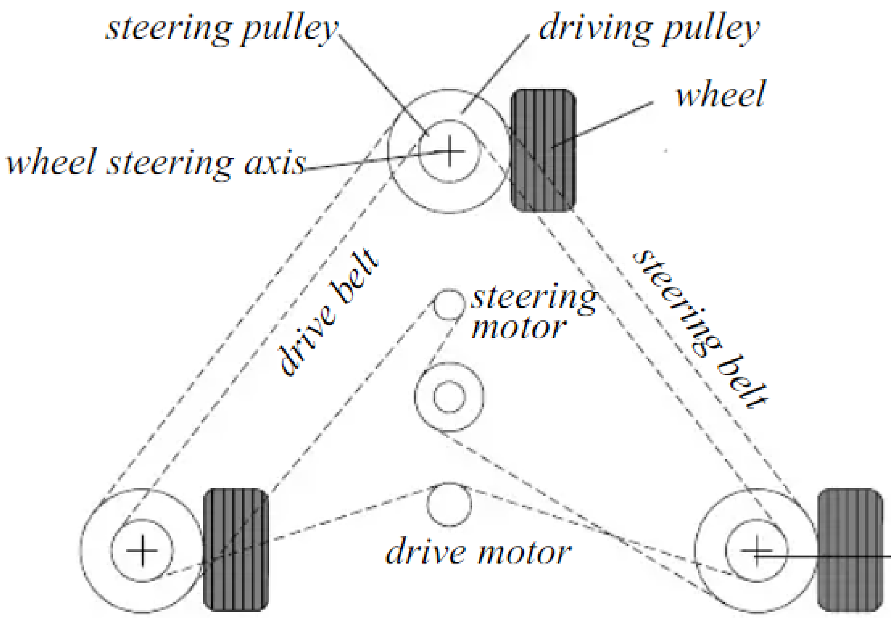


Figure: Synchro drive: The robot can move in any direction; But the orientation of the chassis is not controllable

Robotic Systems
Synchro Drive

https://youtu.be/nurCA5Q4_hw?si=voGohwB1yxPOYHFK

Robotic Systems
Synchro Drive

- Advantages of Synchro Drive for Omnidirectionality:
  - Synchro drive excels when omnidirectional movement is required.
  - By aligning each vertical steering axis with the contact path of the tires, the robot can reorient its wheels without changing its footprint.

- Comparison in Dead Reckoning:
  - Synchro drive outperforms true omni-directional setups but falls behind differential-drive and Ackerman steering systems.
  - This is primarily due to the use of a single belt to drive three wheels, leading to slop and backlash in the drivetrain.

Robotic Systems
Synchro Drive

- Issues with Synchro Drive in Dead Reckoning:
  - Slop and backlash in the drivetrain cause the closest wheel to start spinning before the furthest wheel when the drive motor engages, leading to small changes in chassis orientation.
  - These small angular shifts accumulate with changes in motor speed, resulting in significant orientation errors during dead reckoning.
- Lack of Direct Chassis Orientation Control:
  - The mobile robot lacks direct control over chassis orientation.
  - Depending on chassis orientation, wheel thrust can be highly asymmetric, with two wheels on one side and one wheel alone, or symmetric with one wheel on each side and one wheel straight ahead or behind.

Robotic Systems
Synchro Drive

- Functional Description of Synchro Drive Mechanism:
  - The asymmetric cases result in a variety of errors when tire-ground slippage can occur, again causing errors in dead reckoning of robot orientation.
  - This mechanism consists of three steerable wheels arranged in a triangle. All wheels are driven and connected by a single belt which is actuated by one motor, thereby this single motor sets the speed of all wheels together.
  - A second belt, which is actuated by an additional motor and is connected to the wheels too, is used to spin each wheel around its individual vertical axis.
  - In this way the robot can be driven and steered relatively simple by controlling just two motors.
  - One drawback of this method is that all wheels are steered with respect to the robots' chassis together, so there is no way to reorientate the chassis directly

Legged *Vs* Wheeled Locomotion

Feature	Legged Locomotion	Wheeled Locomotion
Mechanism	Uses legs for movement, mimicking animals and humans.	Uses wheels for rolling movement on surfaces.
Terrain Adaptability	Highly adaptable to uneven, rough, or unstructured terrain. Can climb stairs and obstacles.	Works best on flat, even surfaces. Struggles on highly irregular terrain.
Energy Efficiency	Generally consumes more energy due to complex actuation and balancing.	More energy-efficient on smooth surfaces due to continuous contact with the ground.
Control Complexity	Requires advanced control algorithms for balance, gait planning, and stability.	Easier to control as wheels provide stable motion with fewer degrees of freedom.
Speed	Typically slower than wheeled systems due to intermittent contact with the ground.	Faster on smooth terrain due to continuous contact and rolling motion.
Stability	May require active stabilization (especially for bipedal robots). Quadrupeds and hexapods are more stable.	Inherently more stable, especially with three or more wheels.
Applications	Used in rough terrains, exploration (e.g., Mars rovers with legs), rescue missions, humanoid robots, and biological research.	Common in vehicles, industrial robots, and indoor service robots where smooth mobility is needed.
Complexity & Cost	More complex in design, control, and actuation, making it expensive.	Simpler and cheaper to design and implement.

LOCOMOTION MECHANISMS FOR ROBOTS

Types of Locomotion: Aerial Mobile Robots

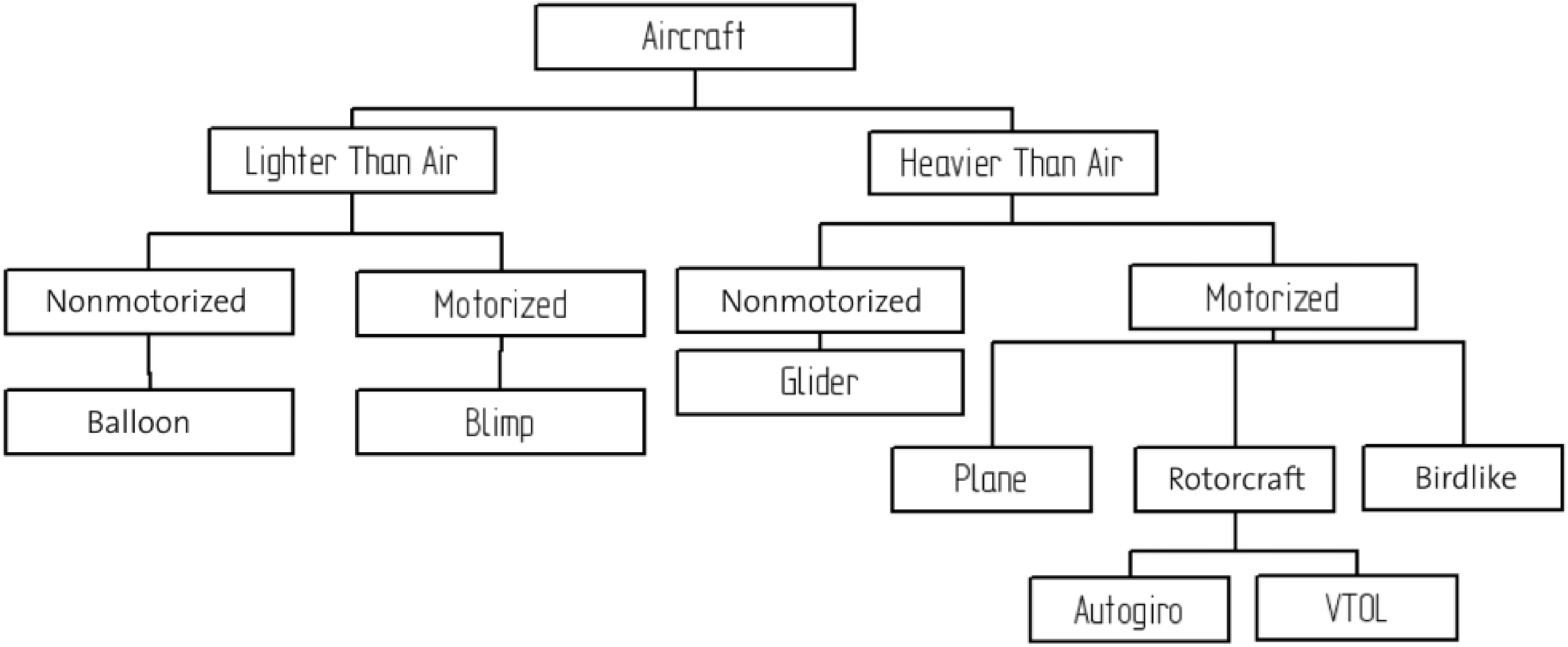
Locomotion Mechanisms for Robots

Aerial Mobile Robots

- Aerial Mobile Robots, often referred to as aerial drones or unmanned aerial vehicles (UAVs), are robotic systems designed to navigate and operate in the air.
- Unlike ground-based robots, aerial mobile robots can achieve three-dimensional movement, enabling them to cover large areas, access challenging terrains, and perform a variety of tasks from an elevated perspective.
- These robots have gained significant attention and applications across various industries due to their versatility and ability to gather data, perform surveillance, and execute tasks in aerial environments.

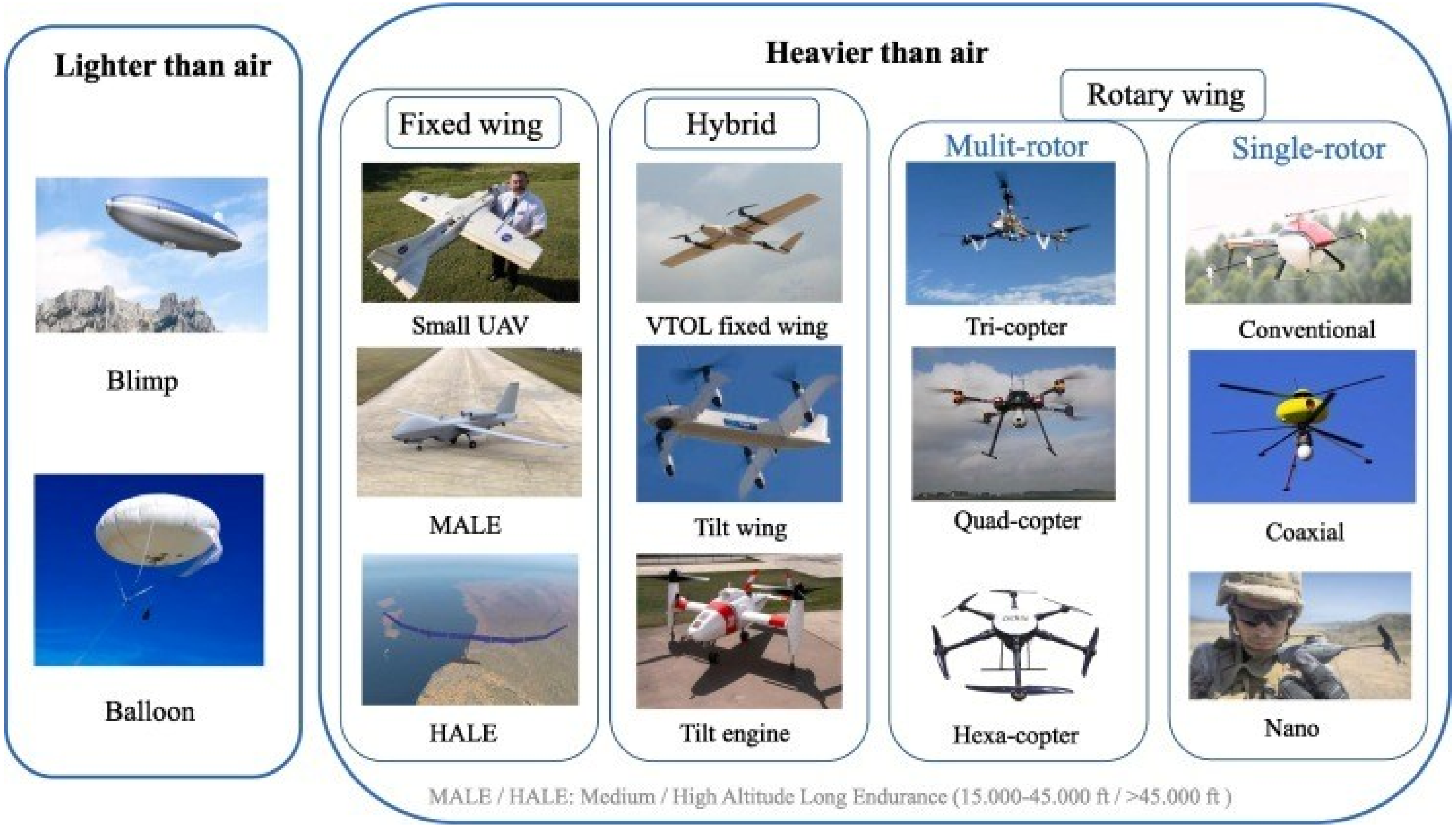
Locomotion Mechanisms for Robots

Aerial Mobile Robots



Locomotion Mechanisms for Robots

Aerial Mobile Robots



Locomotion Mechanisms for Robots

Aircraft configurations

Aerial vehicles can be divided into two categories:

- Lighter Than Air (LTA)
- Heavier Than Air (HTA)

- Vertical Take-Off and Landing (VTOL) systems such as helicopters or blimps have an unquestionable advantage compared with the other concepts.
- This superiority is owed to their unique ability for vertical, stationary, and low-speed flight.
- The key advantage of blimps is the autolift and simplicity of control, which can be essential for critical applications, such as aerial surveillance and space exploration.
- However, VTOL vehicles in different configurations represent today one of the most promising flying concepts seen in terms of miniaturization.

Locomotion Mechanisms for Robots

Key Features and Characteristics of Aerial Mobile Robots:

1. **Autonomous Flight:** Aerial mobile robots are equipped with navigation systems and sensors that enable autonomous or semi-autonomous flight. They can navigate through predefined paths or respond to real-time environmental changes.

1. **Multisensor Integration:** To gather information about the surroundings, aerial robots are equipped with various sensors such as cameras, lidar, GPS, IMU (Inertial Measurement Unit), and other specialized sensors. This sensor integration allows them to perceive and interact with the environment.

2. **Communication and Connectivity:** Aerial robots often have communication systems for data transmission, enabling remote control or real-time data sharing. This is crucial for maintaining connectivity and control over the robot, especially when operating at a distance.

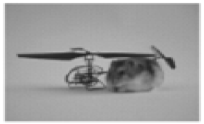
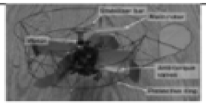
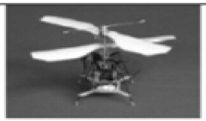
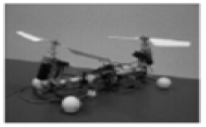
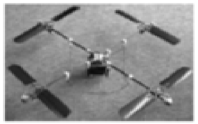
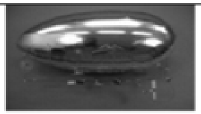
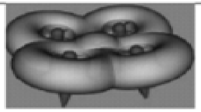
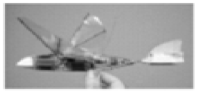
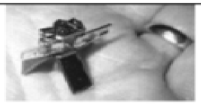
Locomotion Mechanisms for Robots

4. **Payload and Application Variety:** Aerial mobile robots are designed for a wide range of applications, including:
   - **Surveillance and Monitoring**: Monitoring large areas for security or environmental purposes.
   - **Mapping and Surveying**: Generating high-resolution maps and 3D models of terrain or structures.
   - **Search and Rescue**: Locating and assisting in the rescue of people in challenging or disaster-stricken environments.
   - **Delivery Services**: Transporting goods or medical supplies to remote or inaccessible locations.
   - **Agriculture**: Crop monitoring, precision agriculture, and pesticide spraying.

5. **Energy Efficiency:** Aerial mobile robots need to be energy-efficient to achieve longer flight durations. Battery technologies, propulsion systems, and lightweight materials are essential considerations to optimize energy consumption.

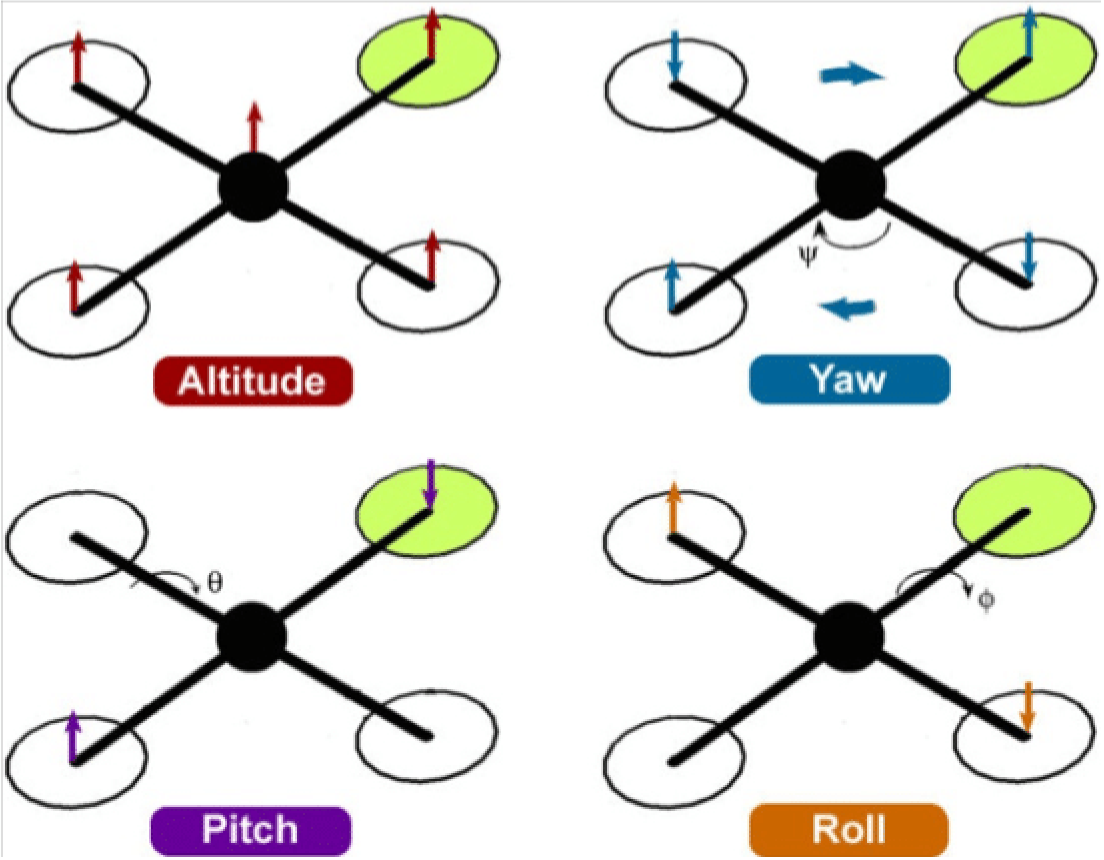
Locomotion Mechanisms for Robots

Common MAV configurations

Configuration e.g.	Advantages	Drawbacks	Picture
Fixed-wing (AeroVironment)	Simple mechanics, silent operation	No hovering	
Single rotor (A. V de Rostyne)	Good controllability, good maneuverability	Complex mechanics, large rotor, long tail boom	
Axial rotor (Univ. of Maryland)	Simple mechanics, compactness	Complex aerodynamics	
Coaxial rotors (EPSON)	Simple mechanics, compactness	Complex aerodynamics	
Tandem rotors (Heudiasyc)	Good controllability, simple aerodynamics	Complex mechanics, large size	
Quadrotor (EPFL-ETHZ)	Good maneuverability, simple mechanics, increased payload	High energy consumption, large size	
Blimp (EPFL)	Low power, long flight operation, auto-lift	Large size, weak maneuverability	
Hybrid quadrotor-blimp (MIT)	Good maneuverability, good survivability	Large size, weak maneuverability	
Birdlike (Caltech)	Good maneuverability, compactness	Complex mechanics, complex control	
Insectlike (UC Berkeley)	Good maneuverability, compactness	Complex mechanics, complex control	
Fishlike (US Naval Lab)	Multimode mobility, efficient aerodynamics	Complex control, weak maneuverability	

Locomotion Mechanisms for Robots

Common MAV configurations



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Mobile and Autonomous Robots

UE23CS343BB7

Locomotion Mechanisms for Robots

Degrees of Freedom (DoF)

Degrees of Freedom (DOF)

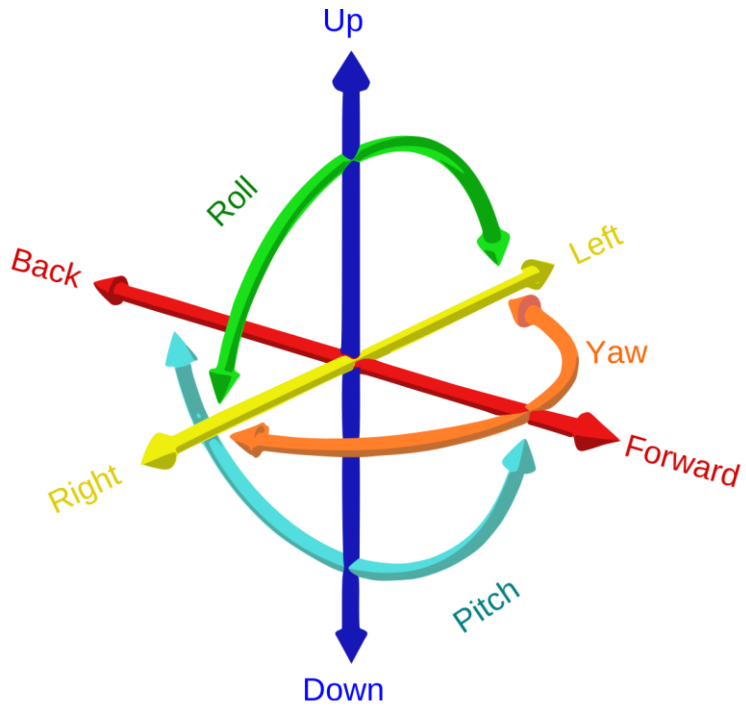
- The concept of degrees-of-freedom, often abbreviated as DOF, is important for defining the possible positions and orientations a robot can reach.

- An object in the physical world can have up to six degrees of freedom, namely forward/backward, sideways, and up/down as well as rotations (Roll, Pitch, Yaw) around those axes.

- The number of degrees of freedom is also known as its mobility M.

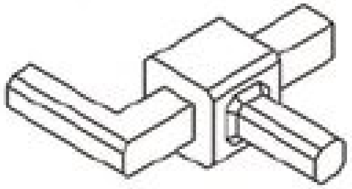
$$M = \sum_{i=1}^{n} f_i$$

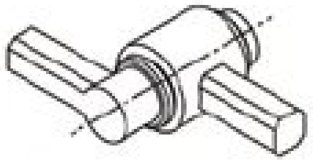
where there are n joints and f_i is the mobility of the i^{th} joint, with f_i=1 for revolute, prismatic, and helical joints, and f_i=3 for spherical joints.



Degrees of Freedom (DOF)

- In the case of a serial or branched fixed base mechanism, the degrees of freedom are the union of all individual joint degrees of freedom, and the mobility is the sum of the mobilities of all individual joints.
- The degrees of freedom for a single joint are expressed as the offset of the two attached links from their layout in a given reference frame.
- For revolute joints, the one dof is a joint angle defining the offset from a joint's zero position along its axis of rotation.
- For prismatic joints, the one dof is a translation along the axis relative to its zero position.
- Spherical joints allow for rotations around three perpendicular axes. The degrees of freedom for a spherical joint are represented by Euler angles.


Prismatic (P)


Revolute (R)


Spherical (S)

Degrees of Freedom (DOF)

- Only robots that use exclusively wheels with three degrees-of-freedom (3-DOF wheels) will be able to freely move on a plane.
- This is because the pose of a robot on a plane is fully given by its position (two values) and its orientation (one value).
- Robots that don’t have wheels with three degrees of freedom will have kinematic constraints that prevent them from reaching every possible point at every possible orientation.

Degrees of Freedom (DOF)

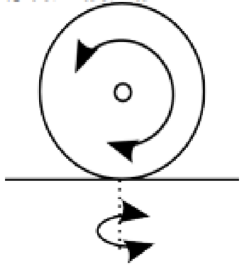
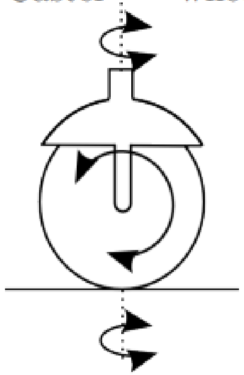
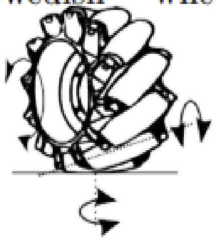
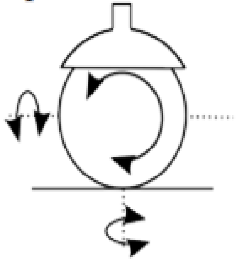
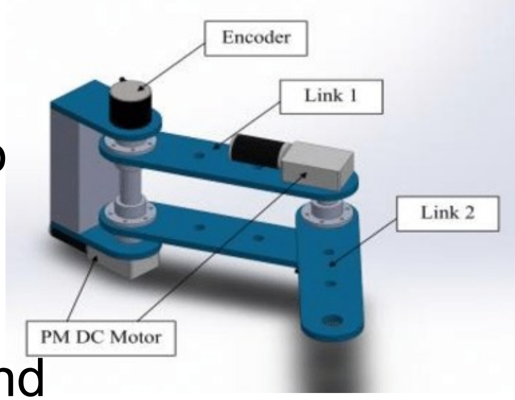
Wheel type	Example	Degrees-of-Freedom
Standard 	Front-wheel of a wheelbarrow	Two • Rotation around the wheel axle • Rotation around its contact point with the ground
Caster wheel 	Office chair	Three • Rotation around the wheel axle • Rotation around its contact point with the ground • Rotation around the caster axis
Swedish wheel 	Standard wheel with non-actuated rollers around its circumference	Three • Rotation around the wheel axle • Rotation around its contact point with the ground • Rotation around the roller axles
Spherical wheel 	Ball Bearing	Three • Rotation in any direction • Rotation around its contact point

Table 2.1.: *Different types of wheels and their degrees of freedom. Adopted from Siegwart et al. (2011),*

Degrees of Freedom (DOF)

2 DOF Robots

A 2-DOF robot has two independent degrees of freedom, meaning it can move in two different directions or rotate around two distinct axes. These movements can include translation along a plane and rotation around an axis.

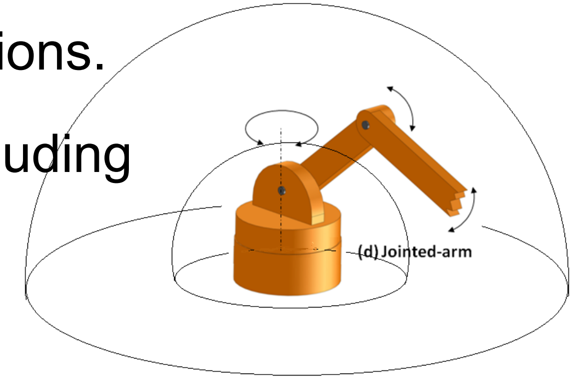


Eg: **Planar Robot Arm**: A simple robot arm that can move in two dimensions (x, y) and rotate around the vertical axis (z).

3-DOF Robot

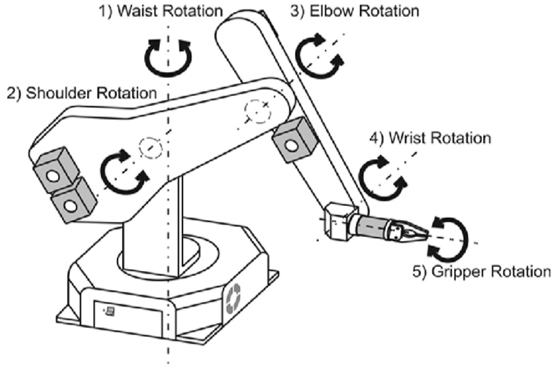
A 3-DOF robot has three independent degrees of freedom, allowing it to move or rotate in three different directions or axes. This includes translational and rotational motions.

Eg: **Spherical Robot:** A robot that can move in any direction within a sphere, including translations and rotations.



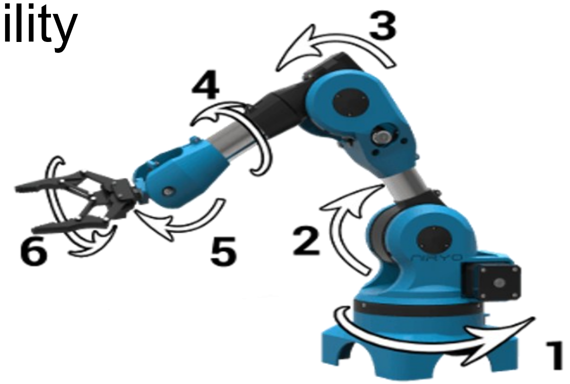
Degrees of Freedom (DOF)

<u>5-DOF Robot:</u>
A 5-DOF robot has five independent degrees of freedom, allowing it to move or rotate in five different directions or axes. These movements can include translations and rotations in three-dimensional space.



Eg: Serial Manipulator: A robotic arm with five joints, each contributing to the robot's overall movement. This might include translation along three axes (x, y, z) and rotation around two additional axes.

<u>6-DOF Robot:</u>
A 6-DOF robot has six independent degrees of freedom, providing more flexibility in movement and rotation. This allows the robot to position itself and orient its end-effector in six different ways in three-dimensional space.



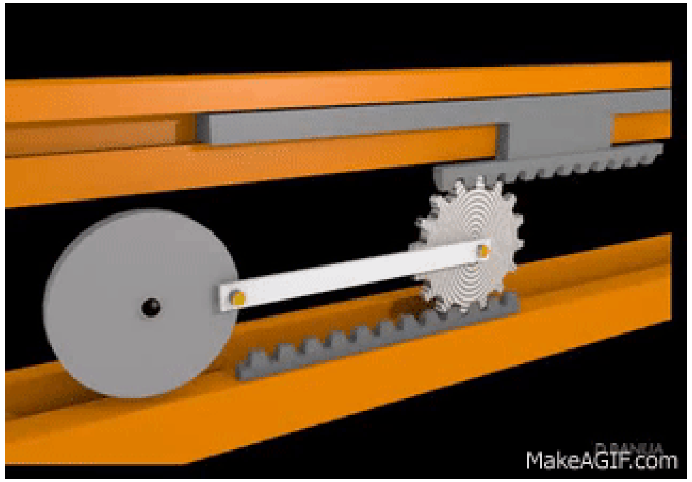
Eg:Industrial Robot Arm: Many industrial robotic arms are 6-DOF robots, enabling them to reach different positions and orientations accurately.

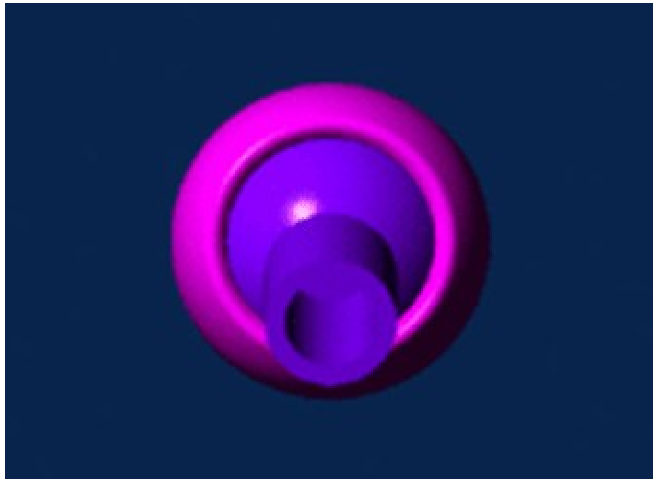
Degrees of Freedom (DOF)

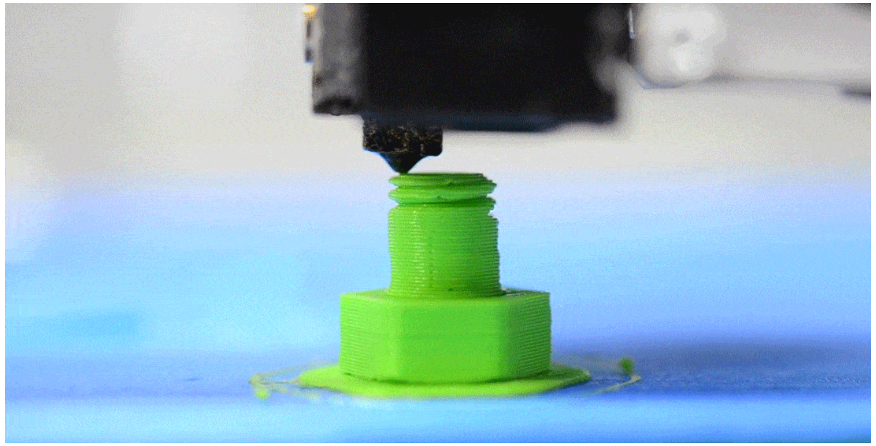
Applications

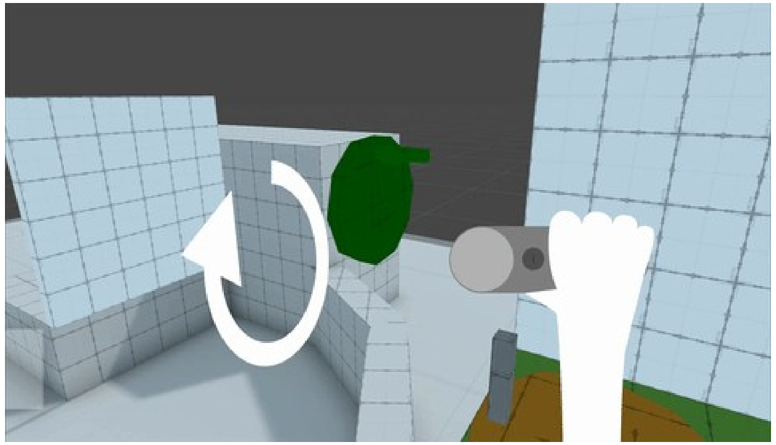
- 2-DOF: Often used in simpler robotic systems or scenarios where two-dimensional movement is sufficient.
- 3-DOF: Suitable for a wider range of applications, including tasks that require more complex and flexible movements.
- 5-DOF: Suitable for applications where moderate flexibility and precision are required, such as certain assembly tasks or material handling.
- 6-DOF: Widely used in industrial automation, manufacturing, and other applications requiring precise control over the position and orientation of the end-effector.

Identify the Degree of Freedom

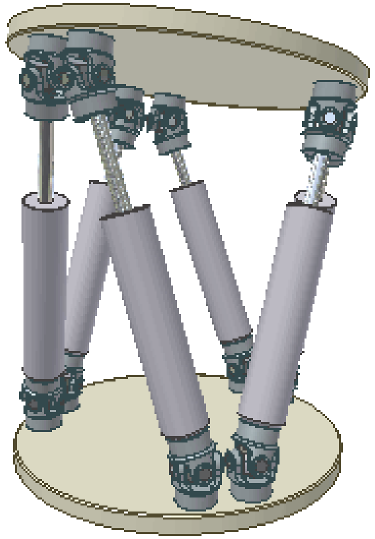








Identify the Degree of Freedom



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Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu

Mobile and Autonomous Robots

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Locomotion Mechanisms for Robots

Robot Kinematic Models and Constraints:

Forward and Inverse Kinematics

Kinematics

- The word **kinematics** is derived from a Greek term meaning **motion**.

- Kinematics is defined as the study of motion without considering its causes.

  - This can be illustrated by considering a bird in flight: kinematics describes how it turns, rises, and descends, without regard to the factors responsible for the bird's motion, such as the lift provided by its wings, the effects of wind, or the force of gravity. Kinematics only describes the bird's motion.

  - Although we live in three dimensions, many situations involving motion can be addressed in terms of one or two dimensions. For instance, a train traveling on a long, straight railway is essentially confined to one- dimensional motion since it can only move backwards or forwards.

Kinematics

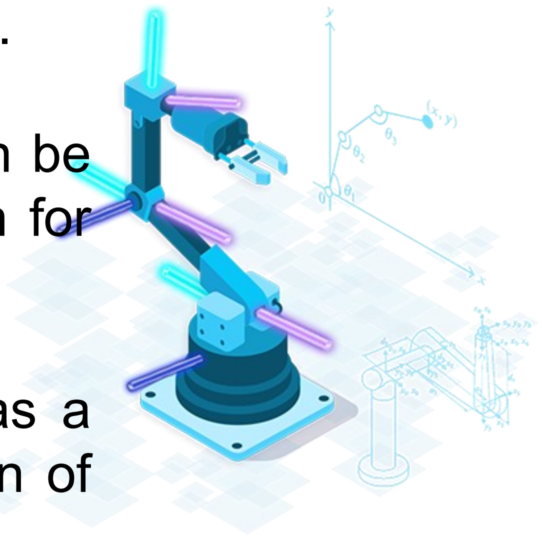
- Kinematics - the analytical study of the geometry of motion of a robot arm:
    - with respect to a fixed reference co-ordinate system
    - without regard to the forces or moments that cause the motion.
- Kinematics, a foundational concept in robotics, explores how a robot's joint coordinates relate to its spatial arrangement.
- In order to **control and programme** a robot we must have knowledge of both it's **spatial arrangement** and a means of **reference to the environment.**

Kinematics

- By employing kinematics, precise calculations can be achieved for various tasks, such as positioning a gripper in space, devising mechanisms to move tools between specific points, and assessing potential collisions during robot motion.

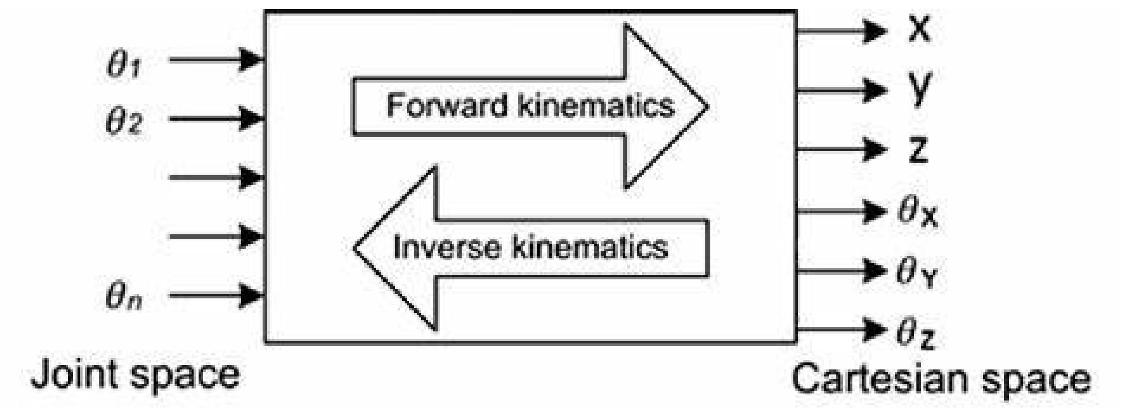
- Assuming each joint operates with a single degree of freedom, its action can be succinctly described by a single numerical value: either the angle of rotation for revolute joints or the displacement for prismatic joints.

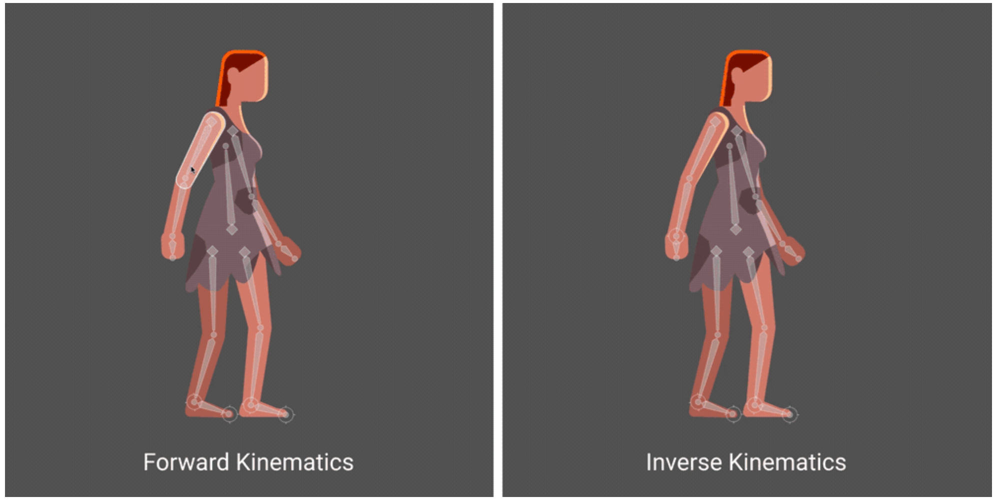
- Kinematics, with its focus on joint coordination and spatial layout, serves as a cornerstone in robotic engineering, enabling accurate analysis and execution of diverse tasks.



Kinematics

- **Forward Kinematics**: Forward kinematic analysis aims to ascertain the combined impact of all joint variables, leading to:
    - Deriving the position and orientation of the end effector based on these joint values.
    - Establishing the frames of a robot's links by considering a configuration and the robot's kinematic structure as inputs.
- **Inverse Kinematics**: Inverse kinematic analysis focuses on:
    - Determining the joint variable values required to achieve a specific position and orientation of the end effector frame.





Forword vs Inverse Kinematics

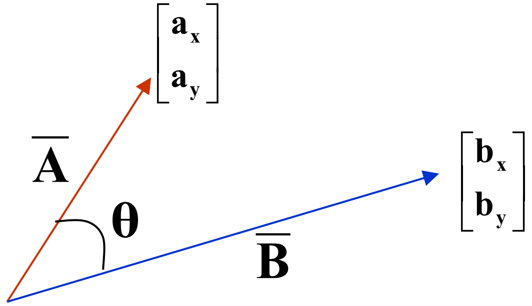
	Forward Kinematics (angles to position)	**Inverse Kinematics (position to angles)**
What you are given:	The length of each link The angle of each joint	The length of each link The position of some point on the robot
What you can find:	The position of any point (i.e. it's (x, y, z) coordinates	The angles of each joint needed to obtain that position

Quick Math Review

Dot Product:

Geometric Representation:

$$\overline{\mathbf{A}} \bullet \overline{\mathbf{B}} = \|\mathbf{A}\|\|\mathbf{B}\|\cos\theta$$



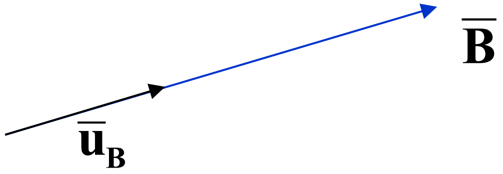
Matrix Representation:

$$\overline{\mathbf{A}} \bullet \overline{\mathbf{B}} = \begin{bmatrix} \mathbf{a}_x \\ \mathbf{a}_y \end{bmatrix} \bullet \begin{bmatrix} \mathbf{b}_x \\ \mathbf{b}_y \end{bmatrix} = \mathbf{a}_x\mathbf{b}_x + \mathbf{a}_y\mathbf{b}_y$$

Unit Vector

Vector in the direction of a chosen vector but whose magnitude is 1.

$$\overline{\mathbf{u}}_B = \frac{\overline{\mathbf{B}}}{\|\mathbf{B}\|}$$



Quick Matrix Review

Matrix Multiplication:

An (m x n) matrix A and an (n x p) matrix B, can be multiplied since the number of columns of A is equal to the number of rows of B.

<u>Non-Commutative Multiplication</u>

AB is NOT equal to BA

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} * \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} (ae+bg) & (af+bh) \\ (ce+dg) & (cf+dh) \end{bmatrix}$$

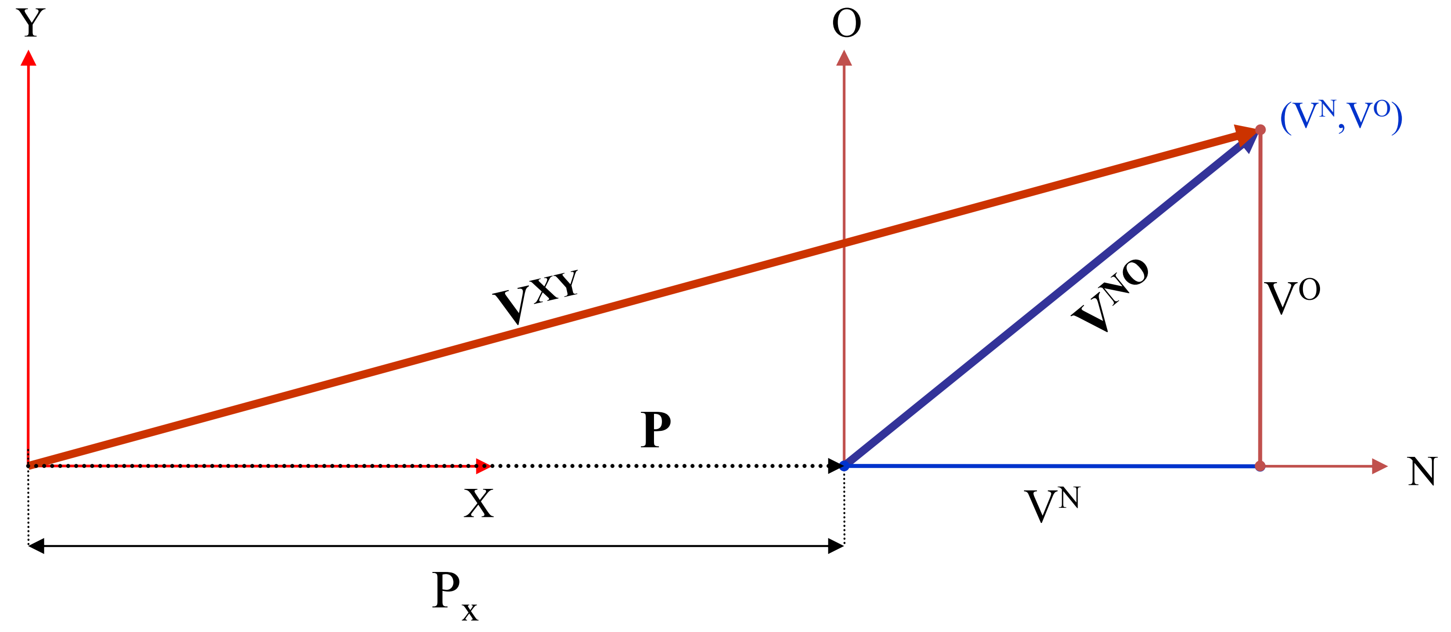
Matrix Addition:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} (a+e) & (b+f) \\ (c+g) & (d+h) \end{bmatrix}$$

Basic Transformations

Moving Between Coordinate Frames

Translation Along the X-Axis

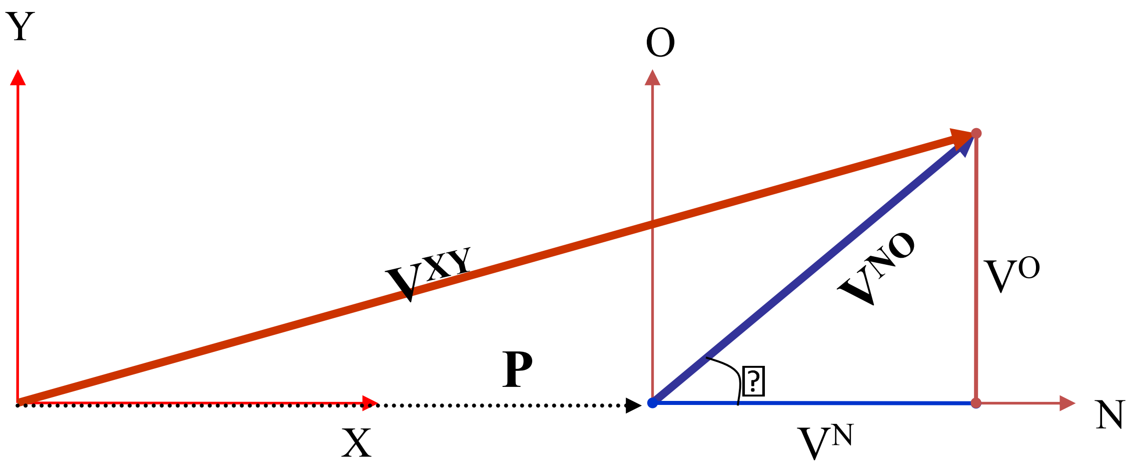


P_x = distance between the XY and NO coordinate planes

Notation: $\overline{V}^{XY} = \begin{bmatrix} V^X \\ V^Y \end{bmatrix}$ $\overline{V}^{NO} = \begin{bmatrix} V^N \\ V^O \end{bmatrix}$ $\overline{P} = \begin{bmatrix} P_x \\ 0 \end{bmatrix}$

Basic Transformations

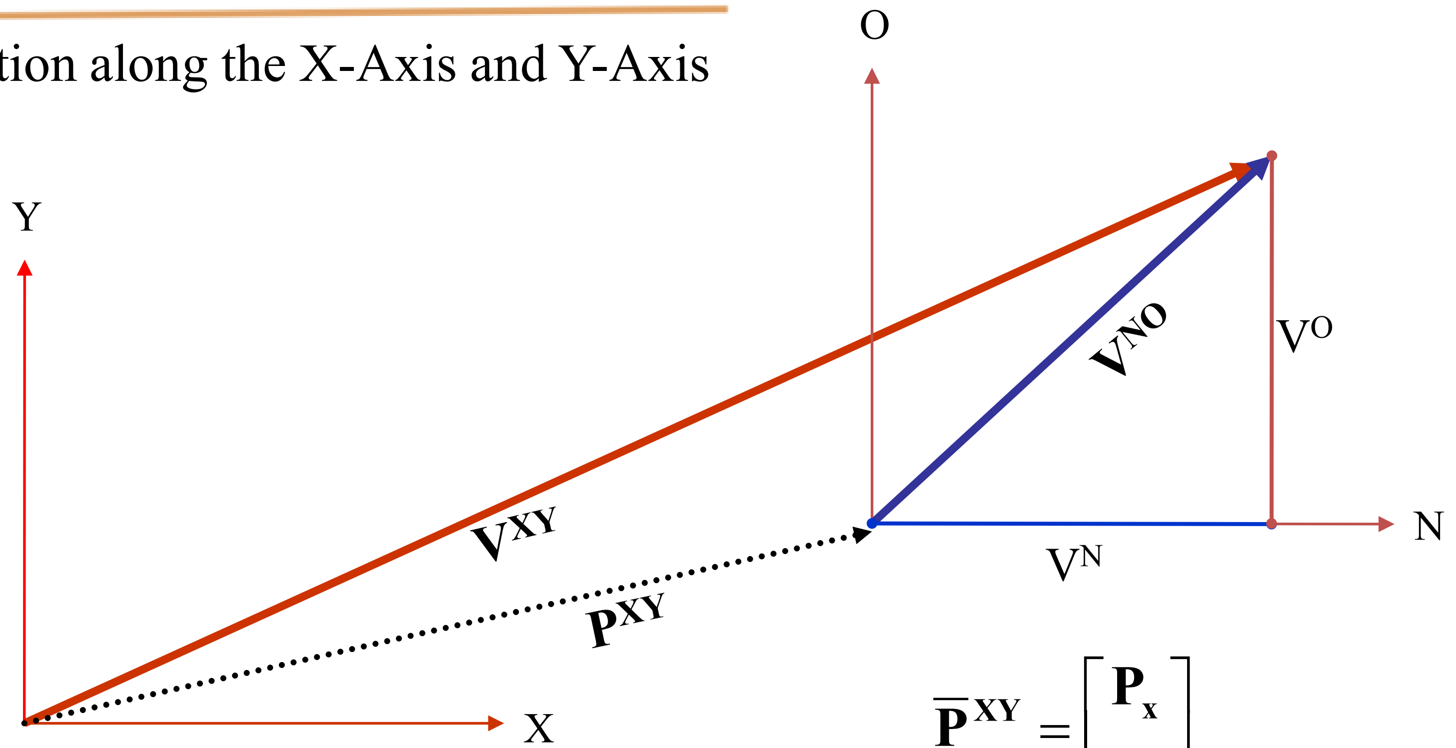
Writing $\overline{\mathbf{V}}^{XY}$ in terms of $\overline{\mathbf{V}}^{NO}$



$$\overline{\mathbf{V}}^{XY} = \begin{bmatrix} \mathbf{P}_X + \mathbf{V}^N \\ \mathbf{V}^O \end{bmatrix} = \overline{\mathbf{P}} + \overline{\mathbf{V}}^{NO}$$

Basic Transformations

Translation along the X-Axis and Y-Axis



$$\overline{P}^{XY} = \begin{bmatrix} P_x \\ P_Y \end{bmatrix}$$

$$\overline{V}^{XY} = \overline{P} + \overline{V}^{NO} = \begin{bmatrix} P_X + V^N \\ P_Y + V^O \end{bmatrix}$$

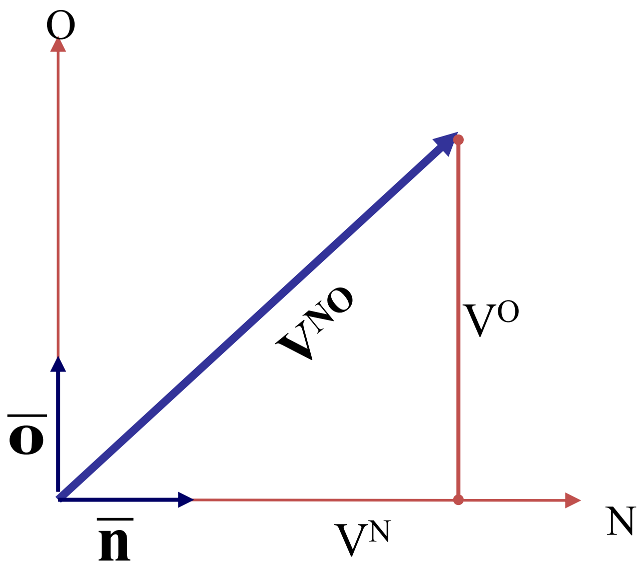
Using Basis Vectors

Basis vectors are unit vectors that point along a coordinate axis

$\overline{\mathbf{n}}$ Unit vector along the N-Axis

$\overline{\mathbf{o}}$ Unit vector along the N-Axis

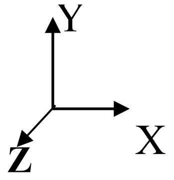
$\left\| \mathbf{V}^{NO} \right\|$ Magnitude of the V^{NO} vector

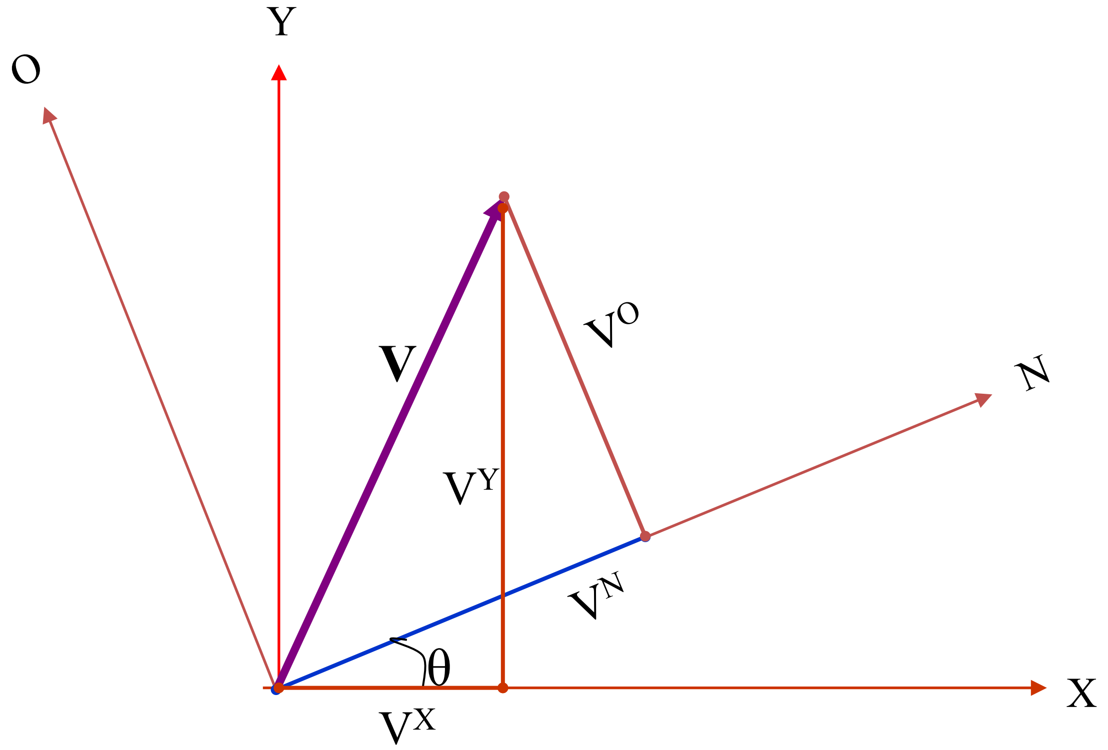


$$\overline{\mathbf{V}}^{NO} = \begin{bmatrix} \mathbf{V}^{N} \\ \mathbf{V}^{O} \end{bmatrix} = \begin{bmatrix} \left\|\mathbf{V}^{NO}\right\| \cos\theta \\ \left\|\mathbf{V}^{NO}\right\| \sin\theta \end{bmatrix} = \begin{bmatrix} \left\|\mathbf{V}^{NO}\right\| \cos\theta \\ \left\|\mathbf{V}^{NO}\right\| \cos(90-\theta) \end{bmatrix} = \begin{bmatrix} \overline{\mathbf{V}}^{NO} \bullet \overline{\mathbf{n}} \\ \overline{\mathbf{V}}^{NO} \bullet \overline{\mathbf{o}} \end{bmatrix}$$

Using Basis Vectors

Rotation (around the Z-Axis)





θ = Angle of rotation between the XY and NO coordinate axis

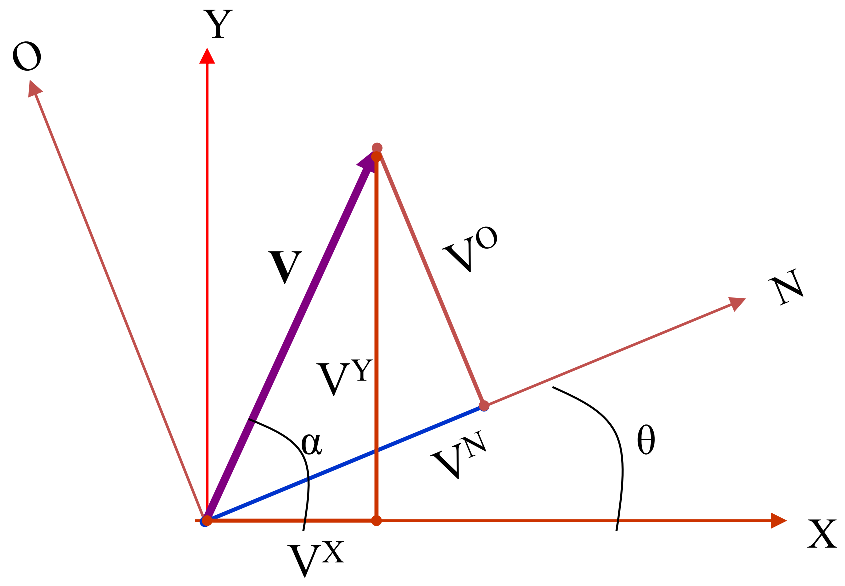
$$\overline{\mathbf{V}}^{XY} = \begin{bmatrix} \mathbf{V}^{X} \\ \mathbf{V}^{Y} \end{bmatrix} \qquad \overline{\mathbf{V}}^{NO} = \begin{bmatrix} \mathbf{V}^{N} \\ \mathbf{V}^{O} \end{bmatrix}$$

Using Basis Vectors

$\underline{\mathbf{x}}$ Unit vector along X-Axis

$\triangle$ Can be considered with respect to the XY coordinates or NO coordinates

$$\left\| \overline{\mathbf{V}}^{XY} \right\| = \left\| \overline{\mathbf{V}}^{NO} \right\|$$



$$\mathbf{V}^X = \left\| \overline{\mathbf{V}}^{XY} \right\| \cos\alpha = \left\| \overline{\mathbf{V}}^{NO} \right\| \cos\alpha = \overline{\mathbf{V}}^{NO} \bullet \overline{\mathbf{x}}$$

$$\mathbf{V}^X = (\mathbf{V}^N * \overline{\mathbf{n}} + \mathbf{V}^O * \overline{\mathbf{o}}) \bullet \overline{\mathbf{x}}$$

(Substituting for V^{NO} using the N and O components of the vector)

$$\begin{aligned} \mathbf{V}^X &= \mathbf{V}^N(\overline{\mathbf{x}} \bullet \overline{\mathbf{n}}) + \mathbf{V}^O(\overline{\mathbf{x}} \bullet \overline{\mathbf{o}}) \\ &= \mathbf{V}^N(\cos\theta) + \mathbf{V}^O(\cos(\theta + 90)) \\ &= \mathbf{V}^N(\cos\theta) - \mathbf{V}^O(\sin\theta) \end{aligned}$$

Similarly….

$$\mathbf{V^Y} = \left\|\overline{\mathbf{V}}^{\mathbf{NO}}\right\|\sin\alpha = \left\|\overline{\mathbf{V}}^{\mathbf{NO}}\right\|\cos(90-\alpha) = \overline{\mathbf{V}}^{\mathbf{NO}} \bullet \overline{\mathbf{y}}$$

$$\mathbf{V^Y} = (\mathbf{V^N} * \overline{\mathbf{n}} + \mathbf{V^O} * \overline{\mathbf{o}}) \bullet \overline{\mathbf{y}}$$

$$\mathbf{V^Y} = \mathbf{V^N}(\overline{\mathbf{y}} \bullet \overline{\mathbf{n}}) + \mathbf{V^O}(\overline{\mathbf{y}} \bullet \overline{\mathbf{o}})$$

$$= \mathbf{V^N}(\cos(90-\theta)) + \mathbf{V^O}(\cos\theta)$$

$$= \mathbf{V^N}(\sin\theta) + \mathbf{V^O}(\cos\theta)$$

So….

$$\mathbf{V^X} = \mathbf{V^N}(\cos\theta) - \mathbf{V^O}(\sin\theta)$$

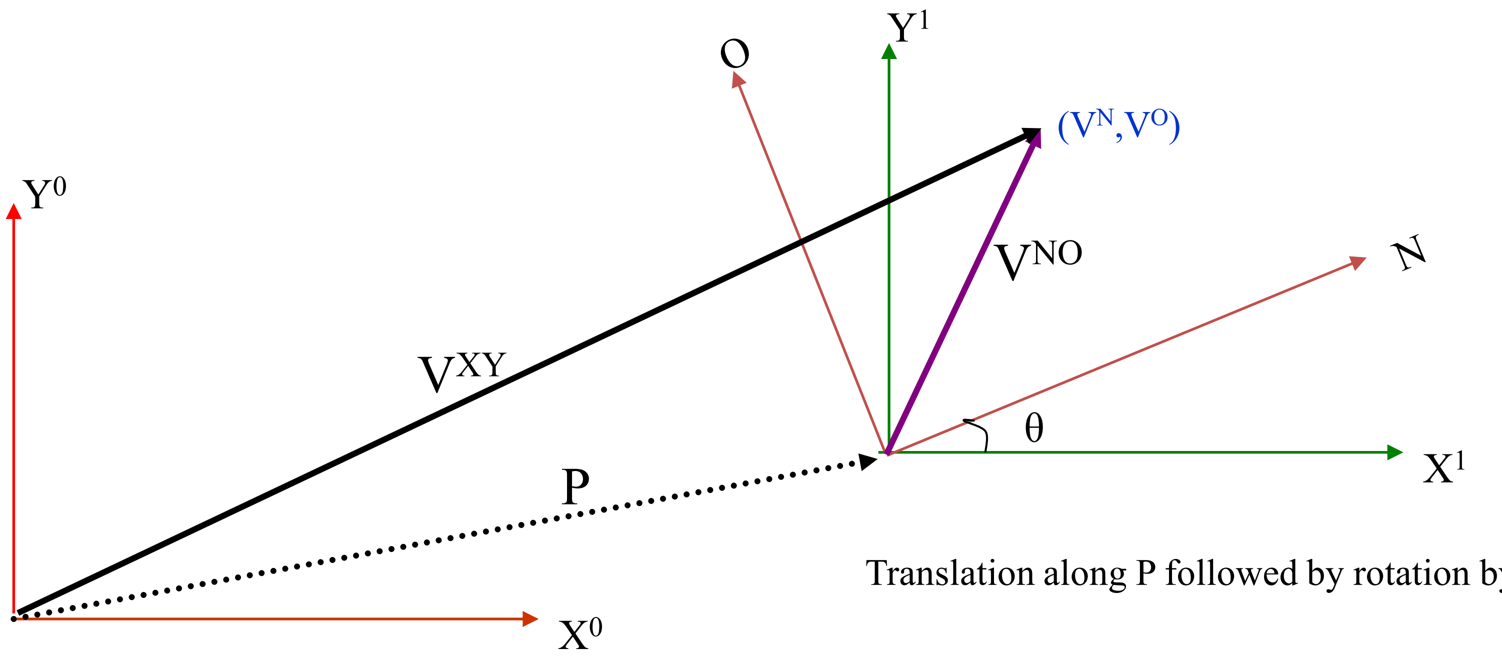
$$\mathbf{V^Y} = \mathbf{V^N}(\sin\theta) + \mathbf{V^O}(\cos\theta)$$

$$\overline{\mathbf{V}}^{\mathbf{XY}} = \begin{bmatrix} \mathbf{V^X} \\ \mathbf{V^Y} \end{bmatrix}$$

Written in Matrix Form

$$\overline{\mathbf{V}}^{\mathbf{XY}} = \begin{bmatrix} \mathbf{V^X} \\ \mathbf{V^Y} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \mathbf{V^N} \\ \mathbf{V^O} \end{bmatrix}$$

Rotation Matrix about the z-axis



Translation along P followed by rotation by θ

$$\mathbf{V}^{\mathbf{XY}} = \begin{bmatrix} \mathbf{V}^{\mathbf{X}} \\ \mathbf{V}^{\mathbf{Y}} \end{bmatrix} = \begin{bmatrix} \mathbf{P_x} \\ \mathbf{P_y} \end{bmatrix} + \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \mathbf{V}^{\mathbf{N}} \\ \mathbf{V}^{\mathbf{O}} \end{bmatrix}$$

(Note : P_x, P_y are relative to the original coordinate frame. Translation followed by rotation is different than rotation followed by translation.)

In other words, knowing the coordinates of a point (V^N, V^O) in some coordinate frame (NO) you can find the position of that point relative to your original coordinate frame (X^0Y^0).

HOMOGENEOUS REPRESENTATION

Putting it all into a Matrix

$$V^{XY} = \begin{bmatrix} V^X \\ V^Y \end{bmatrix} = \begin{bmatrix} P_x \\ P_y \end{bmatrix} + \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} V^N \\ V^O \end{bmatrix}$$

What we found by doing a translation and a rotation

$$= \begin{bmatrix} V^X \\ V^Y \\ 1 \end{bmatrix} = \begin{bmatrix} P_x \\ P_y \\ 1 \end{bmatrix} + \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} V^N \\ V^O \\ 1 \end{bmatrix}$$

Padding with 0's and 1's

$$= \begin{bmatrix} V^X \\ V^Y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & P_x \\ \sin\theta & \cos\theta & P_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} V^N \\ V^O \\ 1 \end{bmatrix}$$

Simplifying into a matrix form

$$H = \begin{bmatrix} \cos\theta & -\sin\theta & P_x \\ \sin\theta & \cos\theta & P_y \\ 0 & 0 & 1 \end{bmatrix}$$

Homogenous Matrix for a Translation in XY plane, followed by a Rotation around the z-axis

Rotation Matrices in 3D –
Lets return from homogenous representation

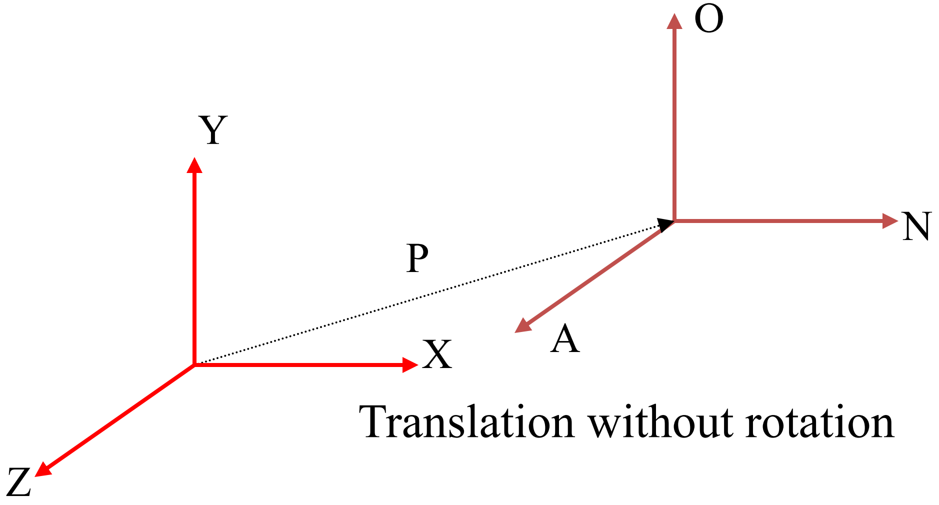
$$\mathbf{R_z} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \longleftarrow \text{Rotation around the Z-Axis}$$

$$\mathbf{R_y} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \longleftarrow \text{Rotation around the Y-Axis}$$

$$\mathbf{R_z} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \longleftarrow \text{Rotation around the X-Axis}$$

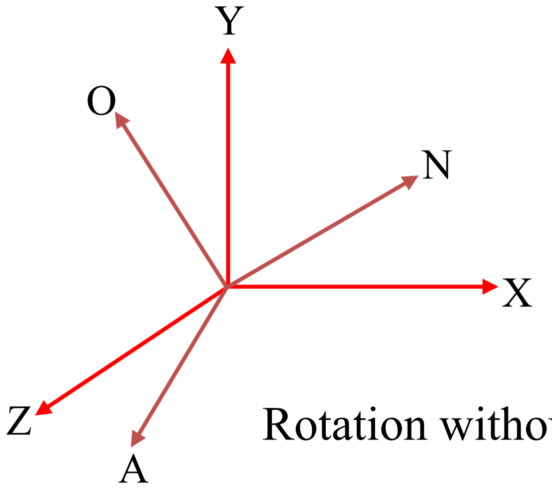
Homogeneous Matrices in 3D

H is a 4x4 matrix that can describe a translation, rotation, or both in one matrix



Translation without rotation

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & P_x \\ 0 & 1 & 0 & P_y \\ 0 & 0 & 1 & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Rotation without translation

$$\mathbf{H} = \begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation part:
Could be rotation around z-axis, x-axis, y-axis or a combination of the three.

Homogeneous Continued….

$$\mathbf{V}^{XY} = \mathbf{H}\begin{bmatrix} \mathbf{V}^{N} \\ \mathbf{V}^{O} \\ \mathbf{V}^{A} \\ 1 \end{bmatrix}$$

The (n,o,a) position of a point relative to the current coordinate frame you are in.

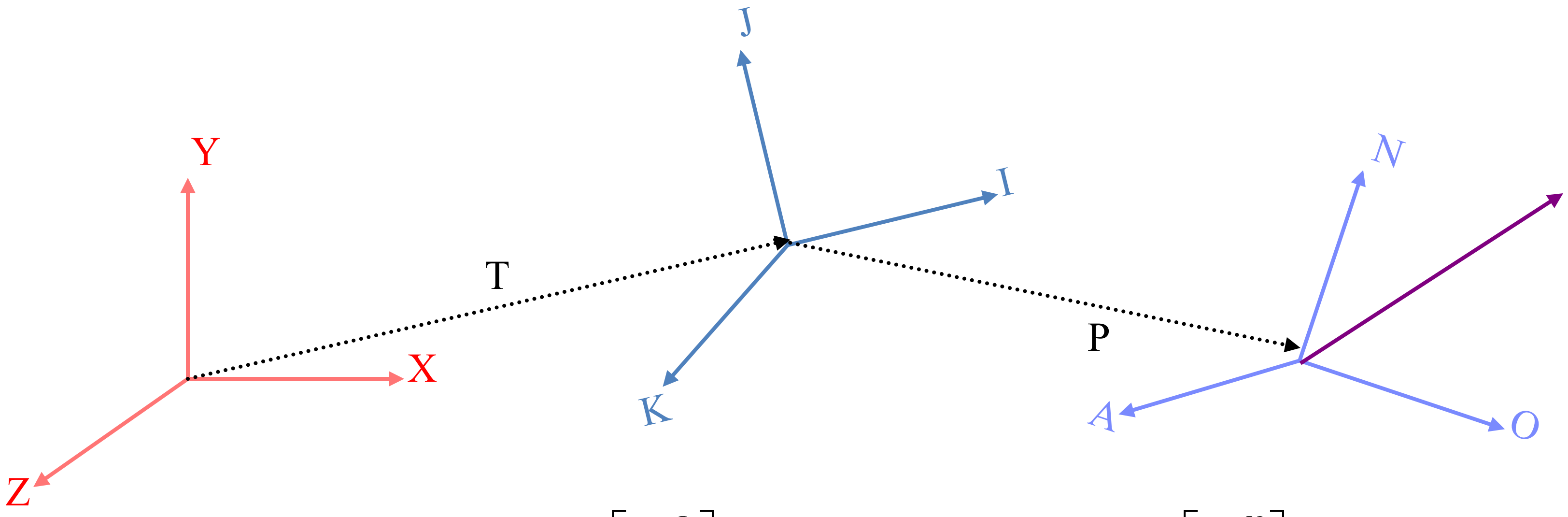
$$\mathbf{V}^{XY} = \begin{bmatrix} \mathbf{n_x} & \mathbf{o_x} & \mathbf{a_x} & \mathbf{P_x} \\ \mathbf{n_y} & \mathbf{o_y} & \mathbf{a_y} & \mathbf{P_y} \\ \mathbf{n_z} & \mathbf{o_z} & \mathbf{a_z} & \mathbf{P_z} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}\begin{bmatrix} \mathbf{V}^{N} \\ \mathbf{V}^{O} \\ \mathbf{V}^{A} \\ 1 \end{bmatrix}$$

$$\mathbf{V}^{X} = \mathbf{n_x}\mathbf{V}^{N} + \mathbf{o_x}\mathbf{V}^{O} + \mathbf{a_x}\mathbf{V}^{A} + \mathbf{P_x}$$

The rotation and translation part can be combined into a single homogeneous matrix IF and ONLY IF both are relative to the same coordinate frame.

Finding the Homogeneous Matrix

EX.



$\begin{bmatrix} W^N \\ W^O \\ W^A \end{bmatrix}$

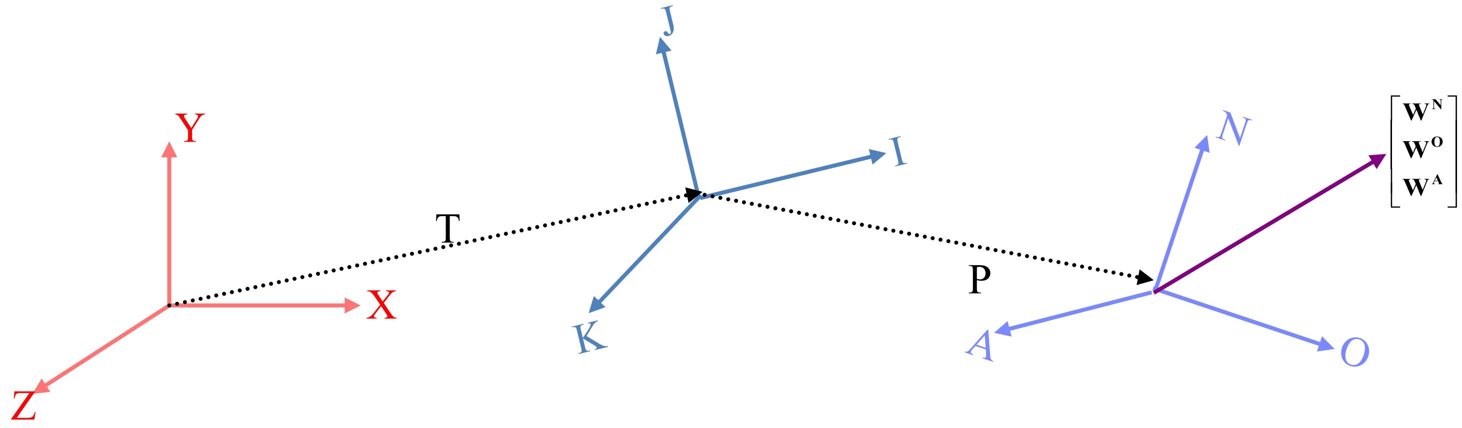
$\begin{bmatrix} W^X \\ W^Y \\ W^Z \end{bmatrix}$ Point relative to the X-Y-Z frame

$\begin{bmatrix} W^I \\ W^J \\ W^K \end{bmatrix}$ Point relative to the I-J-K frame

$\begin{bmatrix} W^N \\ W^O \\ W^A \end{bmatrix}$ Point relative to the N-O-A frame

$$\begin{bmatrix} W^I \\ W^J \\ W^K \end{bmatrix} = \begin{bmatrix} P_i \\ P_j \\ P_k \end{bmatrix} + \begin{bmatrix} n_i & o_i & a_i \\ n_j & o_j & a_j \\ n_k & o_k & a_k \end{bmatrix} \begin{bmatrix} W^N \\ W^O \\ W^A \end{bmatrix}$$

$$\begin{bmatrix} W^I \\ W^J \\ W^K \\ 1 \end{bmatrix} = \begin{bmatrix} n_i & o_i & a_i & P_i \\ n_j & o_j & a_j & P_j \\ n_k & o_k & a_k & P_k \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} W^N \\ W^O \\ W^A \\ 1 \end{bmatrix}$$



$$\begin{bmatrix} W^X \\ W^Y \\ W^Z \end{bmatrix} = \begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} + \begin{bmatrix} i_x & j_x & k_x \\ i_y & j_y & k_y \\ i_z & j_z & k_z \end{bmatrix} \begin{bmatrix} W^I \\ W^J \\ W^k \end{bmatrix} \longrightarrow \begin{bmatrix} W^X \\ W^Y \\ W^Z \\ 1 \end{bmatrix} = \begin{bmatrix} i_x & j_x & k_x & T_x \\ i_y & j_y & k_y & T_y \\ i_z & j_z & k_z & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} W^I \\ W^J \\ W^K \\ 1 \end{bmatrix}$$

Substituting for $\begin{bmatrix} W^I \\ W^J \\ W^K \end{bmatrix}$

$$\begin{bmatrix} W^X \\ W^Y \\ W^Z \\ 1 \end{bmatrix} = \begin{bmatrix} i_x & j_x & k_x & T_x \\ i_y & j_y & k_y & T_y \\ i_z & j_z & k_z & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_i & o_i & a_i & P_i \\ n_j & o_j & a_j & P_j \\ n_k & o_k & a_k & P_k \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} W^N \\ W^O \\ W^A \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} W^X \\ W^Y \\ W^Z \\ 1 \end{bmatrix} = H \begin{bmatrix} W^N \\ W^O \\ W^A \\ 1 \end{bmatrix}$$

$$H = \begin{bmatrix} i_x & j_x & k_x & T_x \\ i_y & j_y & k_y & T_y \\ i_z & j_z & k_z & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_i & o_i & a_i & P_i \\ n_j & o_j & a_j & P_j \\ n_k & o_k & a_k & P_k \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

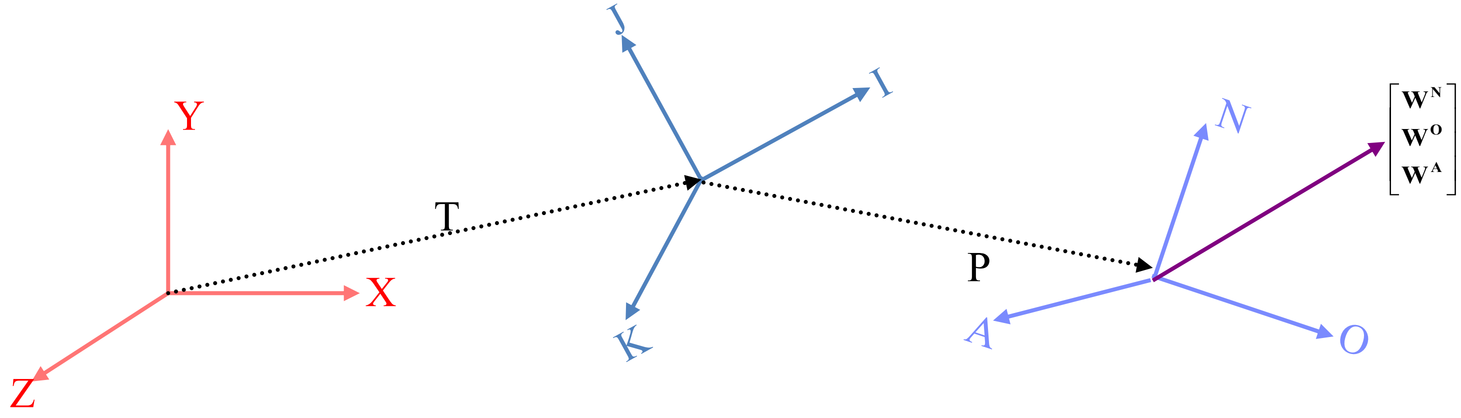
Product of the two matrices

Notice that H can also be written as:

$$H = \begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_x & j_x & k_x & 0 \\ i_y & j_y & k_y & 0 \\ i_z & j_z & k_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & P_i \\ 0 & 1 & 0 & P_j \\ 0 & 0 & 1 & P_k \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_i & o_i & a_i & 0 \\ n_j & o_j & a_j & 0 \\ n_k & o_k & a_k & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

H = (Translation relative to the XYZ frame) * (Rotation relative to the XYZ frame) * (Translation relative to the IJK frame) * (Rotation relative to the IJK frame)

The Homogeneous Matrix is a concatenation of numerous translations and rotations



One more variation on finding H:

H = (Rotate so that the X-axis is aligned with T)

* (Translate along the new t-axis by || T || (magnitude of T))

* (Rotate so that the t-axis is aligned with P)

* (Translate along the p-axis by || P ||)

* (Rotate so that the p-axis is aligned with the O-axis)

This method might seem a bit confusing, but it's actually an easier way to solve our problem given the information we have. Here is an example…

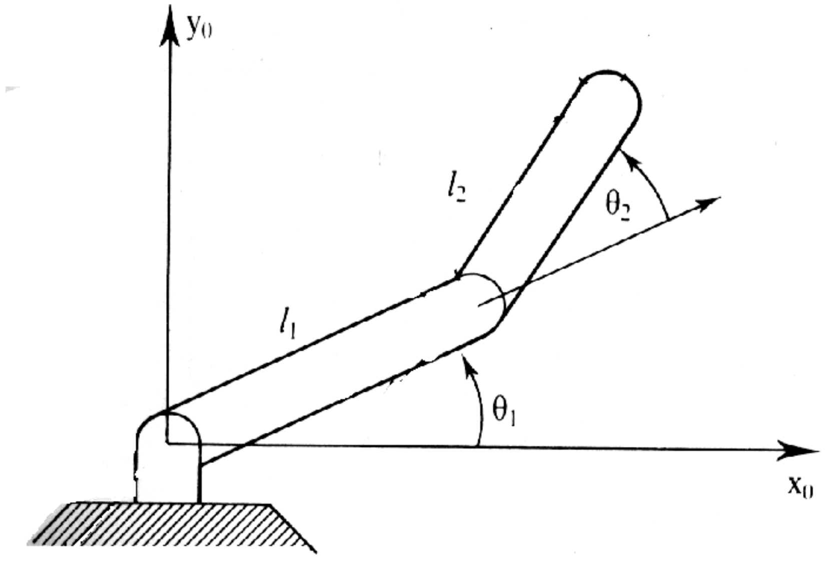
Forward Kinematics

The Situation:

You have a robotic arm that starts out aligned with the x_o-axis. You tell the first link to move by θ_1 and the second link to move by θ_2.

The Quest:

What is the position of the end of the robotic arm?



Solution:

1. Geometric Approach

This might be the easiest solution for the simple situation. However, notice that the angles are measured relative to the direction of the previous link. (The first link is the exception. The angle is measured relative to it's initial position.) For robots with more links and whose arm extends into 3 dimensions the geometry gets much more tedious.

2. Algebraic Approach

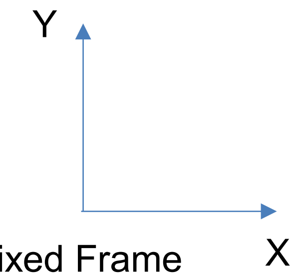
Involves coordinate transformations.

Forward Kinematics

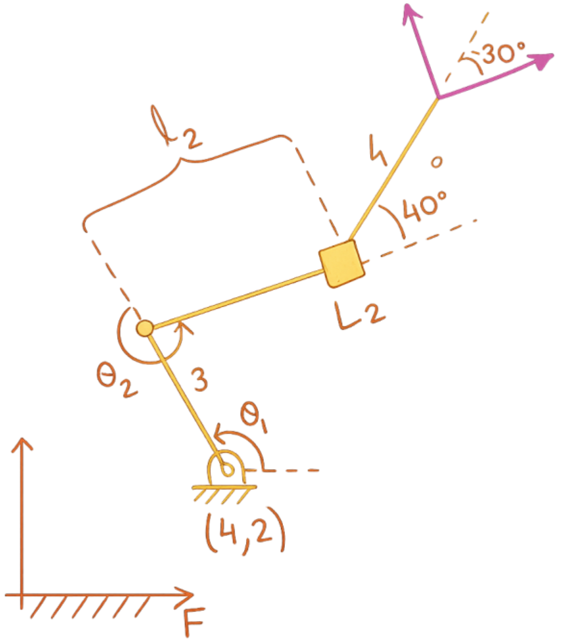
Forward Kinematics — Planar RPR Robot

(a) For the RPR robot shown, determine the location of the moving reference frame by writing the forward kinematics. (No need to multiply out the homogeneous transforms.)

(b) For $\theta_1 = 48^\circ$, $\theta_2 = 112^\circ$, $L_2 = 3.6$, state the values of x, y, and θ of the moving reference frame relative to the fixed reference frame.



Fixed Frame



Forward Kinematics

Part (a)

Step 1: Define the Rotation Matrix

For any planar rotation,

$$A(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

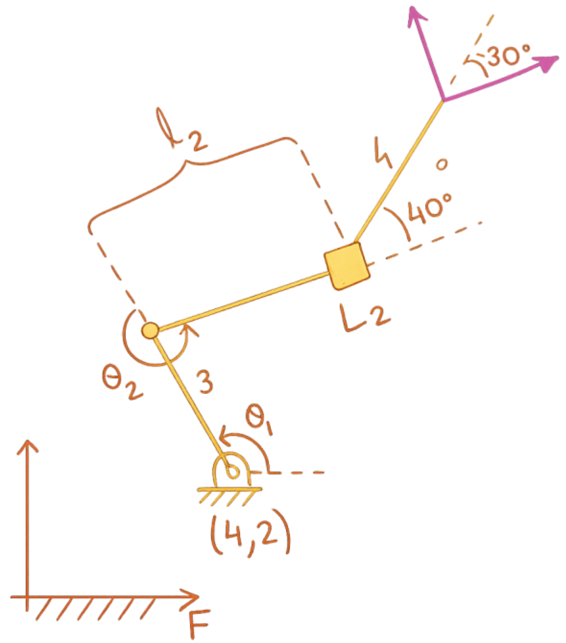
Step 2: Define the Homogeneous Transformation Form

Each link is represented as:

$$H = \begin{bmatrix} A(\theta) & p \\ 0\ 0 & 1 \end{bmatrix}$$

where

$$p = \begin{bmatrix} x \\ y \end{bmatrix}$$



Forward Kinematics

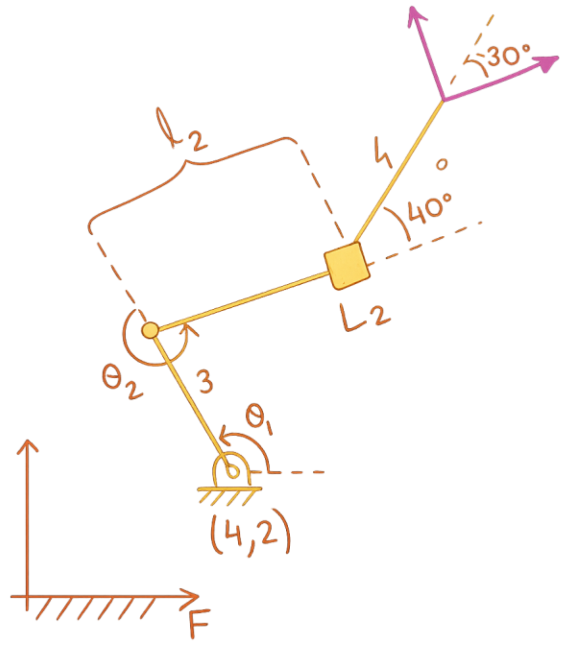
Step 3: Kinematic Chain

The frame sequence is:

$$F \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow M_4$$

Therefore,

$$H = H_1 H_2 H_3 H_4$$



Mobile and Autonomous Robots

Forward Kinematics

Step 4: Write Each Individual Transform Symbolically

Link 1

$$H_1 = \begin{bmatrix} A(\theta_1) & \begin{bmatrix} 4 \\ 2 \end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

Link 2

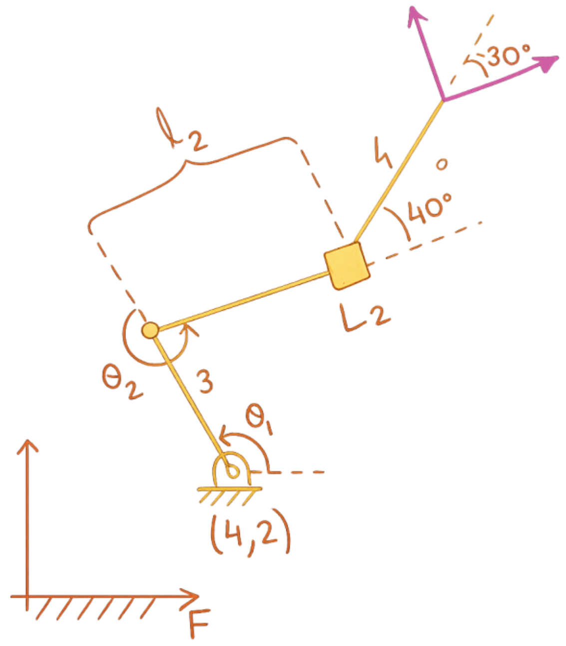
$$H_2 = \begin{bmatrix} A(\theta_2) & \begin{bmatrix} 3 \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

Link 3

$$H_3 = \begin{bmatrix} A(40^\circ) & \begin{bmatrix} L_2 \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}$$

Link 4

$$H_4 = \begin{bmatrix} A(-30^\circ) & \begin{bmatrix} 4 \\ 0 \end{bmatrix} \\ 0 & 1 \end{bmatrix}$$



Mobile and Autonomous Robots

Forward Kinematics

Part (b)

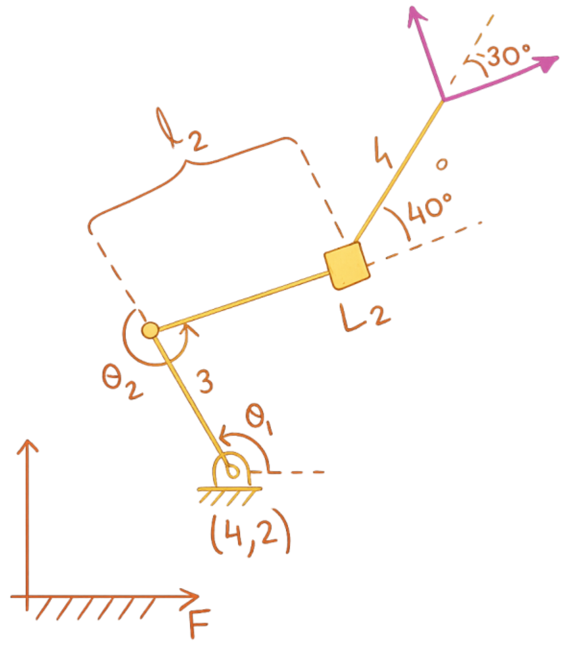
Step 5: Expand Rotation Matrices

$$A(\theta_1) = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{bmatrix}$$

$$A(\theta_2) = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 \\ \sin\theta_2 & \cos\theta_2 \end{bmatrix}$$

$$A(40^\circ) = \begin{bmatrix} \cos 40^\circ & -\sin 40^\circ \\ \sin 40^\circ & \cos 40^\circ \end{bmatrix}$$

$$A(-30^\circ) = \begin{bmatrix} \cos 30^\circ & \sin 30^\circ \\ -\sin 30^\circ & \cos 30^\circ \end{bmatrix}$$



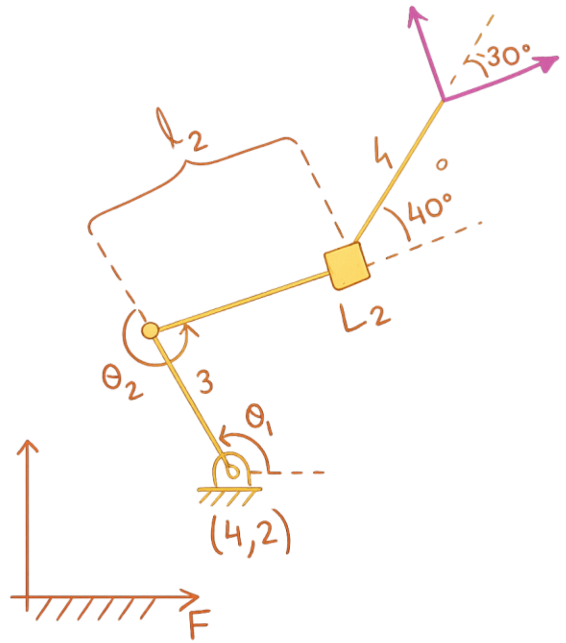
Forward Kinematics

Step 6: Substitute Numerical Values

Given:

$$\theta_1 = 48^\circ, \quad \theta_2 = 112^\circ, \quad L_2 = 3.6$$

$$\cos 48^\circ = 0.6691, \quad \sin 48^\circ = 0.7431$$
$$\cos 112^\circ = -0.3746, \quad \sin 112^\circ = 0.9272$$
$$\cos 40^\circ = 0.7660, \quad \sin 40^\circ = 0.6428$$
$$\cos 30^\circ = 0.8660, \quad \sin 30^\circ = 0.5$$



Forward Kinematics

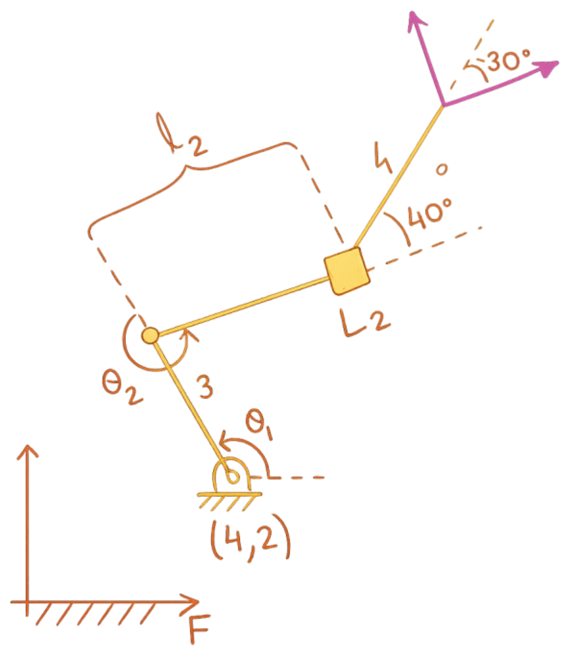
Step 7: Write Numeric Matrices

$$H_1 = \begin{bmatrix} 0.6691 & -0.7431 & 4 \\ 0.7431 & 0.6691 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$H_2 = \begin{bmatrix} -0.3746 & -0.9272 & 3 \\ 0.9272 & -0.3746 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$H_3 = \begin{bmatrix} 0.7660 & -0.6428 & 3.6 \\ 0.6428 & 0.7660 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$H_4 = \begin{bmatrix} 0.8660 & 0.5 & 4 \\ -0.5 & 0.8660 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



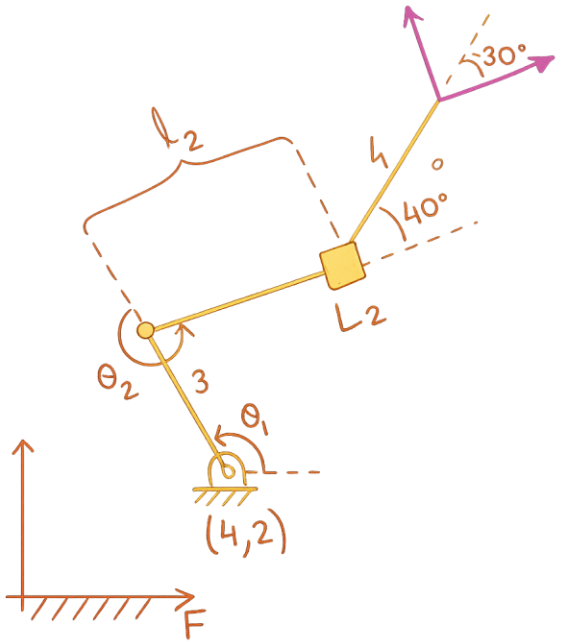
Forward Kinematics

Step 8: Multiply the Transforms

$$H = H_1 H_2 H_3 H_4$$

After performing the matrix multiplications,

$$H = \begin{bmatrix} -0.9848 & -0.1736 & -1.1343 \\ 0.1736 & -0.9848 & 4.0926 \\ 0 & 0 & 1 \end{bmatrix}$$



Step 9: Extract Position

$$x = -1.1343$$

$$y = 4.0926$$

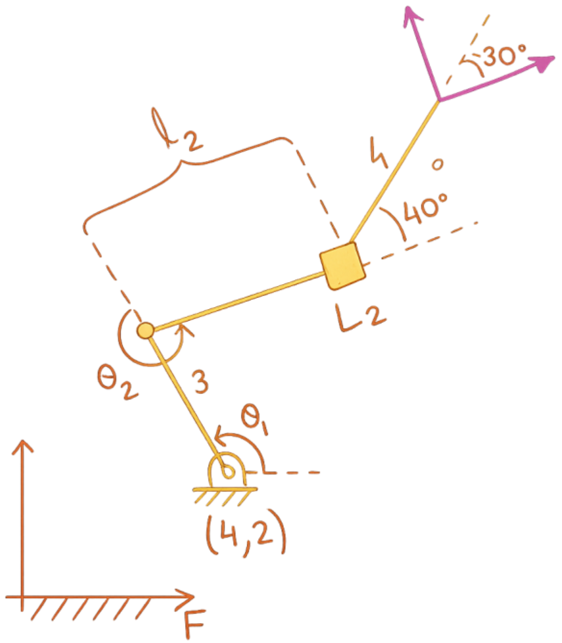
Forward Kinematics

Step 10: Extract Orientation

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{0.1736}{-0.9848}$$

$$\theta = \tan^{-1}\left(\frac{0.1736}{-0.9848}\right)$$

$$\boxed{\theta \approx 170^\circ}$$

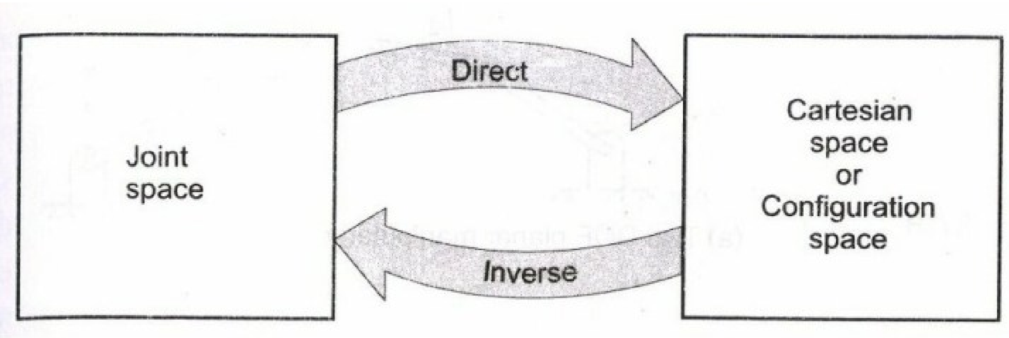


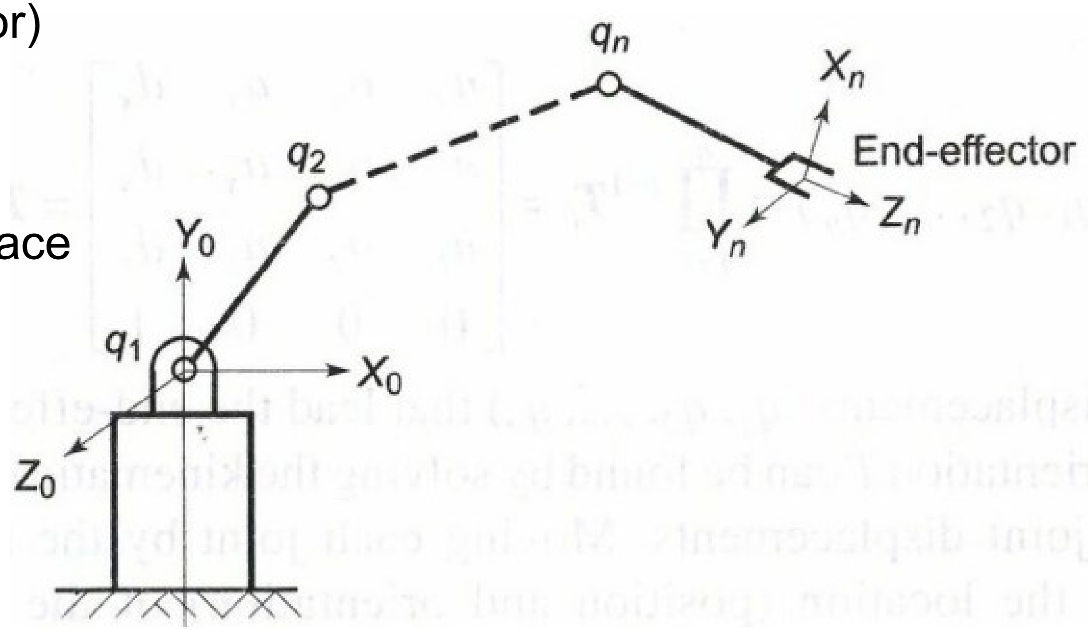
Final Answer

$$\boxed{x = -1.1343, \quad y = 4.0926, \quad \theta = 170^\circ}$$

Inverse Kinematics

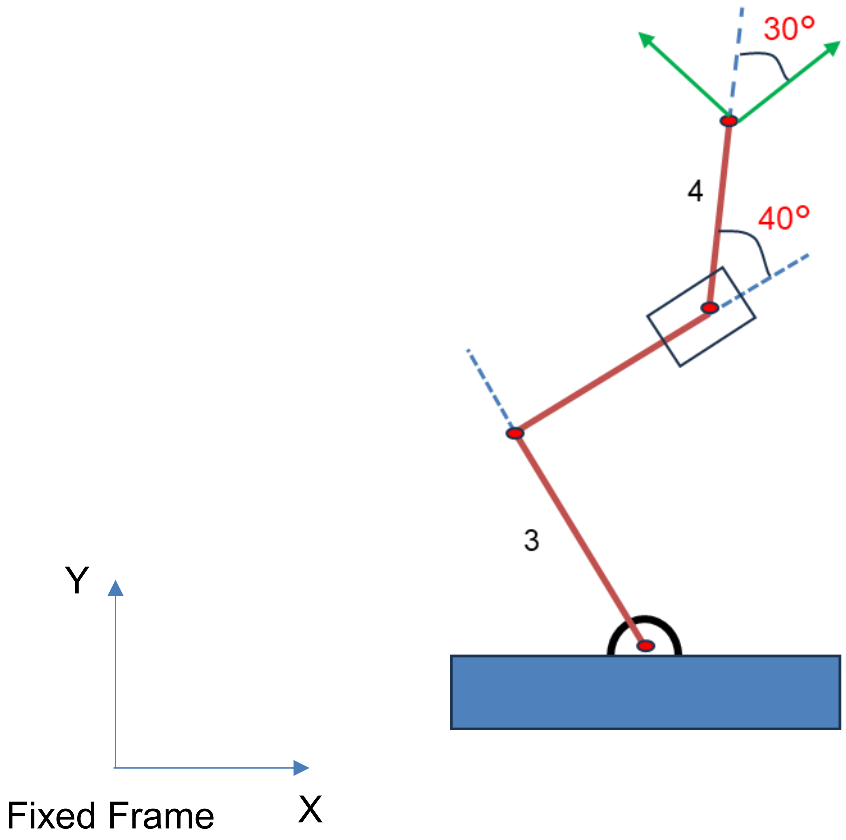
- Inverse kinematics is a mathematical process used to calculate the joint positions that are needed to place a robot's end effector at a specific position and orientation (also known as its "pose").
- The configuration or position and orientation of the end effector is a function of joint displacement variables q1, q2,…. qn.
- The position and orientation of the End Effector is collectively known as configuration.
- For n-DoF manipulator the set of n joint displacement variables is represented as nx1 vector.
- Configuration space or Cartesian Space (space of end effector)
- Joint space (space traced by joints)
- FK model is the mapping of Joint space to Cartesian space.
- IK model is mapping from the Cartesian space to the Joint space





Inverse Kinematics

Example: For the robot's end-effector to be at M (4.7343, -3.8424,160°) where the coordinates given are (x, y, θ) relative to the fixed reference frame F, determine the required robot parameters to achieve it.



Inverse Kinematics

Step 1: Given Data

Desired end-effector pose:

$$(x, y, \theta) = (4.7343,\ -3.8424,\ 160^\circ)$$

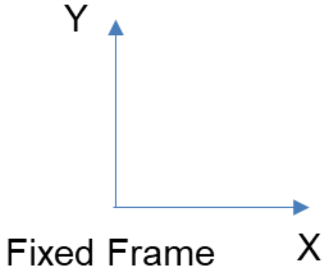
Known parameters:

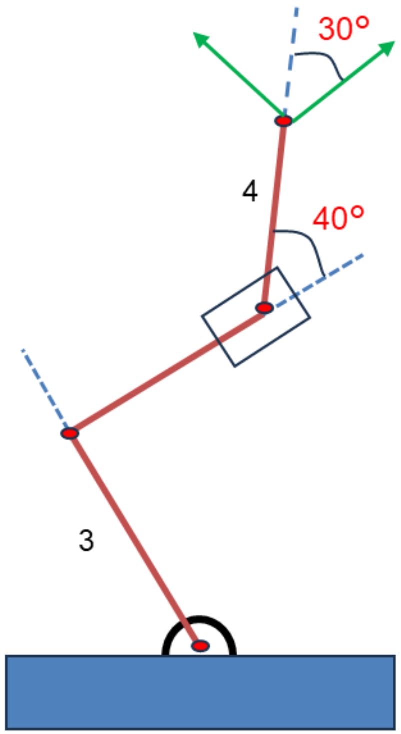
$$L_1 = 3$$

$$\text{offset angle} = 40^\circ$$

Unknowns:

$$\theta_1,\ \theta_2,\ L_2$$





Inverse Kinematics

Step 2: Homogeneous Transformation of Target Pose

General homogeneous transform:

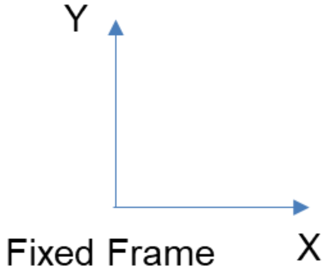
$$H = \begin{bmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{bmatrix}$$

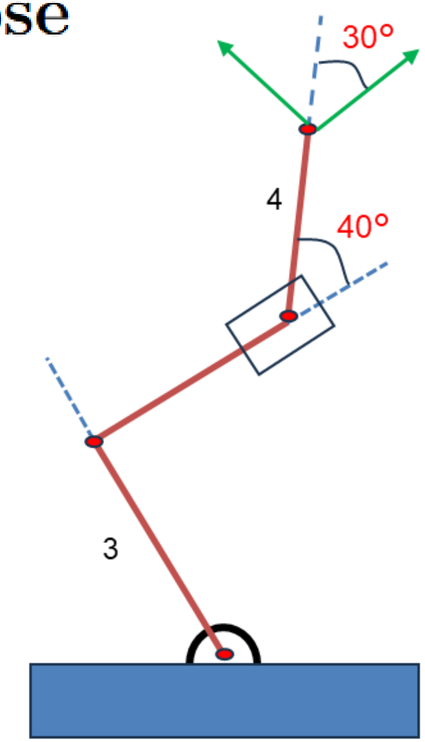
Substitute:

$$\cos 160^\circ = -0.9848$$

$$\sin 160^\circ = 0.1736$$

$$H = \begin{bmatrix} -0.9848 & 0.1736 & 4.7343 \\ -0.1736 & -0.9848 & -3.8424 \\ 0 & 0 & 1 \end{bmatrix}$$





Inverse Kinematics

Step 3: Forward Kinematics Chain

Total forward kinematics:

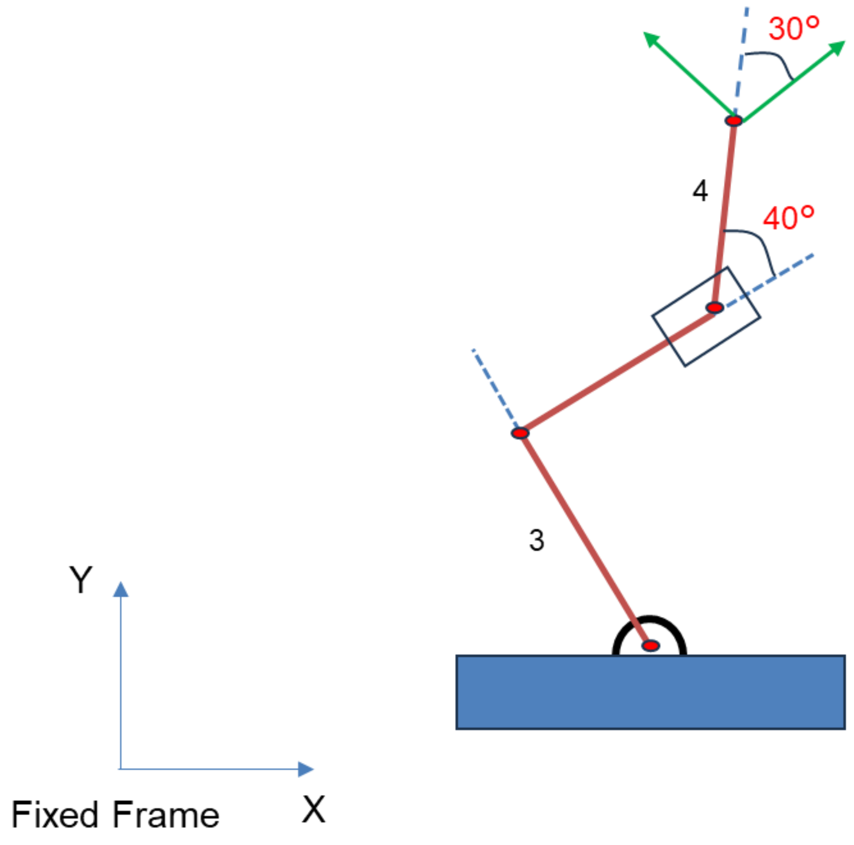
$$H_0 H_1 H_2 H_3 H_4 = H$$

Individual matrices:

$$H_0 = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$H_1 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad H_3 = \begin{bmatrix} \cos 40^\circ & -\sin 40^\circ & L_2 \\ \sin 40^\circ & \cos 40^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$H_2 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 3 \\ \sin\theta_2 & \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad H_4 = \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ & 4 \\ \sin 30^\circ & \cos 30^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Inverse Kinematics

Step 4: Isolate Unknown Joints

Move known transforms to the right:

$$H_1 H_2 H_3 = H_0^{-1}\ H\ H_4^{-1}$$

Inverse of H_0:

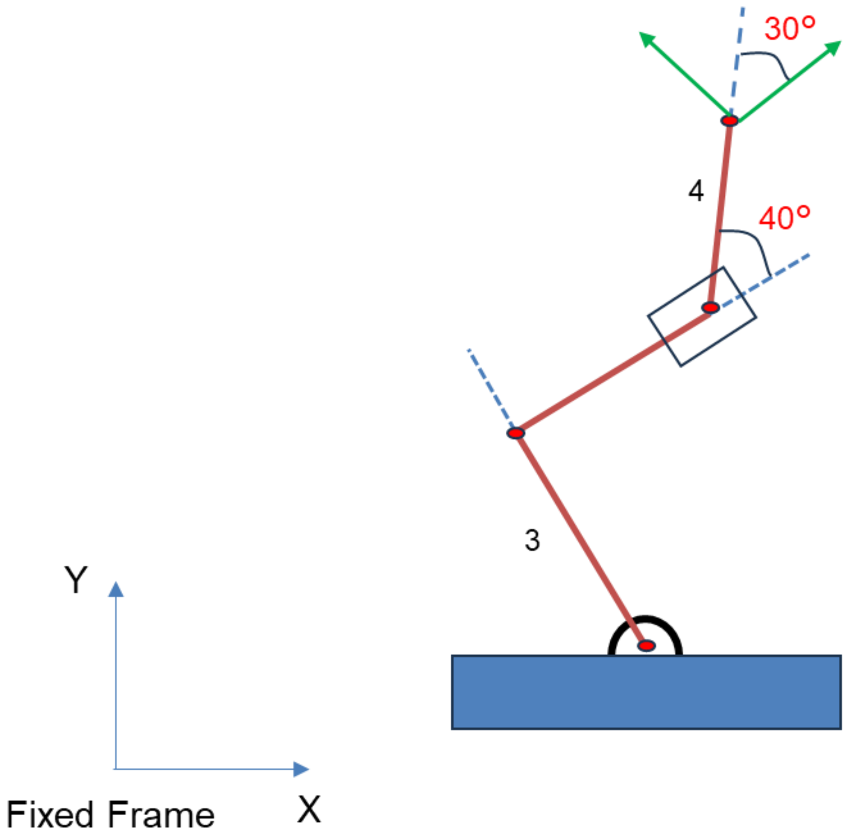
$$H_0^{-1} = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\cos 30^\circ = 0.866, \quad \sin 30^\circ = 0.5$$

$$H_4^{-1} = \begin{bmatrix} 0.866 & 0.5 & -3.464 \\ -0.5 & 0.866 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

After multiplication:

$$H_1 H_2 H_3 = \begin{bmatrix} -0.9848 & 0.1736 & 4.6735 \\ -0.1736 & -0.9848 & -5.1478 \\ 0 & 0 & 1 \end{bmatrix}$$



Inverse Kinematics

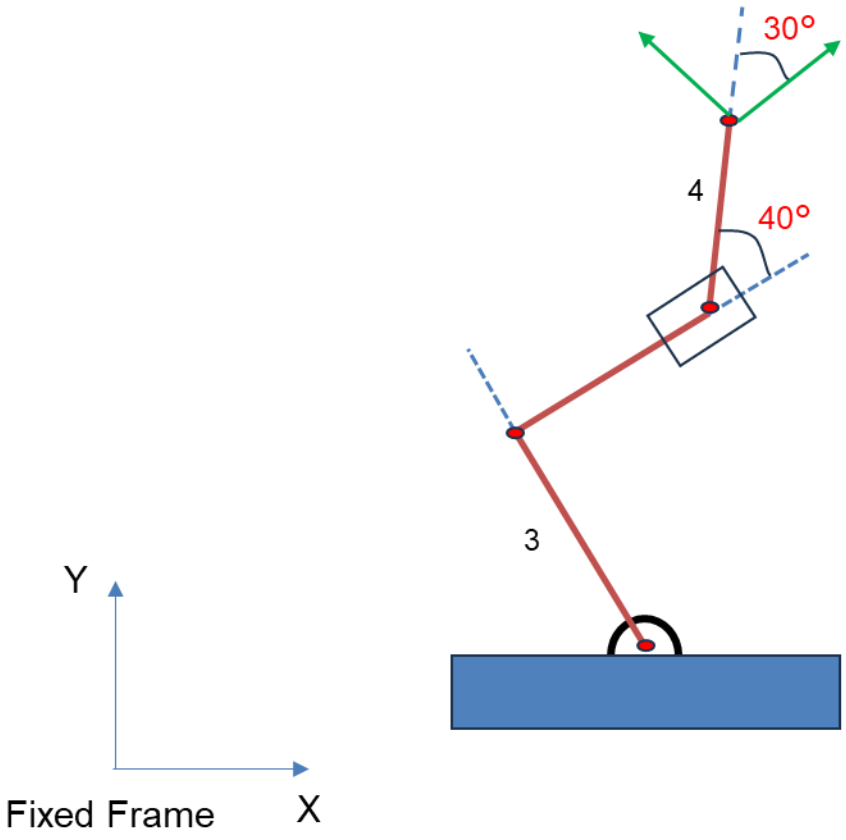
Step 5: Rotation Equations

$$\cos(\theta_1 + \theta_2 + 40^\circ) = -0.9848$$
$$\sin(\theta_1 + \theta_2 + 40^\circ) = -0.1736$$

$$\tan(\theta_1 + \theta_2 + 40^\circ) = \frac{-0.1736}{-0.9848} = 0.1763$$

$$\theta_1 + \theta_2 + 40^\circ = -170^\circ$$

$$\boxed{\theta_1 + \theta_2 = -210^\circ}$$



Inverse Kinematics

Step 6: Position Equations

$$3\cos\theta_1 + L_2\cos(\theta_1+\theta_2) = 4.6735$$

$$3\sin\theta_1 + L_2\sin(\theta_1+\theta_2) = -5.1478$$

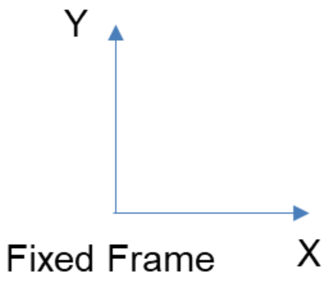
Square and add:

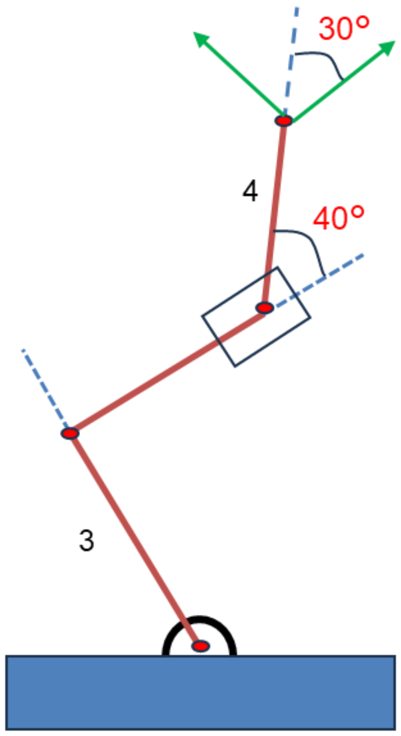
$$9 = (4.6735 - L_2 c)^2 + (-5.1478 - L_2 s)^2$$

Solutions:

$$L_{2a} = -8.74$$

$$L_{2b} = -4.5$$





Inverse Kinematics

Step 7: Solve Joint Angles

Case A:

$$\theta_1 = -165^\circ, \quad \theta_2 = -45^\circ$$

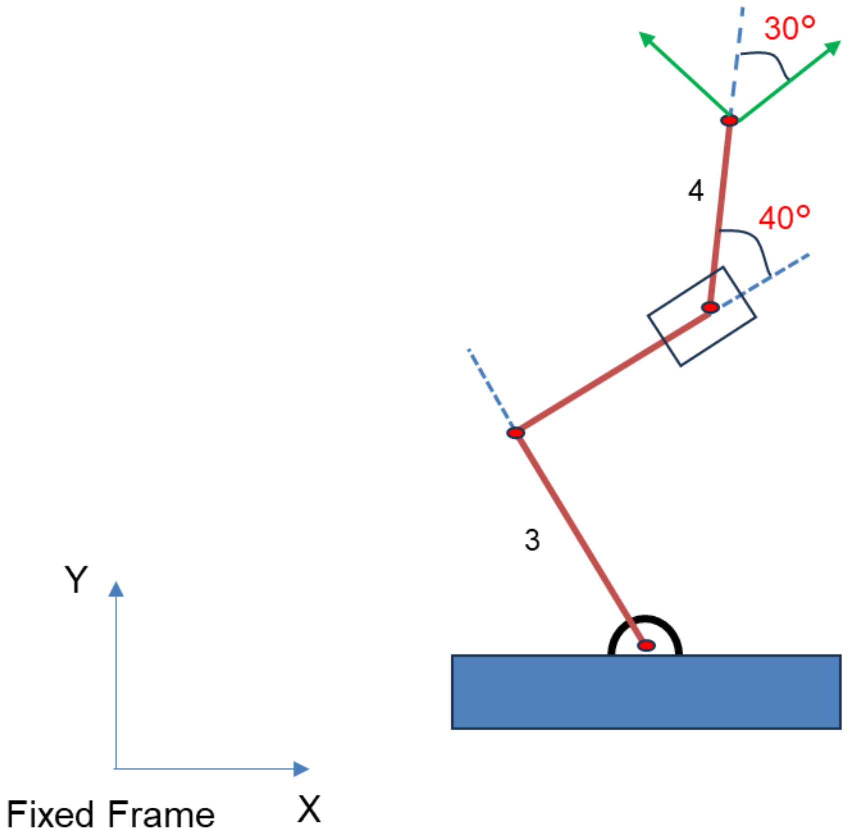
Case B:

$$\theta_1 = -75^\circ, \quad \theta_2 = -135^\circ$$

Final Answers

$$\boxed{\theta_1 = -165^\circ,\ \theta_2 = -45^\circ,\ L_2 = -8.74}$$

$$\boxed{\theta_1 = -75^\circ,\ \theta_2 = -135^\circ,\ L_2 = -4.5}$$



THANK YOU

Dr. Ashok Kumar Patil
Department of Computer Science and Engineering
ashokkumarp@pes.edu